

PM7250B Power Management IC

80-PM008-5. Rev. F

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Revision History

Revision	Date	Description
A	April 2019	Initial release
B	June 2019	■ Slide 38 USB Type-C Debug Accessory Mode (DAM) Support: Updated the GPIO numbers ■ Slide 39 DebugAccessory.SNK Mode Support: Updated the GPIO numbers ■ Slide 79 PMIC GPIO Pins: Updated the custom GPIO features ■ Slide 81 PM7250B GPIO Assignments: Added this slide ■ Slide 90 Unused Pin Termination – PM7250B: Added this slide ■ Slide 91 PM7250B Layout Guidelines: Added this section
C	July 2021	■ Slide 103 Appendix 1 – Powering Schemes for Non-Mobile-Phone Applications: Added this section
D	January 2022	■ Slide 114 Use Case 4: Charging with a Generic DC Charger and USB_IN Input: Added the note ■ Slide 119 Considerations for Swappable Battery Application (1 of 3): Added this slide ■ Slide 120 Considerations for Swappable Battery Application (2 of 3): Added this slide ■ Slide 121 Considerations for Swappable Battery Application (3 of 3): Added this slide
E	January 2024	■ Slide 13 <i>PM7250B Charger Overview</i>: Updated SMB1390 to SMB1393 ■ Slide 14 <i>PM7250B Charger Block Diagram</i>: Updated SMB1390 to SMB1393 ■ Slide 15 <i>Comparison with Charger</i>: Added this slide ■ Slide 49 <i>Parallel Charging with SMB1393</i>: Updated SMB1390 to SMB1393 and updated the image ■ Slide 50 <i>PM7250B Battery Charging: Triple Charge</i>: Added this slide ■ Slide 52 <i>Parallel Charging with SMB1355 or SMB1393 Stand-alone</i>: Updated SMB1390 to SMB1393 ■ Slide 82 <i>PM7250B GPIO Assignments</i>: Updated SMB1390 to SMB1393 and deleted SMB1396 ■ Slide 92 <i>Ship Mode</i>: Added this slide ■ Slide 117 <i>Use Case 4C: Battery Hot Swap Using Secondary Battery</i>: Added this slide ■ Slide 121 <i>Battery Hot Swap and FG reset flow</i>: Added this slide ■ Slide 122 <i>Considerations for Swappable Battery Application</i>: Added QCM4490/QCS4490 and PMK4450 ■ Slide 126 <i>Use Case 6: Charging with Multiple Inputs (cont.)</i>: Added this slide

Revision History

Revision	Date	Description
F	May 2024	Slide 128 VPH_PWR behavior during battery insertion: Added this slide

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Section 1

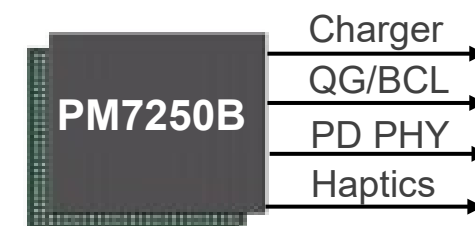
Introduction

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PM7250B Overview

- PM7250B provides best-in-class charging performance while integrating many related functions.

- 12 V charger [28 V overvoltage protection (OVP)]
 - 4 A single-path charge current
 - 8 A multipath charge current
- Qualcomm® Quick Charge™ 2, Quick Charge 3, or Quick Charge 4 + USB Power Delivery (PD) support
- Type-C detection and 22 V configuration channel (CC)/sideband use (SBU) pin protection
- USB PD PHY 3.0 support
- USB host mode support
- Battery interface module (BIM)
- Standalone ADC module
 - Houses two 15-bit sigma delta ADCs that are used by the charger, fuel gauge (FG), and battery current limit (BCL) modules
- Qualcomm battery gauge
 - Internal battery FET (6 mΩ) sensing
 - A hybrid of voltage and current based SoC and voltage loop feedback correction.
- Vibrator driver for ERM applications
- SPMI/I2C dual-mode interface
 - SPMI for Qualcomm® Snapdragon™ and I2C for platforms not based on Snapdragon





Section 2

PM7250B Input Power Management

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Input Power Management

■ Input power management functions include:

- SCHG and battery charging
- Qualcomm battery gauge
- Battery interface module
- Battery current limiting

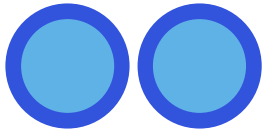
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SCHG Feature Summary

- High efficiency switching charger with power path FETs having extremely low R_{ds} (ON).
- Dual power source support – USB and DC input
 - DC input is joined at MID_CHG node eliminating dissipation across front porch FET.
- Parallel charging support mitigates thermal issues at high charging currents.
- Charge current support of up to 4 A (1.0 mm inductor) stand-alone and 8 A with parallel charging.
- Wide input voltage range of +3.6 V to +13.2 V
- +28 V input over voltage protection.
- USB reverse boost current up to 1.5 A.
- Quick Charge 2.0/3.0/4.0 and 4.0+ support
- JEITA compliance

Battery FET Architecture

- If the battery voltage is above the minimum system voltage, the BATFET fully turns on, and only charging or discharging current is sensed by the BATFET control logic.
- If the battery voltage is between 0 V to 2.1 V, the battery is disconnected from the system. A 75 mA current (trickle charge) source is turned on, which charges the battery up to 2.1 V.
- Battery interface:
 - Current path control allows system operation with a deeply discharged/missing battery.
 - Trickle/precharging using parallel current source(Isrc) with BATFET off
 - Battery supplemental mode (BSM) with BATFET fully on
- Voltage-based battery supplement mode (BSM) detection and entry: an analog comparator triggers BSM when detecting $V_{PH_PWR} < V_{BATT_PWR} - 50 \text{ mV}$ ($\pm 25 \text{ mV}$) for definite amount of time
- During BSM, instead of linearly regulating V_{PH_PWR} 50 mV below V_{BATT_PWR} , BSM fully turns on the BATFET to allow battery to supplement the system current.



Section 2.1

Charger Module

2.1.1 USB APSD

[17](#)

2.1.2 AICL

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2.1.3 JEITA Compliance

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PM7250B Charger Overview

- PM7250B charger block diagram
- USB charging path
 - USB APSD
 - AICL
 - Surge protection
 - Type-C enhancements
 - PD enhancements
 - Type-C moisture/corrosion protection
- DC_IN charging path
 - Interface with a generic DC charger input
- Charger power stage
 - Buck mode efficiency
 - Boost mode efficiency
- Battery interface
 - New battery supplemental mode (BSM)
 - Differential battery sensing
- Parallel charging
 - Parallel charging with SMB1355
 - Parallel charging with SMB1390

PM7250B Charger Block Diagram

- PM7250B incorporates a dual-input, high-efficiency switch-mode battery charger for single-cell lithium batteries.
- Up to 4 A (1 mm height inductor) single-path charging
- It also supports 5 A parallel charging with SMB1355. It supports 6 A parallel charging with 1 × SMB1393 and 8 A parallel charging with 2 × SMB1393.

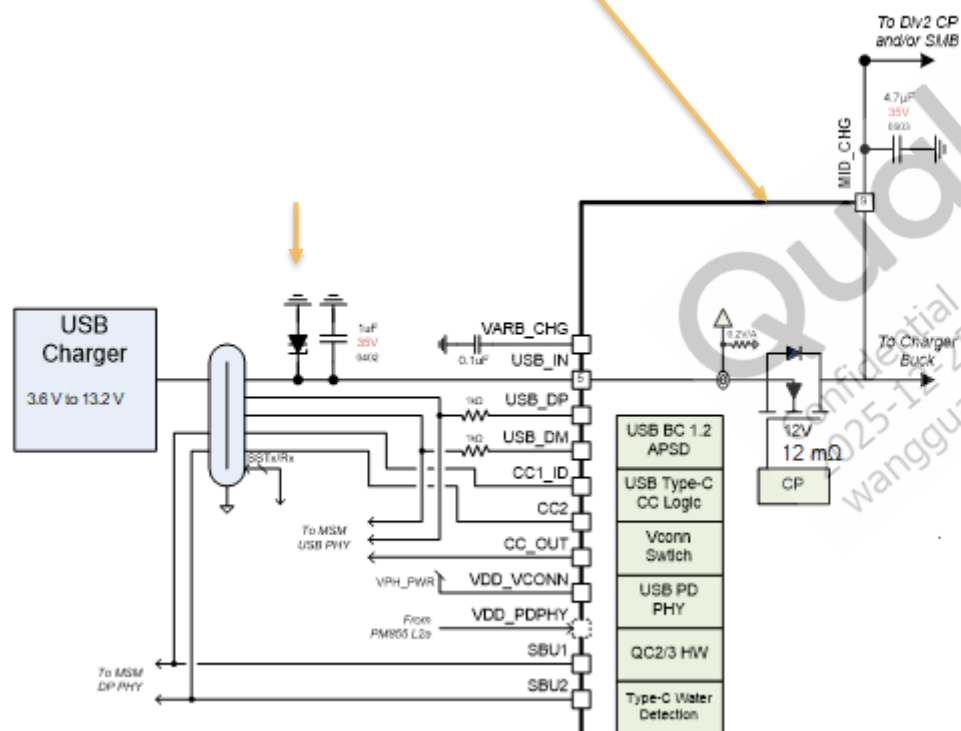
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Comparison with Charger

Parameter	PM660 + PM660A/L	PM7250B
USB_IN Operating voltage	3.6 V to 10 V	3.7 V to 12.6 V
DC_IN/WLS_IN (WLS = Wireless power input path)	DC_IN or WLS_IN	DC_IN through MID_CHG
USB OTG	5 V, 1.5 A	5V, 1.5 A
Adaptor support	Quick charge 2, 3, 4 and USB PD	QC2,3,4+ and USB PD
USB Input OVP abs. max	+16 V	+28 V
Haptics	ERM/LRA.	ERM
Display	AMOLED - PM660A. LCD – PM660L.	-
MISC	- Housekeeping, clocks, output power management (buck, LDO) for system power rails. - Integrated audio codec.	- It is a dedicated charger only. - Separate Qualcomm audio codec (WCD) attaches available.

USB Charging Path

- +3.6 V to +13.2 V operating range on USB_IN with +28 V OVP; input current limit up to 5 A
- 12 mΩ FP FET to minimize power loss during high current charging
- Enhanced FP FET clamping for surge protection (requires an external TVS diode)



- Automatic power source detection (APSD) per the *USB Battery Charging Specification*, Rev. 1.2 (BC1.2). See [USB APSD](#) section.
- Automatic input current limit (AICL) for universal USB adapter compatibility. See [AICL](#) section.
- USB Type-C enhancements:
 - *USB Type-C Specification*, Rev. 1.3 compliant
 - Type-C connector [liquid presence detection](#)
 - [22 V tolerance](#) on CC1/CC2 and SBU1/SBU2 pins
 - Enhanced A-to-C [legacy cable detection](#)
 - [1.5 W VCONN switch](#) to support VCONN-powered USB devices (VPD) and VCONN-powered accessories (VPA)
- USB power delivery PHY and software stack supporting *USB Power Delivery Specification*, Rev. 3.0, including PPS
- Quick Charge 2 and Quick Charge 3 detection in hardware; Quick Charge 4 software support



Section 2.1.1

USB APSD

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USB APSD (1 of 4)

The SCHG is compatible with the *Battery Charging Specification Revisions 1.2* developed by the USB Implementers Forum (USB-IF). Its APSD algorithm determines how much current can be drawn from an upstream USB port and adjusts the input current limit levels accordingly.

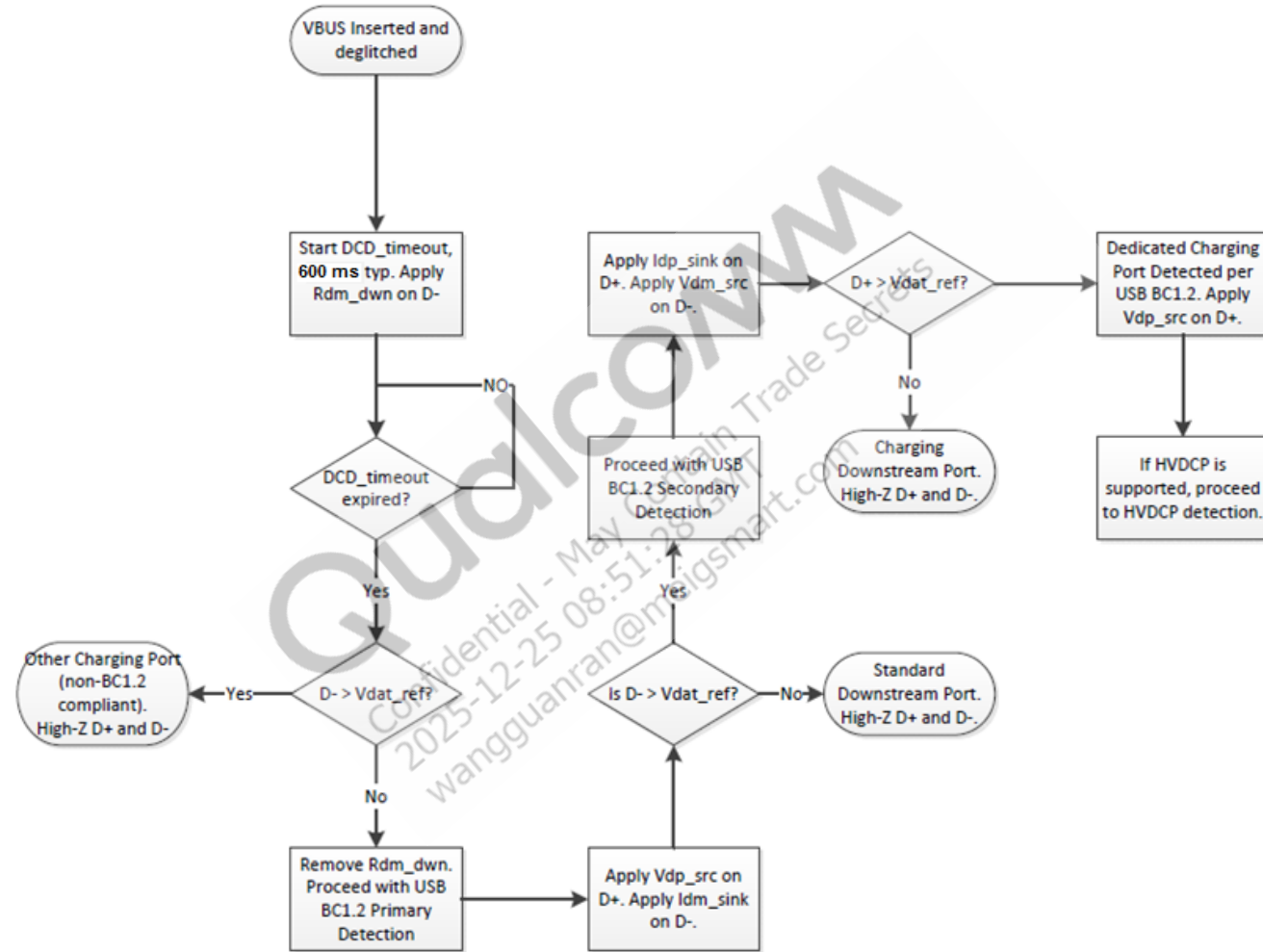
- The PM7250B device automatically detects the type of adapter that is inserted in USBIN by performing D+ and D- lines to set an appropriate input current limit.
- The four power source types that are detected are:
 - Standard downstream port (SDP)
 - This is a computer USB port capable of USB 1.1 (100 mA), USB 2.0 (100/500 mA), or USB 3.0 (150/900 mA). D+ and D- are independently pulled down in the host with a 14.25–24.8 k Ω resistance.
 - Charging downstream port (CDP)
 - This is typically a powered USB hub capable of 1.5 A.
 - Dedicated charging port (DCP)
 - This is a standard wall charger capable of at least 1500 mA. D+ and D- are shorted in the wall adapter with a maximum resistance of 200 Ω . SCHG defaults to high current mode and runs automatic input current limiting (AICL).
 - Other charging port (not covered by *Battery Charging Specification Revisions 1.2*)
 - This is a nonstandard charger with a proprietary D+/D- configuration. Typically, these chargers have similar current capability to normal DCPs. D+ and D- are connected to Vbus via a resistor divider, which causes them to be at specific fixed voltage levels.

USB APSD (2 of 4)

Operational Details

- The algorithm begins as soon as VBUS voltage is detected and is independent of the charge enable status.
- A status register indicates the APSD algorithm results (power source type).
- Input current limits change once the command register is written; before this, the input current limit is determined only by APSD results.
- When the APSD algorithm is completed, the PM7250B device's D+ and D- signal lines enter a Hi-Z state with ~ 4 pF of capacitive load.
- When an SDP is detected with APSD, the default current limit is 500 mA, configured by the register control.

USB APSD (3 of 4)



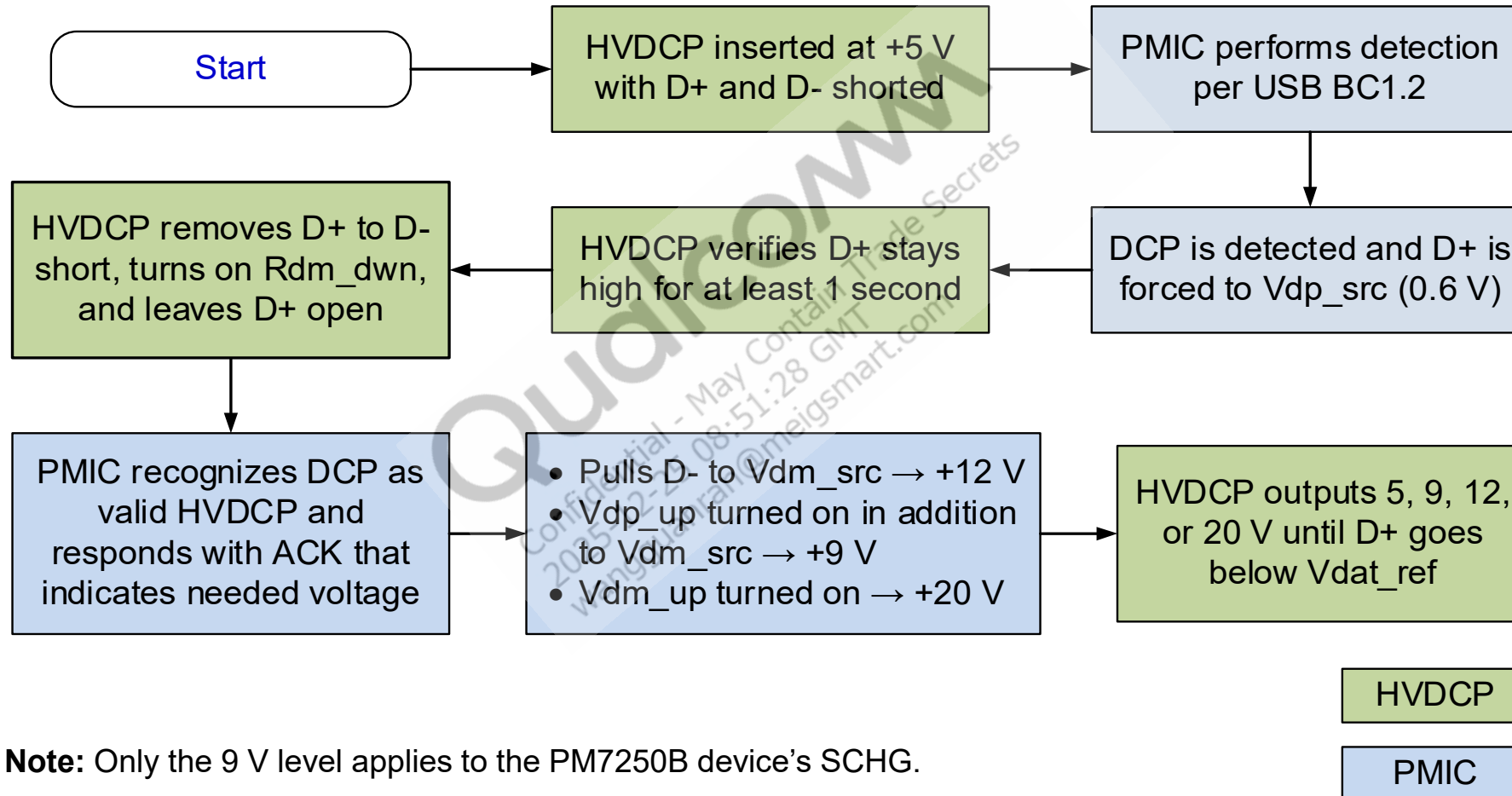
USB APSD (4 of 4)

Dead Battery Provision

- An integrated 30-minute timer ensures that the portable device with a dead battery does not draw 100 mA indefinitely.
- The timer is active when USB_IN is present and a USB host is detected.
- The timer ends under the following conditions:
 - Suspend mode is entered
 - USB 5/USB 9 mode is selected
 - Power source is not an SDP
- During dead-battery charging, the SMBC also brings D+ high to +0.6 V to notify the host of its condition.

High Voltage Dedicated Charging Port (HVDCP)

- The SCHG supports the HVDCP specification that allows charging only from expected voltages – this helps reject noncompliant chargers. It automatically detects HVDCPs using the D+ and D- lines.





Section 2.1.2

AICL

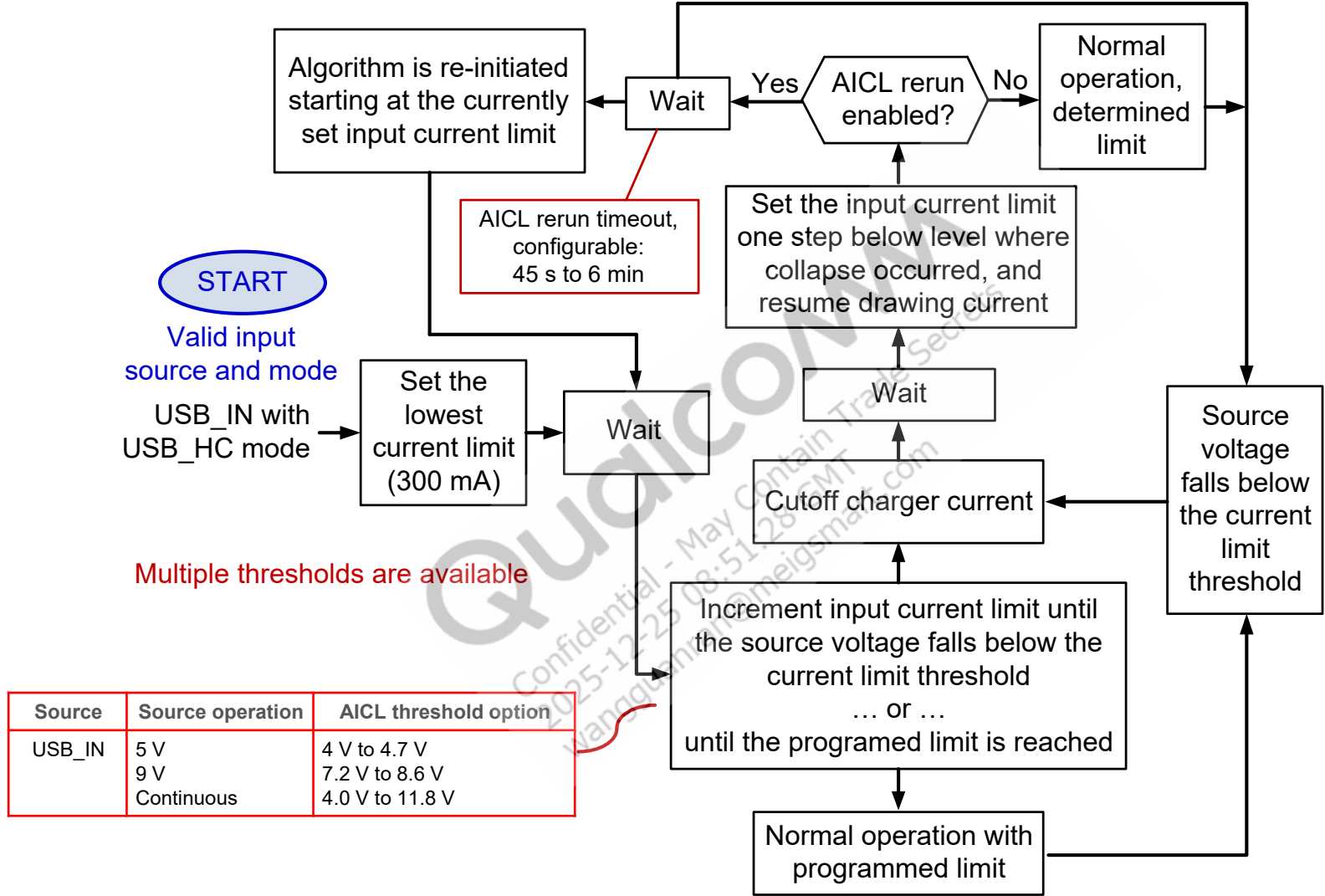
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AICL (1 of 2)

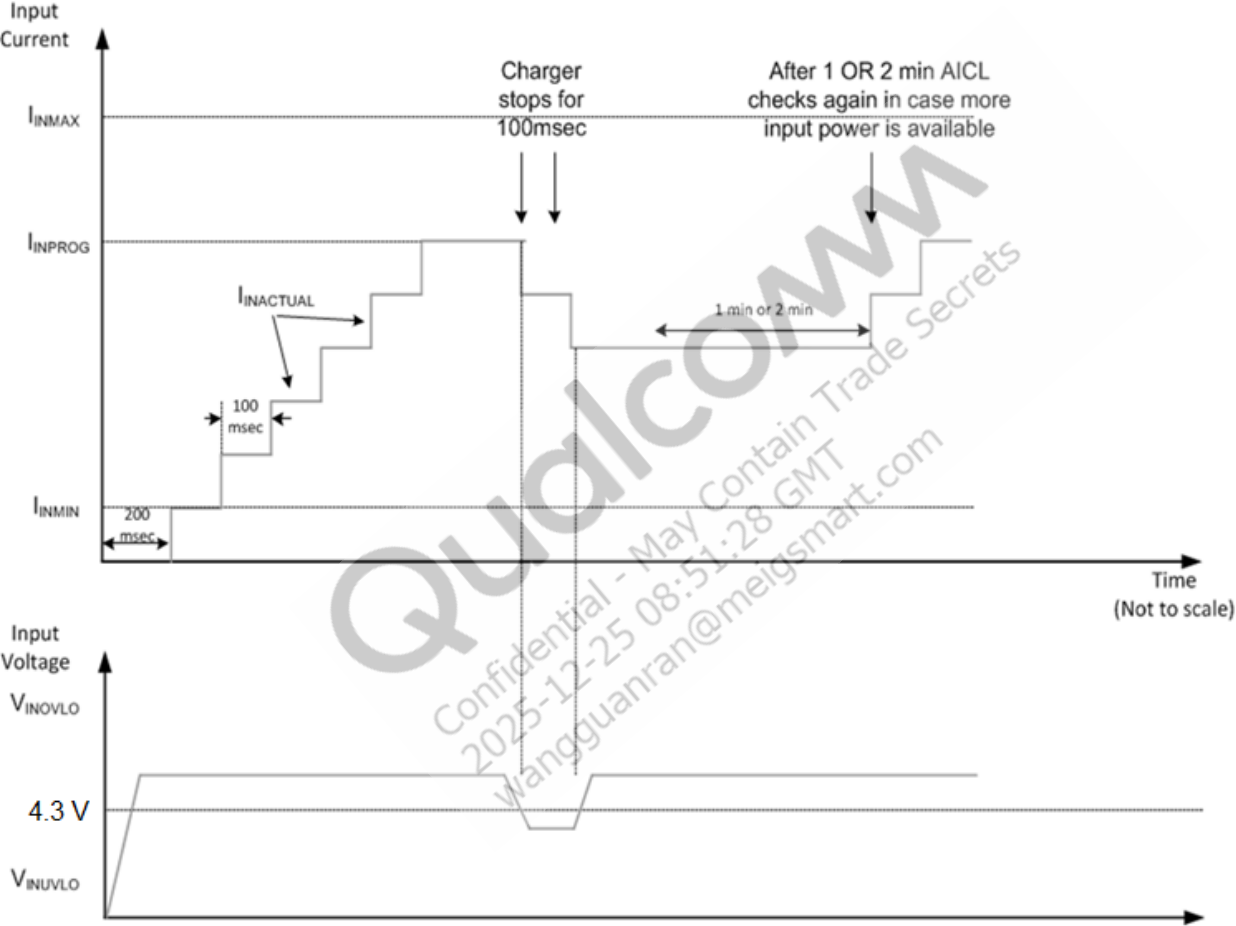
- AICL is an automatic and safe maximization of current from the USB input (only during the USBHC mode).
- The algorithm is only initiated for USBHC mode (default), but the feature can be disabled via hardware register.
- If AICL is enabled, four events can trigger the algorithm. The algorithm is triggered whenever the input voltage collapses.
- If the algorithm must reduce the input current limit, it decrements the input current limit setting until the source voltage is above the AICL threshold.
 - If the input voltage collapses at the lowest current limit setting (300 mA), the charger enters suspend mode.
 - The PM7250B device has two AICL methods (selectable): discrete and continuous.
 - The continuous method (faster) uses a fast ramp and then uses an ADC reading to detect the input current limit.
 - The discrete method (slower – default) uses discrete incremental steps and checks the voltage at every step.
 - Both AICL methods are performed in hardware – minimal software is required.

Input source	Input source voltage operation	AICL threshold option
USB_IN	5 V	4.7 V to 4.0 V
USB_IN	9 V	7.2 V to 8.6 V
USB_IN	Continuous	4.0 V to 11.8 V

AICL (2 of 2)



AICL Discrete Method



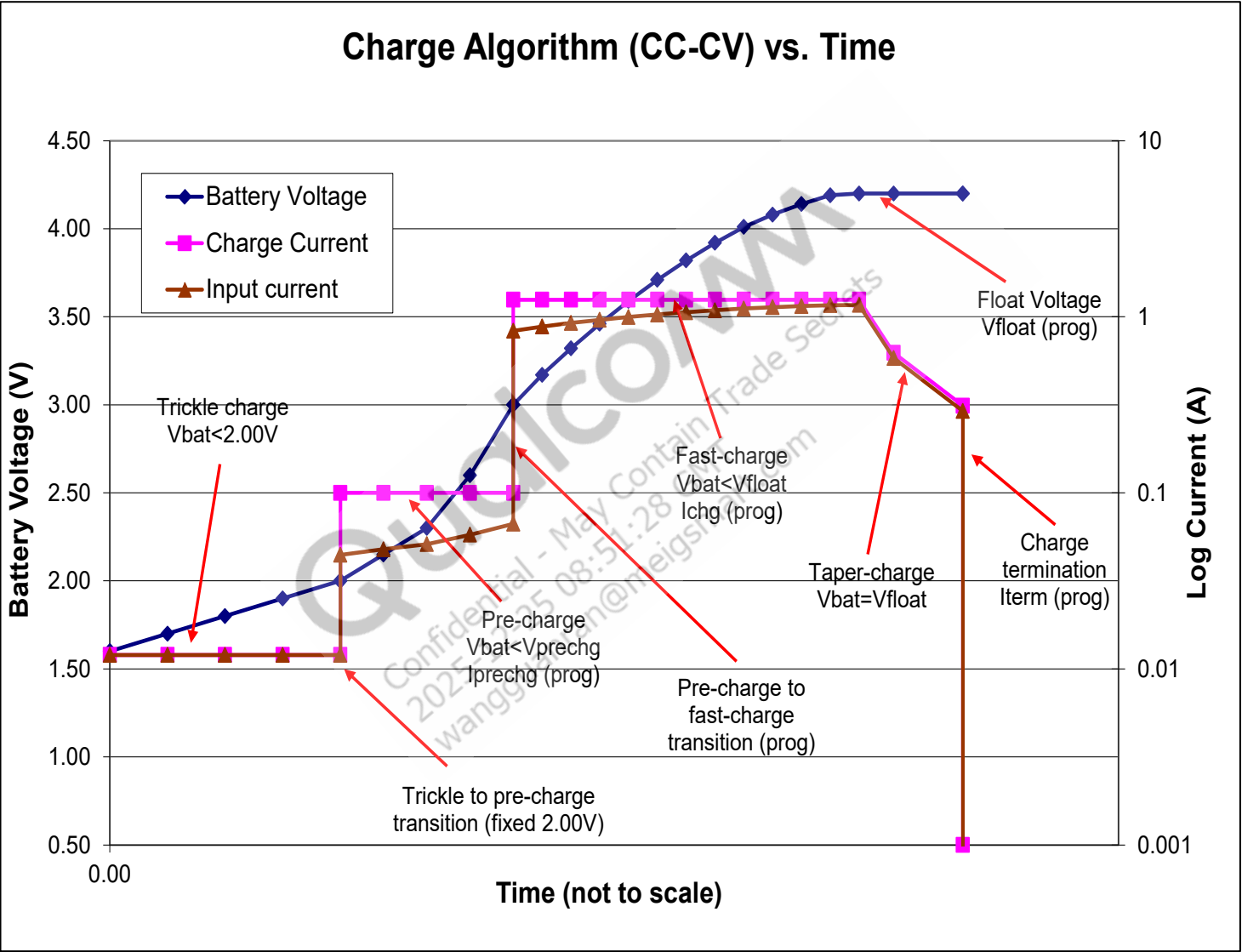
Basic Charging Operation Sequence (1 of 2)

- **Prequalification** – a series of tests after an external source is connected, before initiating the first charge cycle
 - Input voltage level > UVLO threshold, < OVLO threshold, and > battery voltage + 0.1 V.
 - The appropriate ENABLE register SPMI command or EN input must be asserted.
 - Prequalification parameters are continuously monitored, and the charging cycle is suspended when any fall outside their limits.
- **Trickle-charge mode** – The PMIC checks the battery voltage to decide if trickle-charging is required
 - If the battery voltage is below ~ 2.1 V, a charging current of 75 mA (typical) is applied.
 - This charging current resets the battery pack protection circuit and safely raises the battery voltage.
- **Precharge mode** – a new charging category added for this generation PMIC; once the battery voltage exceeds 2.1 V, the PMIC safely performs precharge on deeply discharged cells.
 - The precharge (preconditioning) current is programmable from 100 mA–450 mA, in 50 mA steps.
 - The PMIC remains in this mode until the battery voltage reaches the precharge to fast-charge voltage threshold (programmable from 2.7 V–3.4 V, in 100 mV steps).
 - If this threshold is not exceeded before the precharge timer expires, the charge cycle is terminated and a corresponding timeout fault signal is asserted.
- **Fast charge mode** – when the precharge to fast-charge threshold is crossed, the PMIC enters its constant current (fast charge) mode.
 - The fast charge current level is set by a corresponding register, programmable from 50 mA–8000 mA in 50 mA steps.
 - The fast charge current level (output) is always limited by the input current limit setting.

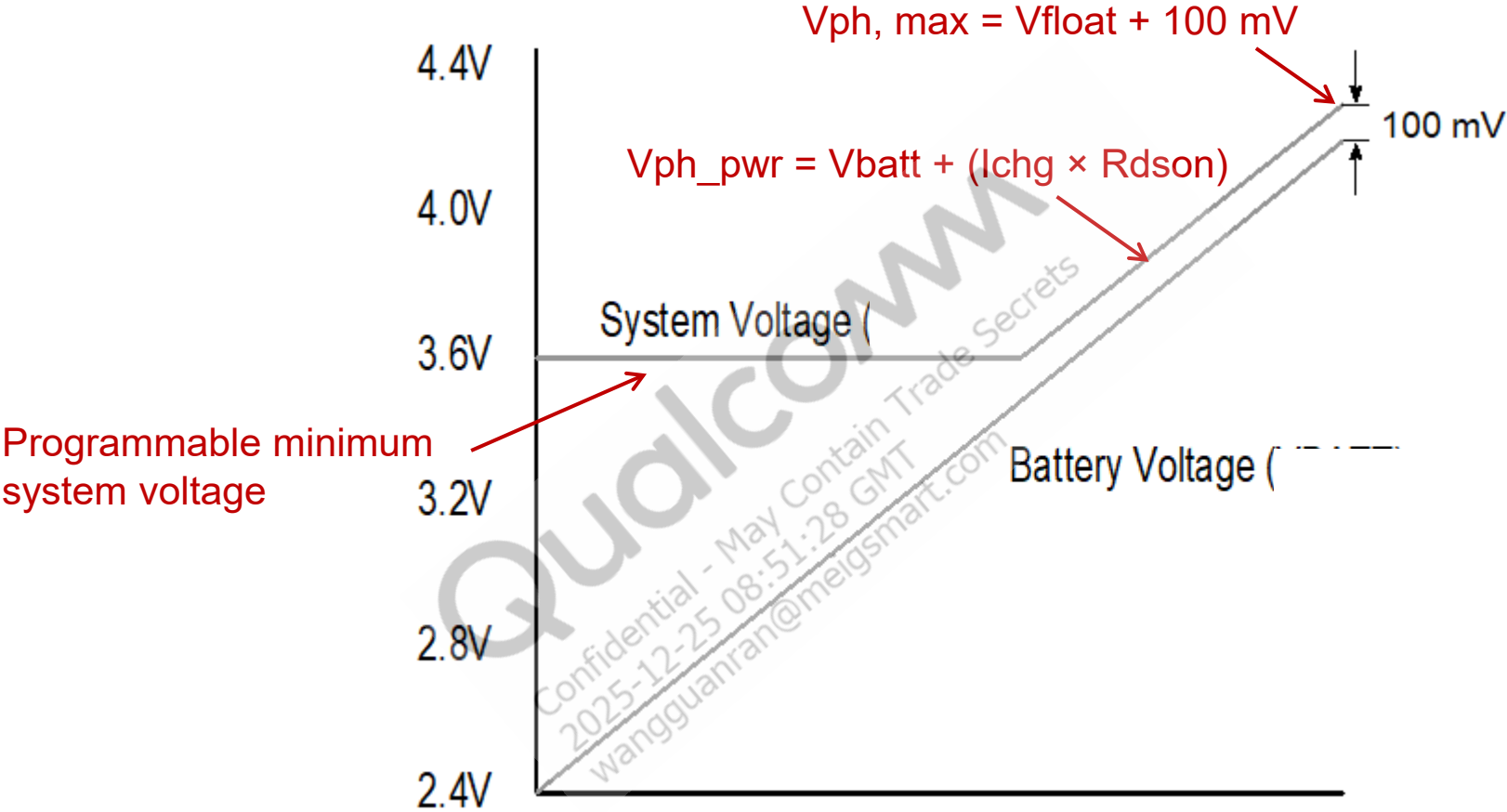
Basic Charging Operation Sequence (2 of 2)

- **Constant voltage mode** – when the predefined *float voltage* is reached, the fast-charge current begins to drop, and the mode transitions into constant voltage mode.
 - Float voltage is programmable from 3.60 V–4.79 V, in 10 mV steps.
 - High float voltage settings support modern battery packs that have float voltages of 3.6 V and 4.35 V.
 - Dynamic float voltage adjustment allows sophisticated battery charging techniques and control algorithms.
- **Charge completion** – the charge cycle is considered complete when the charge current reaches the programmed termination current threshold, assuming current termination is enabled.
 - Termination current is programmable: 100 mA to 450 mA in steps of 50 mA.
 - If the termination current threshold is not met before the charge timer expires, the charge cycle is terminated and a corresponding timeout fault signal is asserted.
- **Automatic battery recharge** – the PMIC automatically recharges the battery when its voltage falls by a value of V_{RECH} below the programmed float voltage; this part of the sequence is also called top off.
 - If 1) the input power supply is still present, 2) charging remains enabled (SPMI command), and 3) all the prequalification parameters are still met, then a new charging cycle is initiated.
 - This charging cycle ensures that the battery capacity remains high, without needing to manually restarting a charging cycle.
 - When the charge inhibit function is enabled, the automatic recharge threshold is overridden to the (higher) charge inhibit voltage threshold.

Voltage and Current Waveforms – Constant Current/Constant Voltage Operation



Basic Charging and Float Voltage



$V_{ph_pwr, charger\ disabled} = V_{ph, max}, \text{ or } V_{batt} + 100\text{ mV}$
The above plot is for $V_{float} = 4.2\text{ V}$

Other Battery and Charging Related Topics (1 of 2)

- **Safety timers** – integrated safety timers protect against a defective battery pack.
 - Both the precharge (24–192 min) and the complete charge timer (192–1536 min) start after the prequalification check is completed (and trickle-charging has started).
 - The precharge timer resets at the constant current mode transition.
 - The charge timer expires and the charging mode is terminated if the termination current level is not reached within the predetermined duration.
 - Both timers can be disabled or enabled by appropriate bit selection.
 - Safety timers do not operate during the USB-OTG mode. They are paused during supplemental mode and during a thermal shutdown event.
- **USB dead battery recovery (automatic battery charging)**
 - The PMIC allows the portable device to automatically start the charging process when the battery voltage is below a low (dead) battery threshold and hence the system is off.
- **Programmable battery charging** – the PMIC is able to modify all the important charger parameters via software.

Other Battery and Charging Related Topics (2 of 2)

■ **Charger inhibit function** – prevents charging initiation under certain conditions

- The charger module provides the option of preventing charging initiation on power cycling or charge enabling/disabling, unless the battery voltage is 50, 100, 200, or 300 mV below the float voltage.
- This prevents extreme stressing of the battery via continuous charging cycles in systems with short run times and frequent power cycling (input power connects and disconnects).
- Since this function is only active during power cycling and manual charge enabling, if the device enters and then exists suspend mode, charging continues even if the battery is above the charge inhibit voltage threshold.
- Note that when this function is enabled, the automatic recharge threshold is overridden to the same threshold. As such, the automatic recharge thresholds can also be used for the charger inhibit function.

■ **Current sinks** – easily recognize a charger removal event

- The USB_IN and USB_IN_MID pins each have a 10 mA current sink that is enabled to discharge the capacitance associated with the pins. This helps protect the device in the case of an overvoltage condition and enables it to recognize an input removal event more quickly.
- The MID_x 10 mA current sinks are enabled if their respective input pin voltage is above ~1 V, below the UVLO threshold, and/or below the V_ASHDN threshold. This current sink also turns on if the input pin is above its OVLO threshold, and remains on for ~100 ms after the input source drops below its OVLO threshold. During this ~100 ms, the input is not recognized as valid.
- The USB_IN 10 mA current sink works identical to the MID_x current sink described in the previous bullet, with the exception that it is disabled whenever USB_IN is above the OVLO threshold.

TurboCharge™ Mode

- When charging, the PMIC exploits the intrinsic ability of the buck architecture to multiply the input current while stepping down the output voltage.
- This property is expressed mathematically in the comparison below and maximizes battery charging from current limited devices and greatly decreases power dissipation and any heat-related dissipation.
- Using TurboCharge, devices can charge the battery with a charging current as high as 750 mA when charging from a port limited to 500 mA.

Linear charging

Equation 1 (linear charge current relationship, not accounting for thermal foldback): $I_{OUT} \approx I_{IN}$

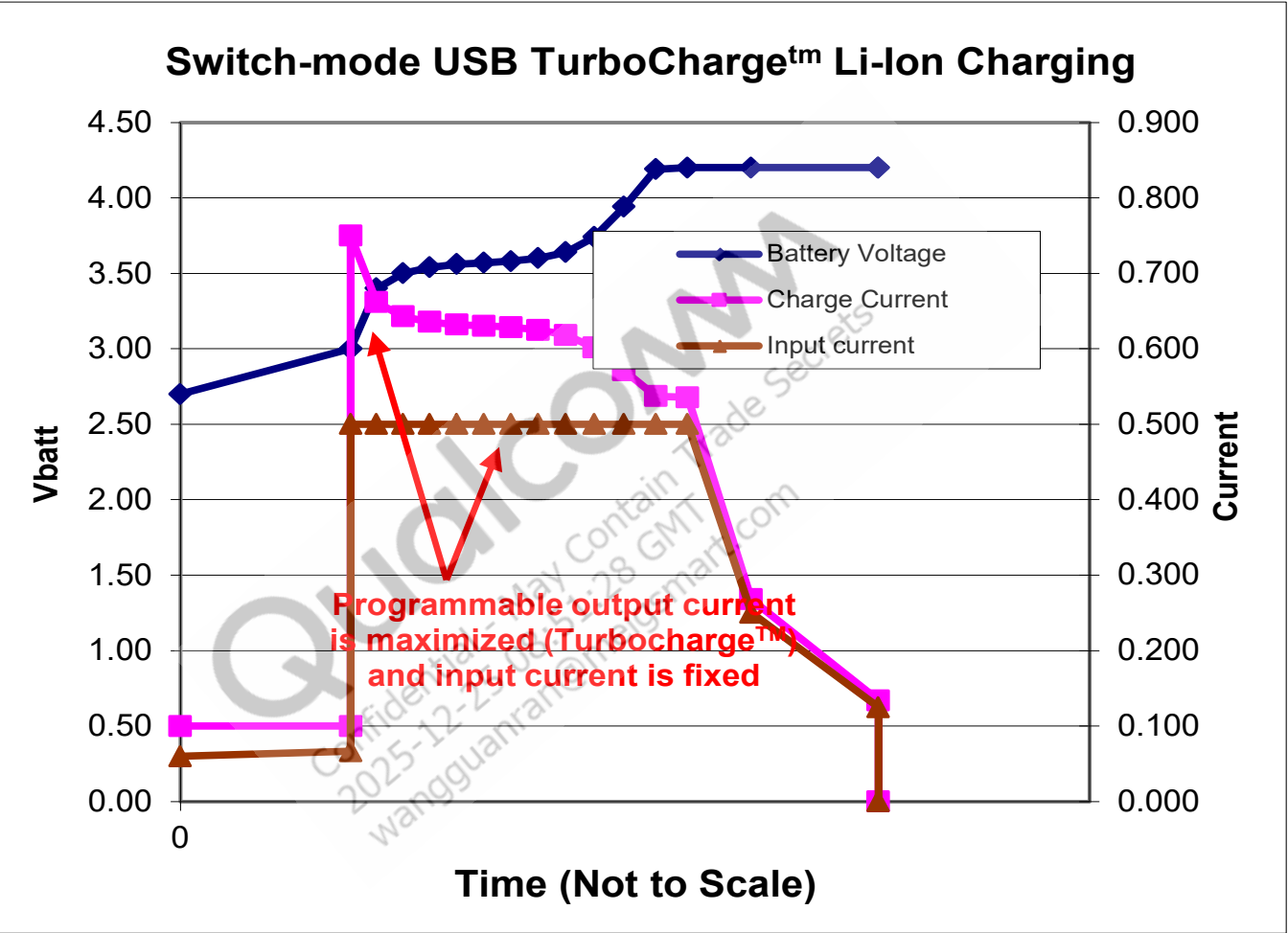
Equation 2 (efficiency of linear charger): $\eta = \frac{V_{BATTERY}}{V_{INPUT}}$

TurboCharge mode – Optimized buck efficiency allows 500 mA at the charging port to deliver the 750 mA output.

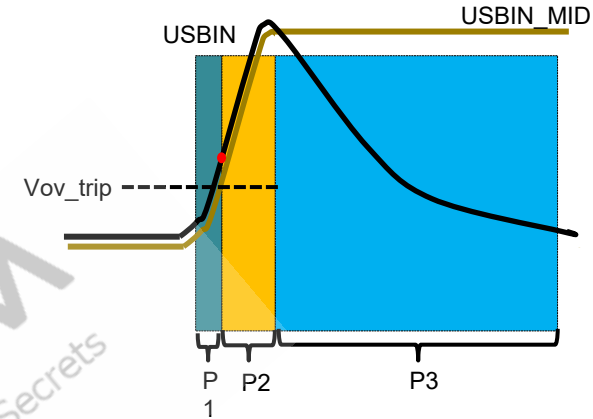
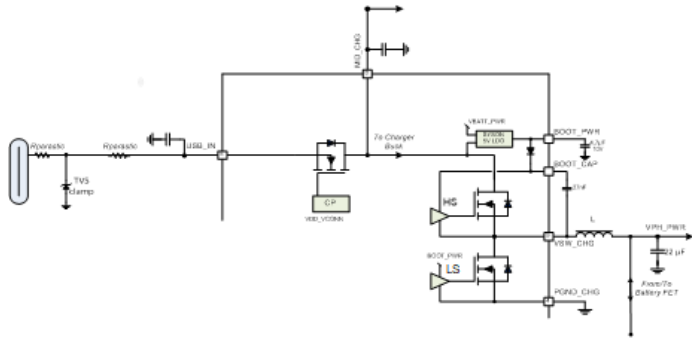
Equation 3 (TurboCharge current relationship): $I_{BAT} = \frac{\eta V_{IN} I_{IN}}{V_{BATT}}$

$\eta = 90\%$

Voltage and Current Waveforms – TurboCharge Operation



USB Surge Protection



■ Three different phases during OVP surge test:

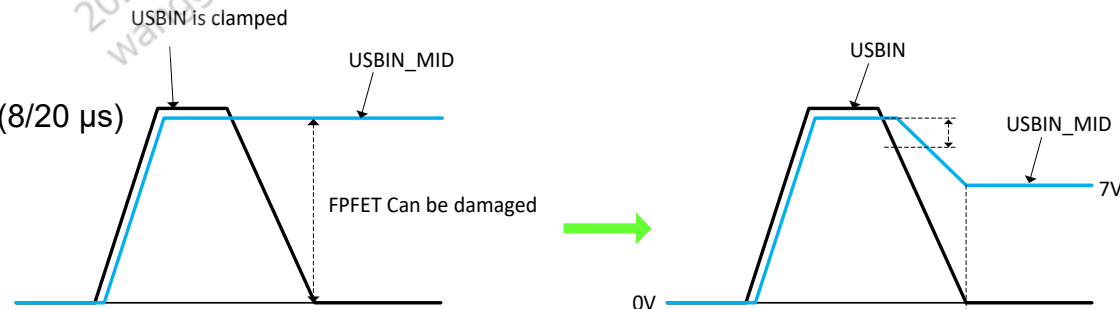
1. Phase 1: Charger switches from **on** to **off** due to input OV trips
2. Phase 2: USB_IN rises to peak and MID_CHG follows
3. Phase 3: USB_IN falls due to pull-down resistance. A new circuit in PM7250B clamps (VMID - VUSB_IN) below 7 V, protecting the FP FET.

■ The PM7250B charger is designed to meet 300 V surge protection, per the IEC 61000-4-5 standard, with an external TVS diode.

□ No external OVP chip is required. Only a TVS clamp is needed.

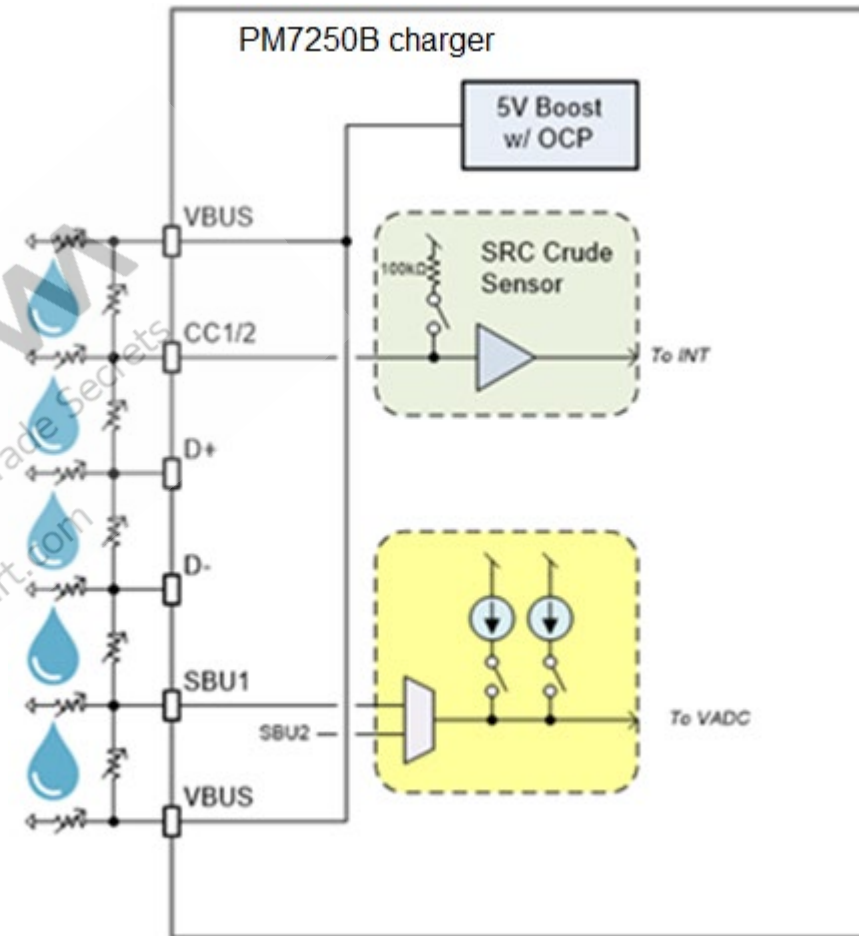
□ OnSemi NSPU5132 surge protection diode

- 13.5 V working peak reverse voltage
- 25 V maximum clamping voltage at 195 A Ipp (8/20 μ s)
- 2x 1.8 x 0.5 mm3



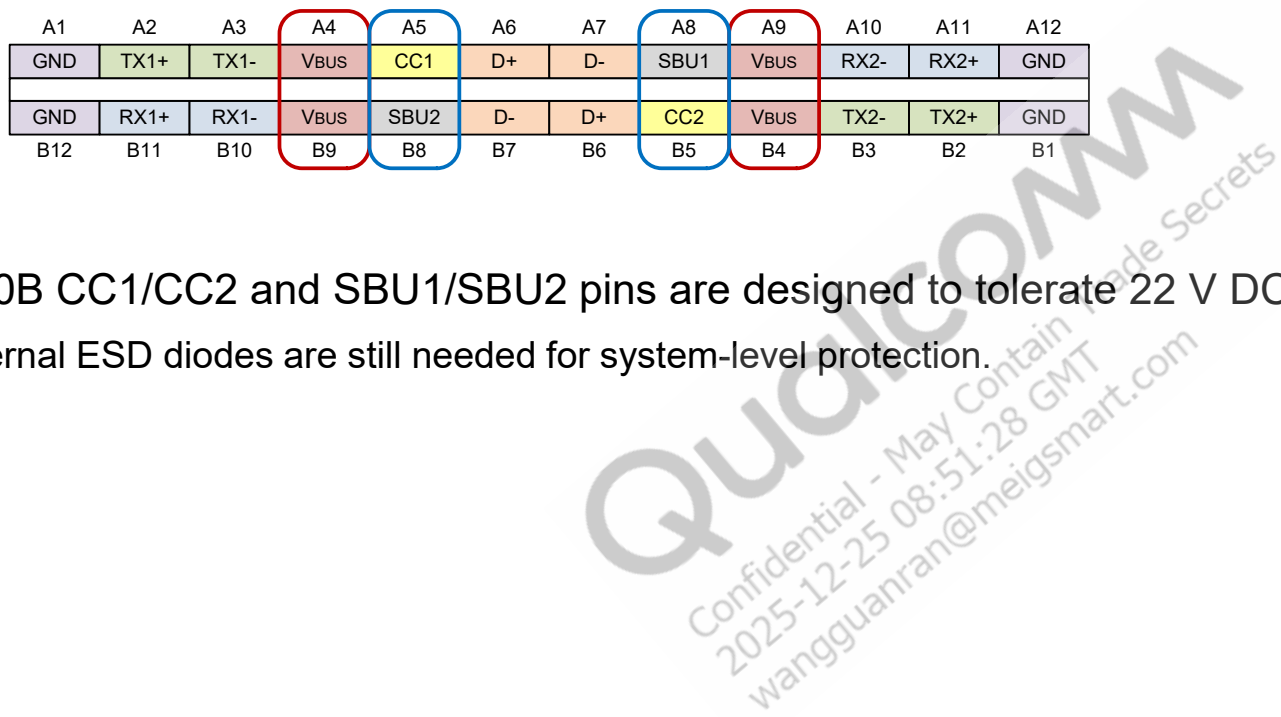
Type-C Liquid Presence Detection

- It is observed that, when exposed to liquid such as water, the constant dual-role power mode (DRP) toggling on the Type-C CC pins expedites the corrosion on these pins, which becomes the weak point of a waterproof design.
- PM7250B aims to detect Type-C liquid presence detection:
 - Moisture detection will only be active in DRP mode
 - While in DRP, the Type-C detection module generates an interrupt when detecting a weak pull-down on CCx.
 - Software invokes a series of hardware and software detections on CCx and SBUx pins, and determines whether the Type-C connector is submerged in water.
 - If the connector *is* submerged, software can:
 - Configure CC logic to SNK mode and stop DRP toggling to prevent CC pin corrosion.
 - Disconnect Rd (Hi-Z CCx) and periodically check until the water is gone.



22 V Tolerance on CC1/CC2 and SBU1/SBU2 Pins

- USB Type-C CC1/CC2 and SBU1/SBU2 pins can be exposed to 20 V VBUS due to:
 - Close proximity to the VBUS pins and connector misalignment, or moisture/debris between the pins
 - Some noncompliant A-to-C cables have the CC pins shorted to VBUS



- PM7250B CC1/CC2 and SBU1/SBU2 pins are designed to tolerate 22 V DC.
 - External ESD diodes are still needed for system-level protection.

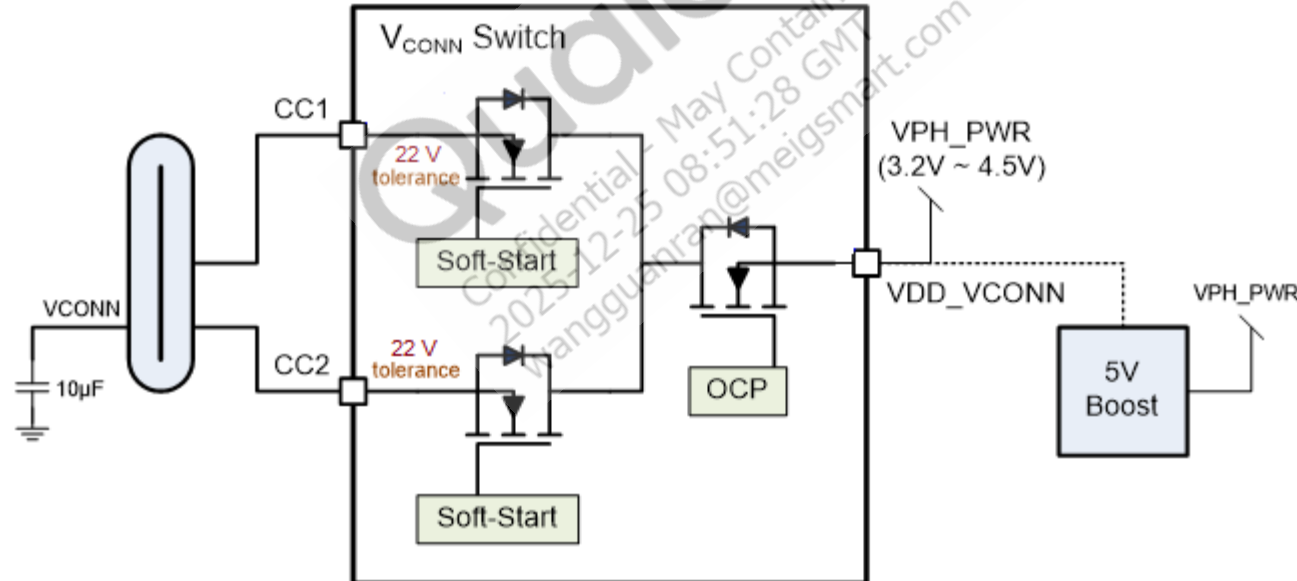
Enhanced A-to-C Legacy Cable Detection

- PM7250B charger has the capability to detect a legacy A-to-C cable and invoke BC1.2 APSD to determine a standard downstream port (SDP), charging downstream port (CDP), dedicated charging port (DCP), or Quick Charge 2 or Quick Charge 3, and apply USB input current limit accordingly.
 - Compliant legacy A-to-C cable: Rp-Default (56 kΩ)
 - Noncompliant legacy A-to-C cable: Rp-1.5A (22 kΩ) or Rp-3A (10 kΩ)
- The PM7250B charger enhances the legacy A-to-C cable detection to detect a noncompliant legacy A-to-C cable with CC shorted to VBUS:
 - If this type of cable is detected, Quick Charge can be automatically disabled; there is also an option to prohibit battery charging.

Cables	Rp	TYPEC_LEGACY_CABLE_STATUS	TYPEC_NONCOMP_LEGACY_CABLE_STATUS	TYPE_C_SNK_STATUS	USB ICL	Quick Charge
Standard Type-C cable (C-to-C)	N/A (Rp in adapter)	0	0	Depending on the source Rp value	Depending on the source Rp value	Quick Charge operates normally
Compliant legacy cable (A-to-C)	Rp-Default (56 kΩ)	1	0	SNK_RP_STD	Use the BC1.2 result	Quick Charge operates normally
Noncompliant legacy cable (A-to-C)	Rp-1.5 A (22 kΩ)	1	1	SNK_RP_1P5	- If an SDP or CDP, use the BC1.2 result - If a DCP, float, OCP, or high-voltage DCP (HVDCP) adapter, use the BC1.2 result or Rp value	Quick Charge operates normally
	Rp-3 A (10 kΩ)	1	1	SNK_RP_3P0		
	0 Ω (CC shorted to VBUS)	1	1	SNK_RP_SHORT	Same as above, except that there is an option to prohibit charging.	Quick Charge can be automatically disabled.

1.5 W V_{CONN} Switch to Support VPD and VPA

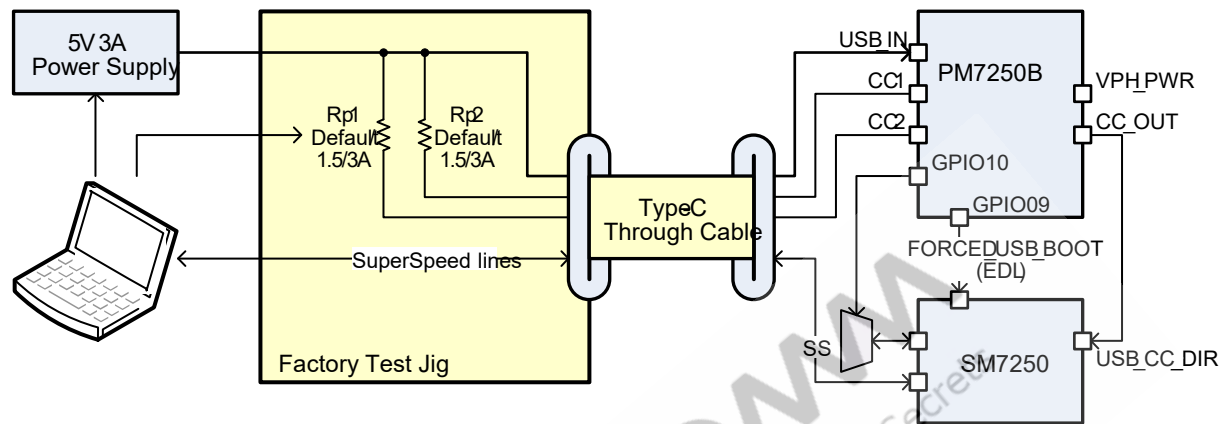
- The PM7250B charger includes a V_{CONN} load switch.
 - Capable of delivering 1.5 W (3 V, 0.5 A) load with 3.2 V $V_{\text{PH_PWR}}$
 - Soft-start when turning on the V_{CONN} switch; two levels of OCP: 100 mA (suitable for e-marker) and 500 mA (suitable for active cables or alternate mode)
 - Designed to support active cables, VPDs up to 1 W, or VPAs, such as DisplayPort adapters, 1.5 W maximum) as defined in the *USB Type-C Specification*, Rev. 1.3.
 - Certain old VCONN adapters work with 4.75 V–5.5 V V_{CONN} only. If these adapters must be supported, an external 5 V boost is required.



USB Type-C Debug Accessory Mode (DAM) Support

- PM7250B supports DebugAccessory.SNK mode as defined in the *USB Type-C Specification*, Rev. 1.3, Appendix B.
 - When in DebugAccessory.SNK mode, PM7250B monitors Rp/Rp on CC1/CC2 for attachment, optional orientation, and current-level detection. Battery charging is enabled and AICL is disabled by OTP default.
 - Based on the Rp combination, GPIO09 is driven high; GPIO09 is connected to SM7250 and forces it into emergency download (EDL) mode.
 - On PM7250B, by default, the SYS_OK power-on trigger is generated only after tCCDebounce (PM7250B enters DebugAccessory.SNK mode and drives GPIO09 to the proper state).
 - Based on the Rp combination, GPIO10 is driven to Hi-Z or low. This can be used to drive external switches to route UART signals in DAM.

DebugAccessory.SNK Mode Support



CC1	CC2	DAM	CC_OUT	ICL_MAX	GPIO09 ⁵	GPIO10 ⁶
Open	Open	Inactive	Hi-Z	N/A	0	0
Rp	Open		0 (CC1)	0.5 A/1.5 A/3 A	0	0
Open	Rp		1 (CC2)	0.5 A/1.5 A/3 A	0	0
Rp-3A	Rp-1.5A	Active	0 (CC1)	0.5 A ³	0	Hi-Z (UART)
Rp-1.5A	Rp-Default			1.5 A ³	1 (EDL)	Hi-Z (UART)
Rp-3A	Rp-Default			3 A ⁴	1 (EDL)	0
Rp-1.5A	Rp-3A		1 (CC2)	0.5 A ³	0	Hi-Z (UART)
Rp-Default	Rp-1.5A			1.5 A ³	1 (EDL)	Hi-Z (UART)
Rp-Default	Rp-3A			3 A ⁴	1 (EDL)	0
Rp-Default	Rp-Default					
Rp-1.5A	Rp-1.5A			0.5 A ³	0	Hi-Z
Rp-3A	Rp-3A			1.5 A ³	0	Hi-Z
				3 A ⁴	0	Hi-Z

1. The table assumes EN_TYPEC_FMB = 1 and EN_DEBUG_ACCESS_SNK = 1 as in the OTP default.
2. When in DebugAccessory.SNK mode, charging is enabled (EN_CHG_ON_DEBUG_ACCESS_SNK = 1) and AICL is disabled (DAM_DIS_AICL = 1) by the OTP default.
3. The ICL_MAX value can be overridden to 3 A when the DAM_SEL_ICL_3A_OR_FMB bit is set.
4. The ICL_MAX value can be overridden to 5 A when the DAM_SEL_ICL_3A_OR_FMB bit is set.
5. GPIO09 is set to: DIG_OUT, SF1 (FMB1), No_Invert, CMOS
6. GPIO10 is set to: DIG_OUT, SF1 (FMB2), No_Invert, open drain NMOS

Factory Mode Enhancement

	Factory mode requirements	PM660 charger	PM7250B charger
1	On USB insertion, the boot system, with or without the battery	Yes. Overrides USB ICL, based on QTI proprietary factory mode detection	Yes. Overrides USB ICL, based on Type-C DAM detection
2	During boot, force SDM into EDL mode for firmware update	No. Timing issue between the SYS_OK PON trigger and driving the EDL GPIO	Yes. The SYS_OK PON trigger is generated only after tCCDebounce (PM7250B enters DebugAccessory.SNK mode and drives GPIO09 to the proper state)
3	Drive external FETs to route UART signals	N/A	GPIO10

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Charging with a Generic DC Charger Input

- The PM7250B charger can also work with a generic DC charger input with an external B2B FET.

- DC input and USB charging paths are joined at the MID_CHG node, eliminating a DC FP FET or external PMUX.

PM7250B's DC_IN_PSNS regulation loop is used to regulate the current drawn from an external DC source:

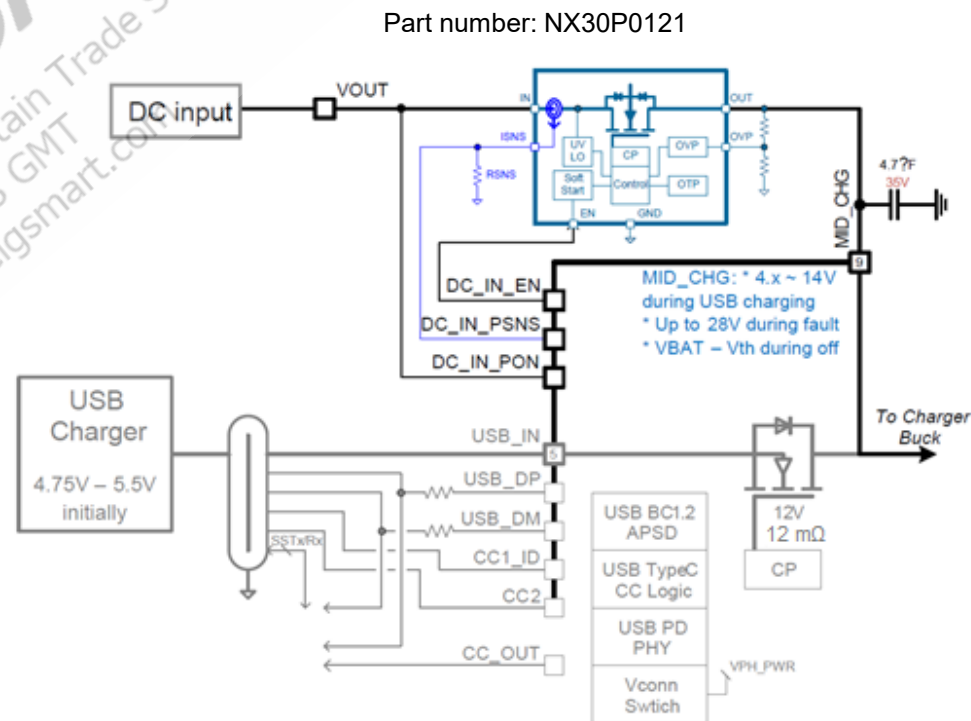
- The B2B FET outputs a scaled version of the input current. The voltage across the RSNS resistor is fed to PM7250B DC_IN_PSNS regulation loop. The DC_IN_PSNS regulation controls the input current.
- PSNS reference can be adjusted from [0V:40mV:1.2V] in 30 steps. For ISNS ratio of 2500 and RSNS value of 2 kΩ, this results in input current control of [0 A: 50 mA: 1.5 A].

1 A of input current corresponds to ~ 0.8 V PSNS

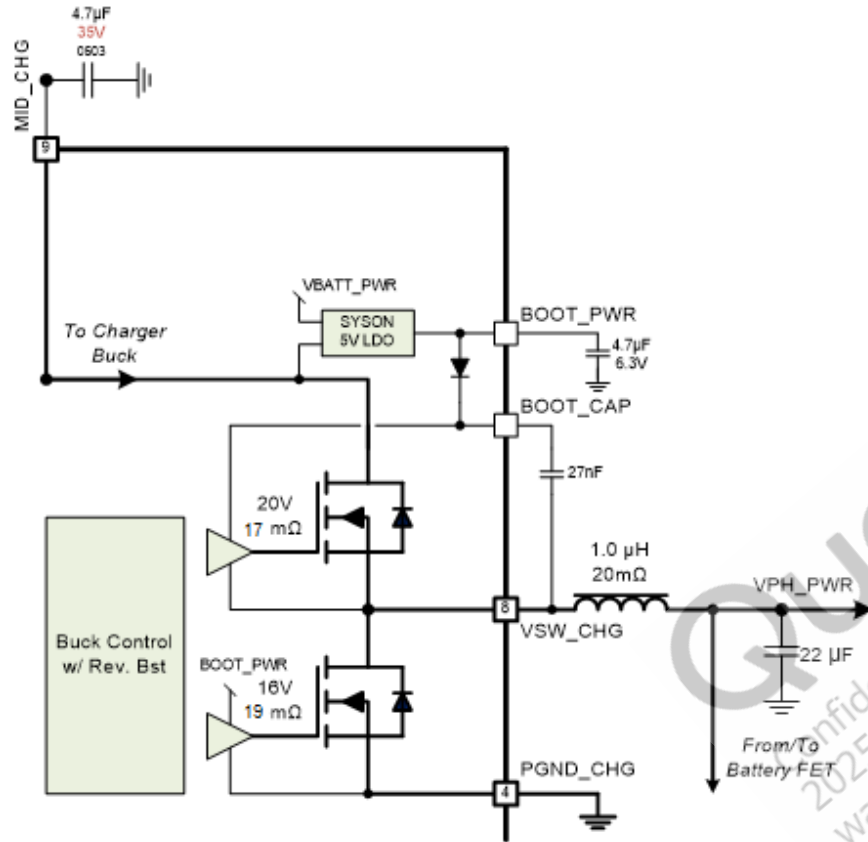
- Use of DC_IN_PSNS regulation offers the following benefits:

- ❑ Prevents the external source from hitting OCP or output UVLO during load attack
- ❑ Allows PM7250B to have thermal regulation of DC charging path and control input power drawn from DC charger

- * During phone off:
 - No DC input DCIN_EN low to turn off B2B FET and block MID_CHG → DCIN_PON.
- * During DC_IN PON:
 - VOUT → DCIN_PON as PON trigger and VARB source
 - DCIN_EN high to connect VOUT to MID_CHG
- * During DC_IN charging:
 - Charger buck regulates MID_CHG above VIN_MIN
- * During USB charging, DC_EN low to block VOUT ↔ MID_CHG
- * During USB fault, B2B FET OVP kicks in and turns off B2B FET

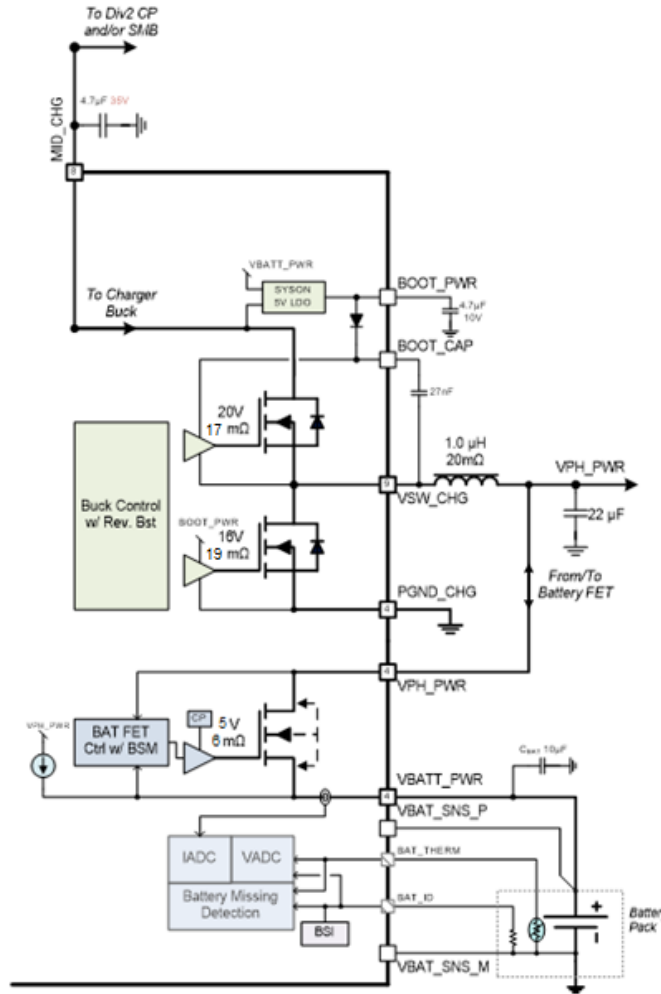


PM7250B Charger Power Stage



- High-efficiency switch-mode Li-ion battery charger
 - Extremely low $R_{ds(on)}$ MOSFETs, minimizing power dissipation during charging:
 - 17 m Ω HS FET, 19 m Ω LS FET
 - 12 m Ω USB FP FET, 6 m Ω battery FET
 - 600 kHz to 1.2 MHz programmable F_{sw}
 - Allows an optimizing tradeoff between power-train efficiency and external component selection
 - Enables 1.0 mm and 0.8 mm high BOMs
- Up to 1.5 A reverse boost

Charger Battery Interface



Charger battery interface

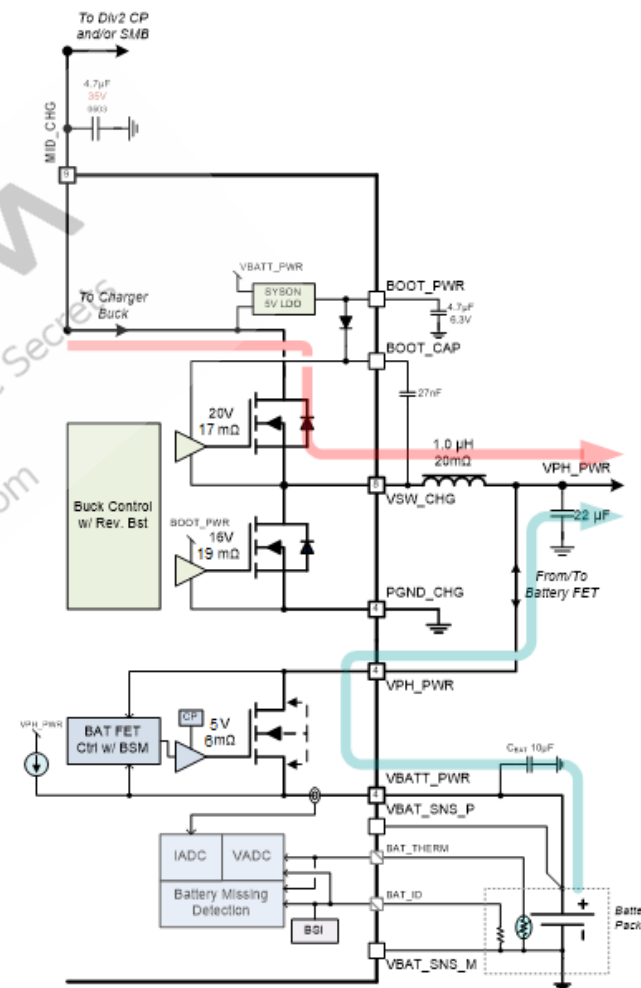
- CurrentPath control allows system operation with a deeply discharged/missing battery.
- Trickle/Precharging using parallel Isrc with BATFET off
- BSM with BATFET fully on
- Differential battery voltage Kelvin sensing to compensate for IR drop caused by the phone PCB and battery pack protection FETs
 - Option to enable AFVC for battery packs without Kelvin sensing pins
- Battery temperature monitoring with JEITA and JISC 8714 support
- Battery missing detection options
- Ship mode and AFP mode

New Battery Supplemental Mode (BSM)

Battery supplemental mode

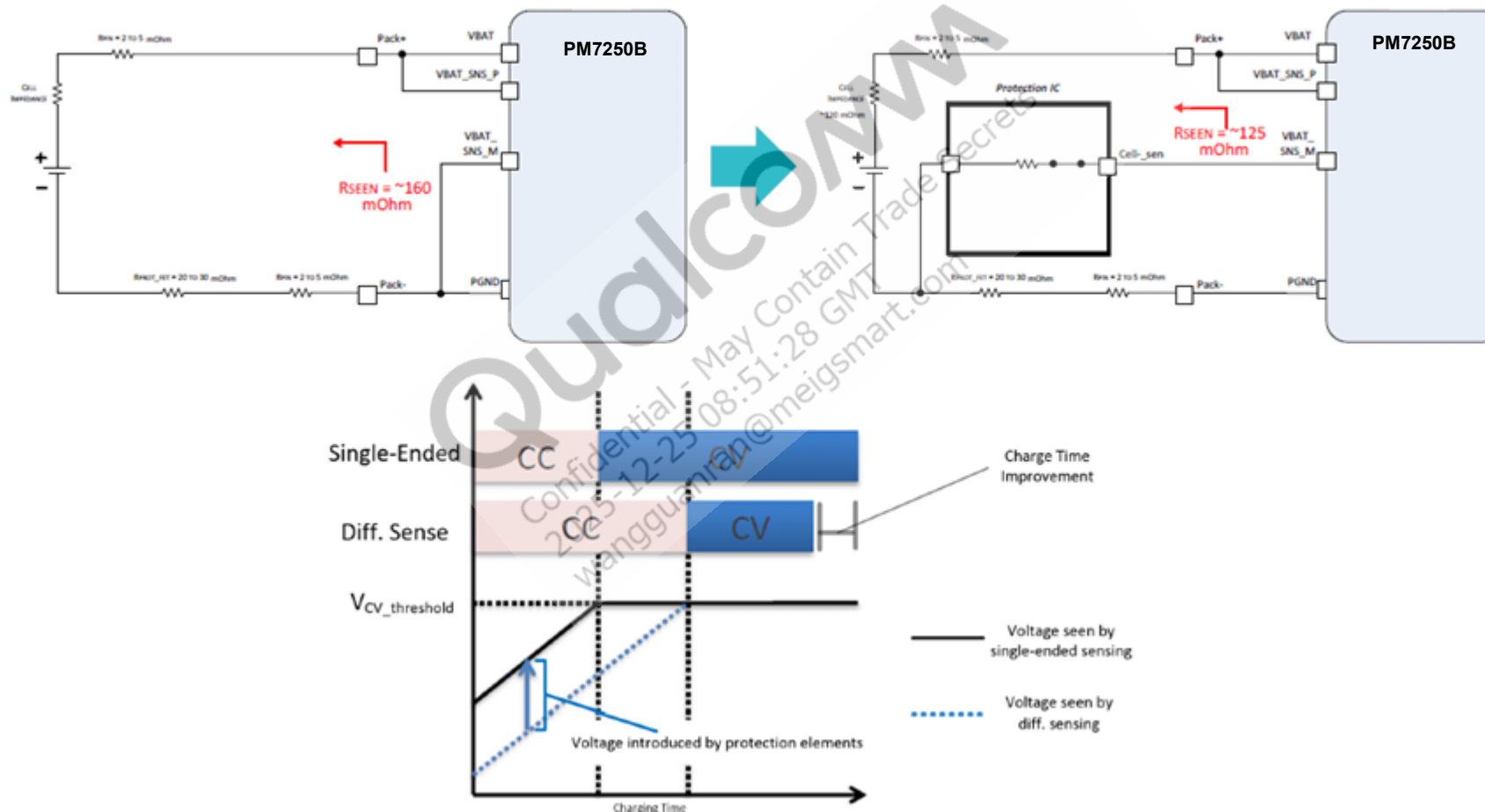
(also known as *VPH collapse protection* or *VCP*):

- When BAT FET is open and I_{SYS} exceeds what the charger buck can provide due to I_{USB_LIM} , the charger turns on BAT FET to allow the battery to supplement I_{SYS} .
- On PM7250B: The charger fully turns on BAT FET.
 - BSM mode is exited when a positive current $> I_{TERM}$ is detected flowing into the battery.



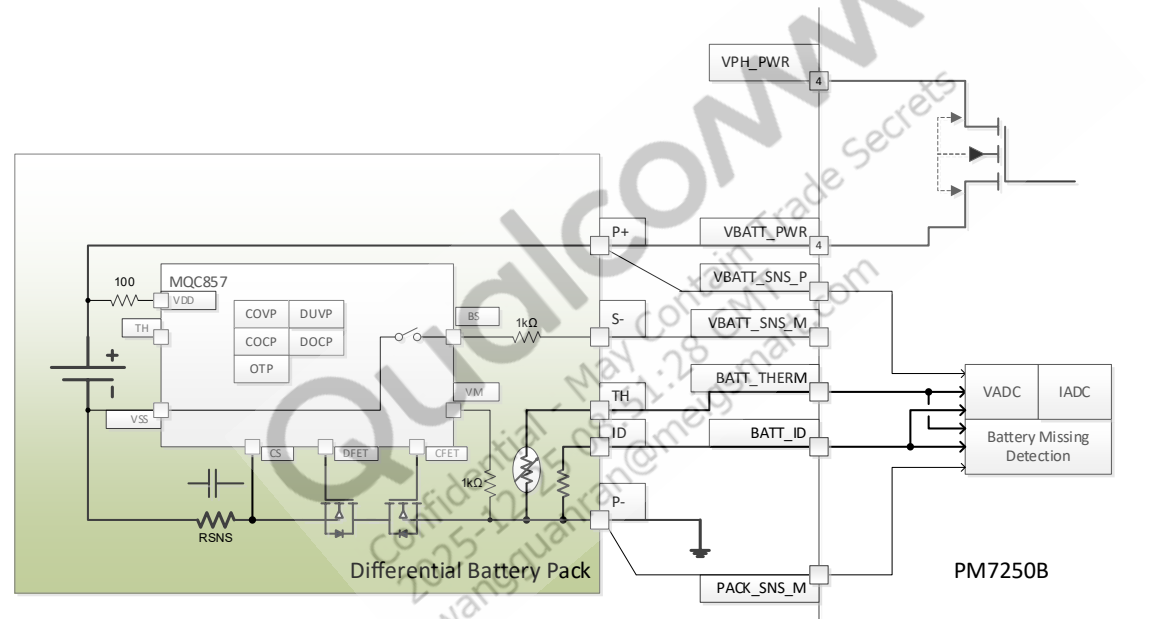
Differential Battery Sensing (1 of 2)

- Charging batteries with differential sensing yields faster charging time.
 - Bypasses the impedance introduced by the protection elements
 - Extends CC times and reduces the CV period



Differential Battery Sensing (2 of 2)

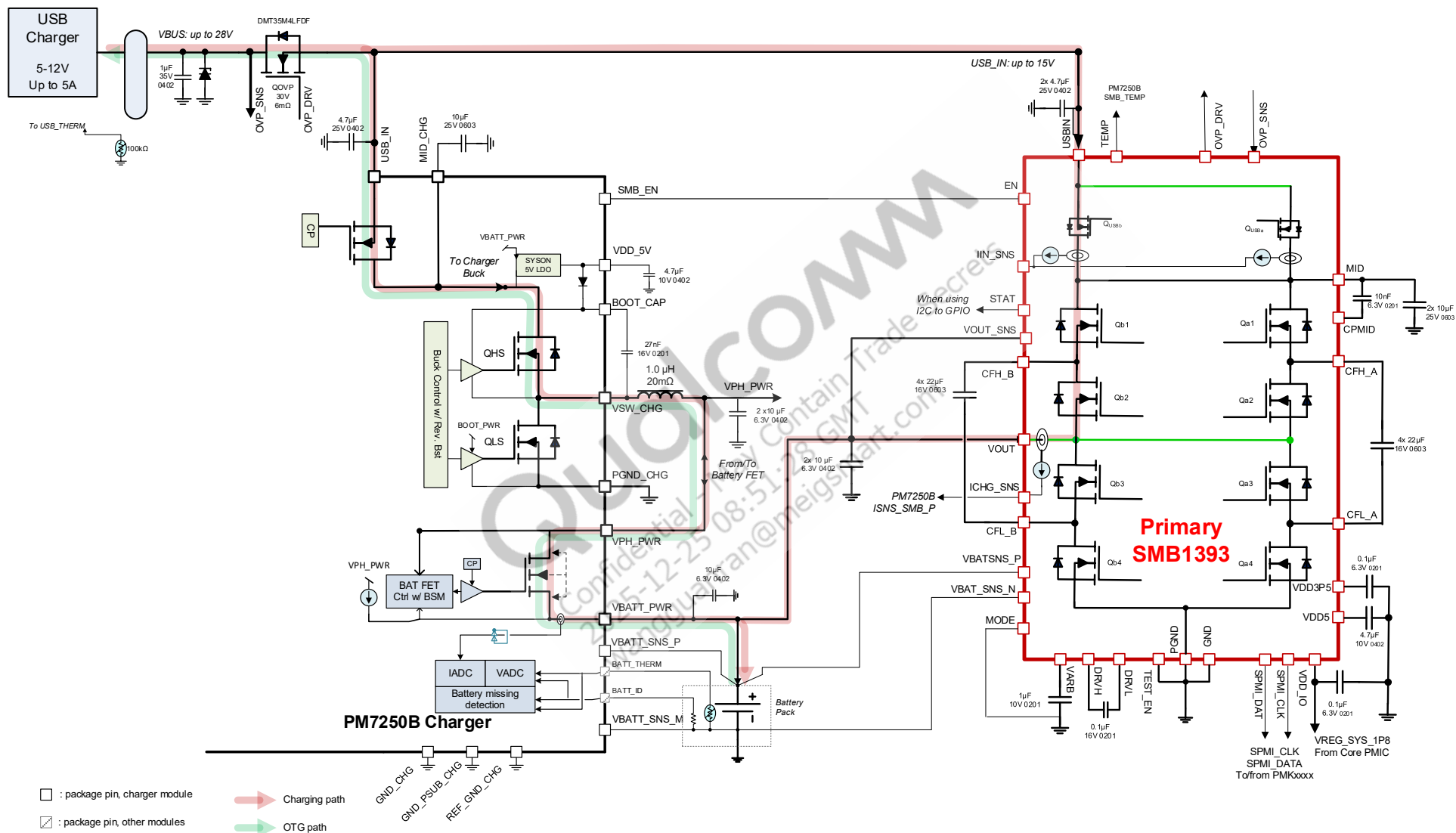
- PM7250B supports differential “in-pack” sensing with the Mitsumi MQC857 family of protection ICs.
 - Below diagram shows how a differential battery pack is connected to PM7250B.
 - When the battery hits its under voltage threshold or any other fault conditions and the protection FETs open, Mitsumi IC disconnects sense- to prevent any leakage to take place inside the PM7250B device.

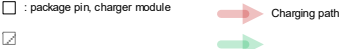


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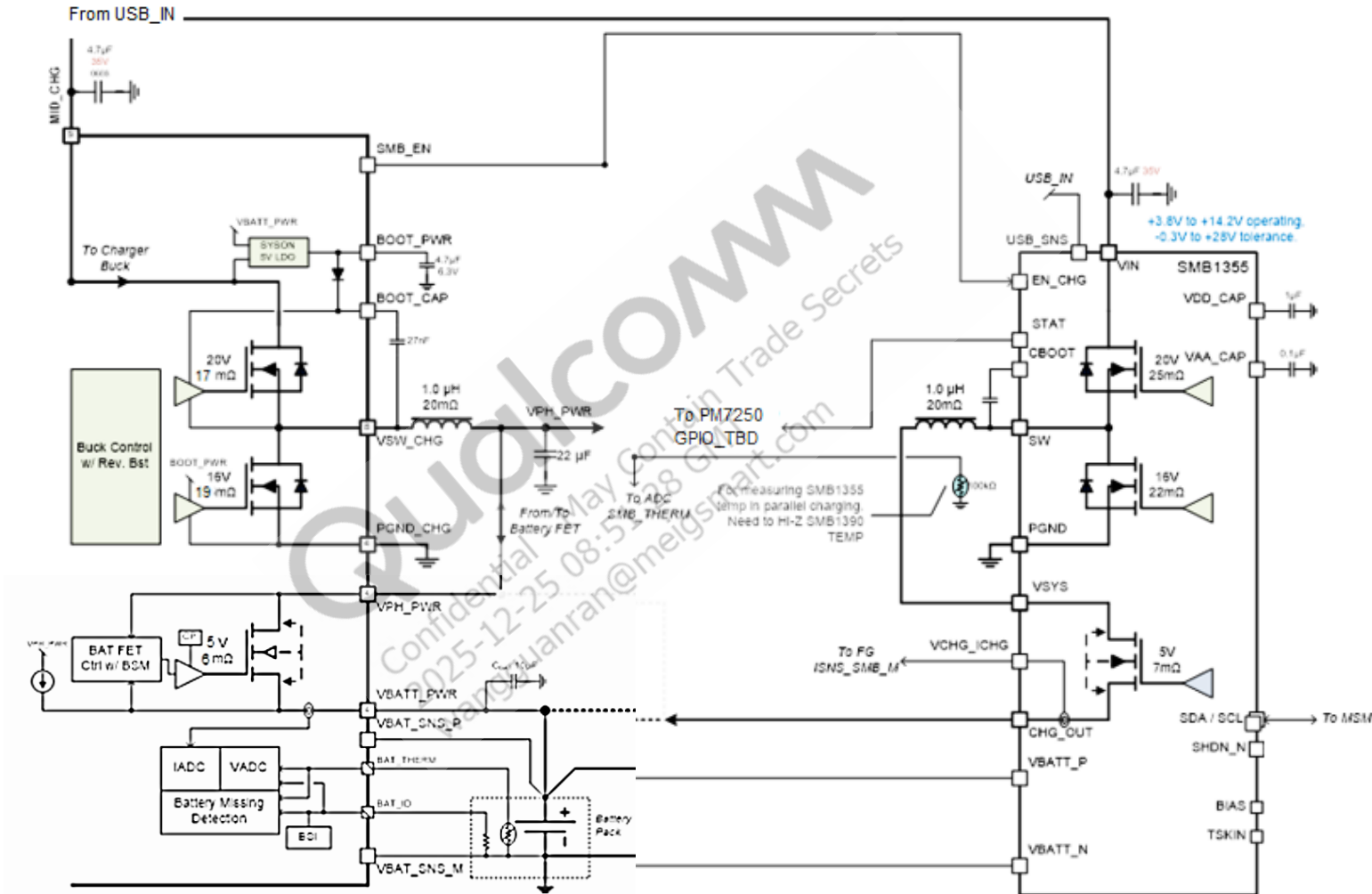
- Do not use resistor in series with VBATT_SNS_P. This net is shared by charger and ADC. Adding a series resistor will cause error in VBATT ADC measurement due to IR drop. DC shift on VBATT_SNS_P during VBATT ADC measurement may also cause problem with charger regulation.
- Do not use a series resistance between VBATT_SNS_N and PACK_SNS_M as it can interfere with negative cell sense accuracy and battery pack protection IC fault handling.

Parallel Charging with SMB1393





Parallel Charging with SMB1355



Parallel Charging with SMB1355 or SMB1393 Stand-alone

- SMB1393 charge pump: 6 A charging with 3 A cable
 - Suitable for Quick Charge 4/USB PD PPS
 - 6 A charging at ~97.2% efficiency with a standard 3 A Type-C cable (a thinner, cheaper cable with no e-marker)
 - Supports 2x SMB1393 operation for a 5 A cable
- Parallel charging with SMB1355 up to 5 A
 - Suitable for all types of adapters: Quick Charge 2 or Quick Charge 3, PD; Quick Charge 4 (PPS)
 - SMB1355 CHG_OUT connects to VBATT_PWR, avoiding BAT FETs in series and extra P_{diss}
 - Sensed ICHG to Qualcomm battery gauge ISNS_SMB_M and ISNS_SMB_P to count SMB1355 charging current

Hardware configurations	Quick Charge 4 (PD PPS)	Quick Charge 3	Quick Charge 2/PD/DCP	Note
PM7250B stand-alone	PM7250B			
PM7250B + SMB1355	PM7250B + SMB1355			
PM7250B + SMB1393	PM7250B + SMB1393	PM7250B	PM7250B	For up to 6 A charging
PM7250B + 2x SMB1393	PM7250B + 2x SMB1393	PM7250B	PM7250B	For 8 A charging with 5 A cable



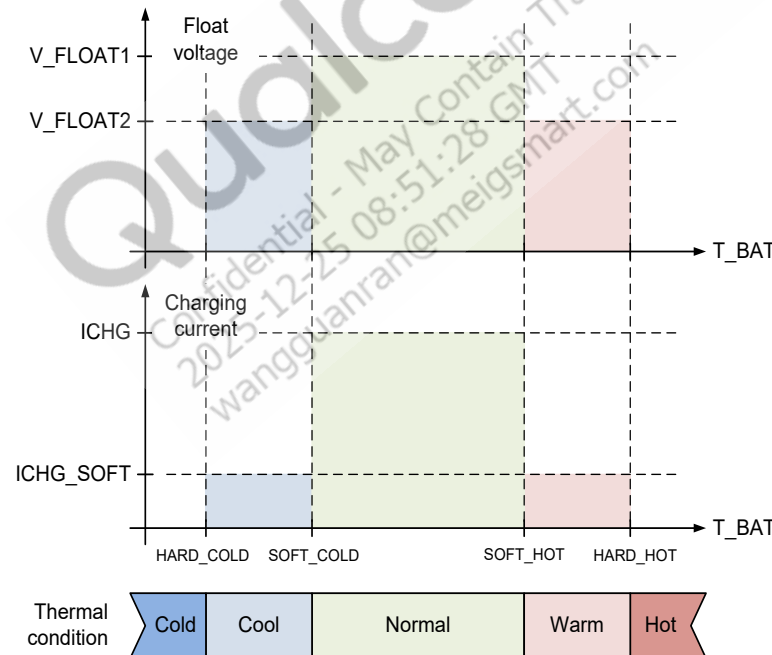
Section 2.1.3

JEITA Compliance

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JEITA Compliance (1 of 3)

- The JEITA standard allows battery charging with reduced charging voltage and/or current outside the conventional battery temperature range (illustrated below).
 - Temperature threshold points (HARD_COLD, SOFT_COLD, HARD_HOT, and SOFT_HOT) are determined by battery chemistry and vary with different battery vendors; the thermistor temperature coefficient also varies.
 - As a result, the BATT_THERM voltages corresponding to the temperature points vary.
- JEITA compliance is mandatory in Japan after November 2011.
- See *A Guide to the Safe Use of Secondary Lithium Ion Batteries in Notebook-type Personal Computers*.

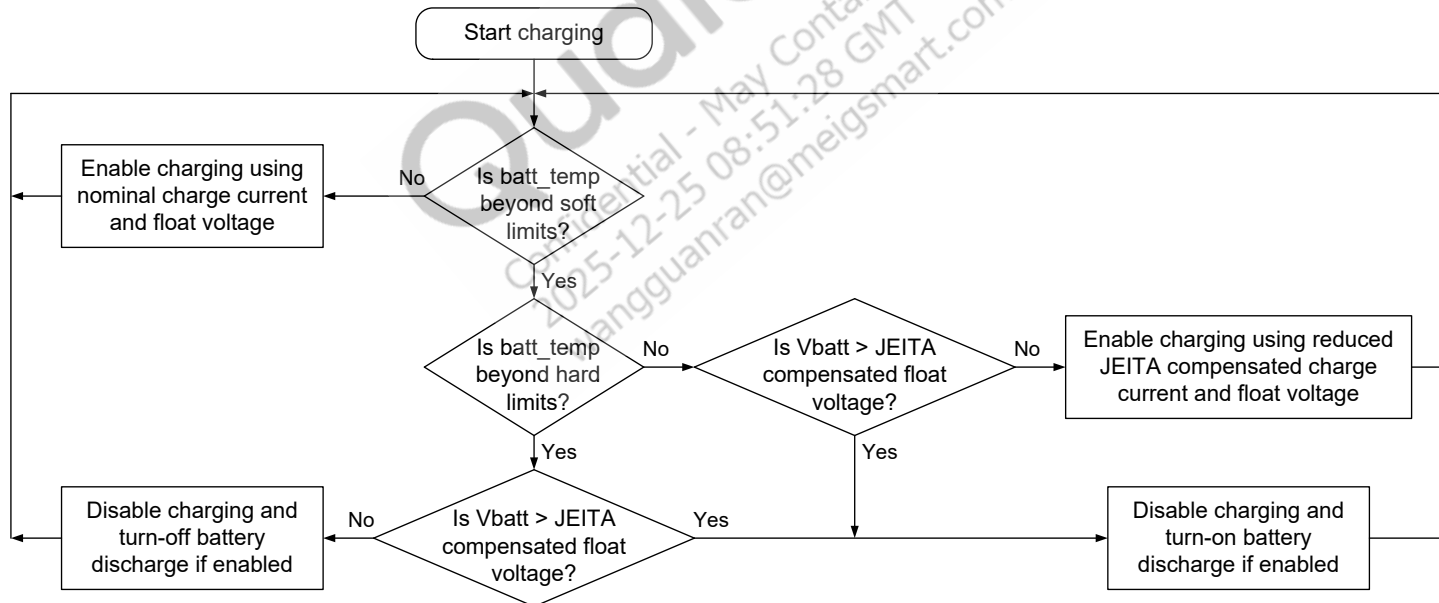


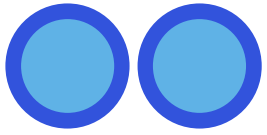
JEITA Compliance (2 of 3)

- The SCHG battery temperature monitor (BTM) has threshold configurability that helps cover both the traditional battery charging temperature range and the extended JEITA temperature range.
- The HARD_COLD, SOFT_COLD, HARD_HOT, and SOFT_HOT thresholds are fully software-programmable over a wide temperature range.
 - JEITA thresholds are programmed into a 16-bit unsigned register.
- The Qualcomm battery gauge uses an automated digital BTM routine to monitor the battery HARD_COLD, SOFT_COLD, HARD_HOT, and SOFT_HOT conditions.
- The battery temperature measurement result is compared with programmable HARD_COLD, SOFT_COLD, HARD_HOT, and SOFT_HOT thresholds; interrupts are generated if any of the thresholds are exceeded.
 - The charger module automatically adjusts the charge current to the JEITA-compensated charge current and adjusts the float voltage to the JEITA-compensated float voltage.
 - Battery discharge commences, if configured.
 - A JEITA compliance flowchart is shown in the next slide.

JEITA Compliance (3 of 3)

- When the battery temperature is outside the soft temperature limits and the battery voltage is above the JEITA-compensated float voltage, a 10 mA current source can discharge the battery voltage to the new JEITA-compensated float voltage.
- In this mode:
 - Charging is disabled
 - The system voltage remains 100 mV above the battery voltage and is supplied by the input source.
 - A supplemental battery current is provided if system loads increase beyond the capability of the input source.
- This function is optional and can be disabled, if required.





Section 2.2

Qualcomm Battery Gauge

2.2.1 Qualcomm Battery Gauge Block Diagram and Features [58](#)

2.2.2 Qualcomm Battery Gauge State of Charge Estimation [63](#)

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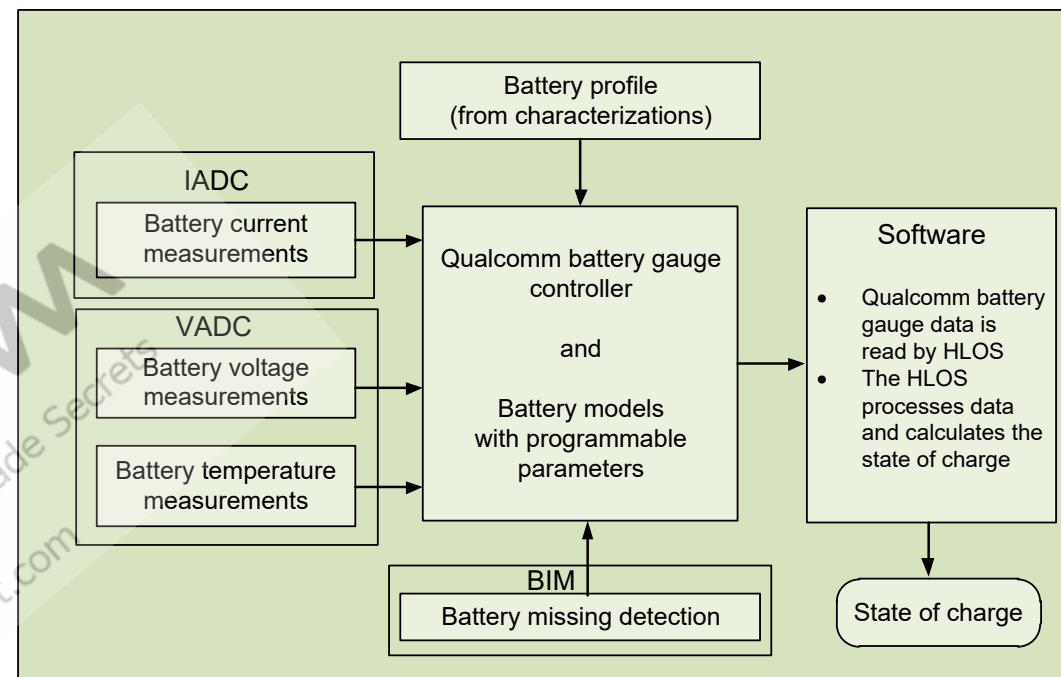
Section 2.2.1

Qualcomm Battery Gauge Block Diagram and Features

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Qualcomm Battery Gauge Feature Summary

- Qualcomm battery gauge is a hybrid of voltage and current based state of charge.
- Separate VADC and IADC for simultaneous voltage and current measurements.
- Estimate of state of charge is purely based on the software. The hardware gives Vbatt and Ibatt values to the software.
- Ibatt sensing is through BATFET.
- Accurate convergence to 0% state of charge at cutoff voltage by using Vbatt is reported by the hardware.
- Accurate end of charge by charge termination is based on the measured Ibatt.
 - Digital detection of the termination current improves accuracy.
- Online battery ESR estimation.
- Provide current now values for system power estimation.
- Digital temperature detection for JEITA compliance.
- Automatic recharge using state of charge threshold.
- Detection of supplemental mode, when the battery is supplementing charger to power the system.
- Avoids monotonicity problem.
- Feature improvements/updates can be done with OTA software updates.

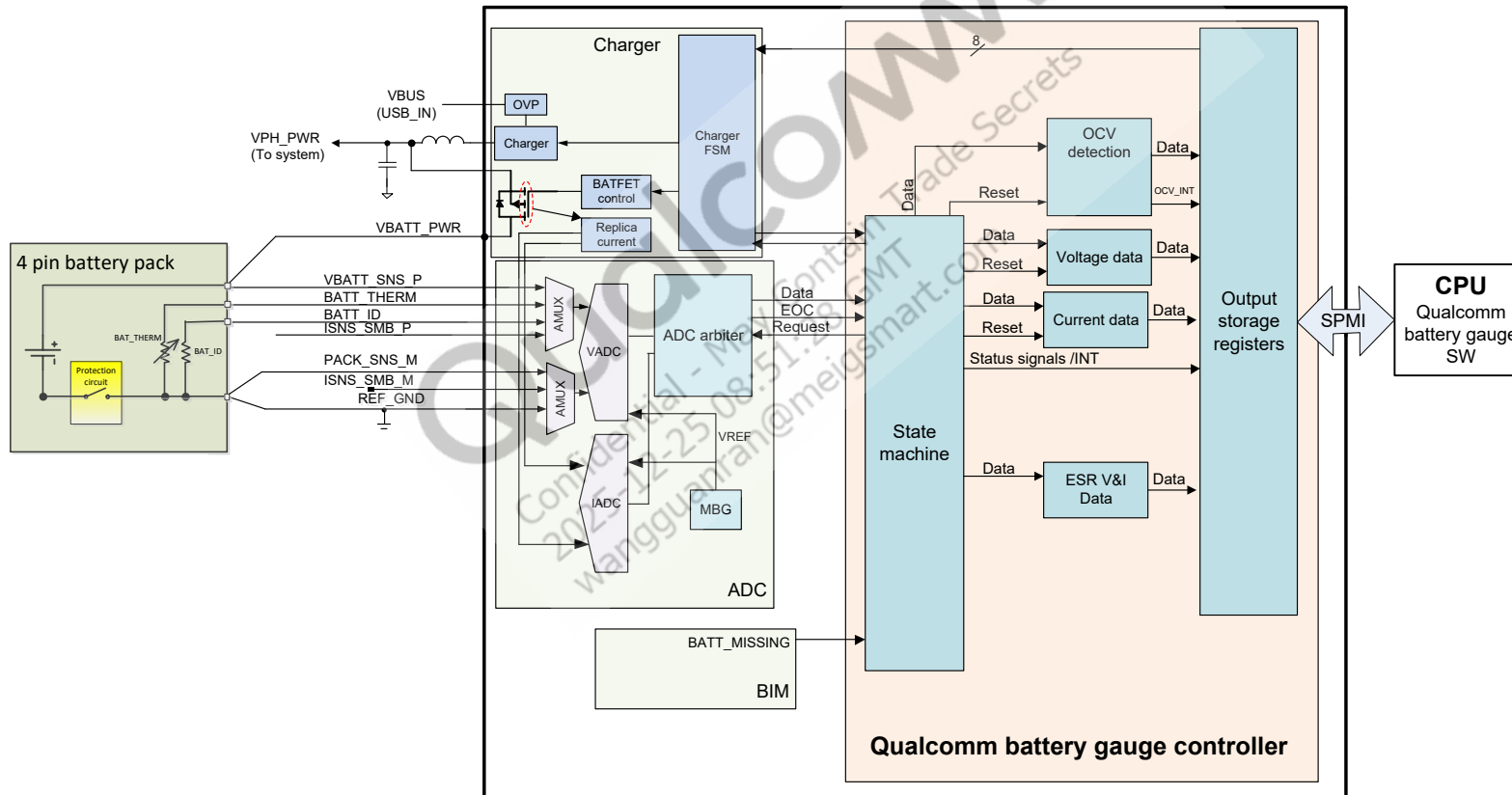


Comparison with Fuel Gauge

Parameter	PM660 (fuel gauge)	PM7250B (Qualcomm battery gauge)
Feature update flexibility	No	Yes
Online ESR estimation	Yes	Yes
Capacity learning	Yes	Yes
Accurate Ibatt-based end of charge	Yes	Yes
Converge to 0% at cutoff voltage	Yes	Yes
SoC update cycle	1.5 sec	12 sec (software programmable)
Battery characterization at sub zero temperature	Yes	Yes
Parallel charging support	Yes	Yes
Battery profile interpolation	Yes	Yes (improved)
Battery ID detection and profile loading	Yes	Yes
Internal Ibatt sensing block	Yes	Yes
Ibatt sensing range	-8.5 A to +8.5 A	-10 A to +10 A

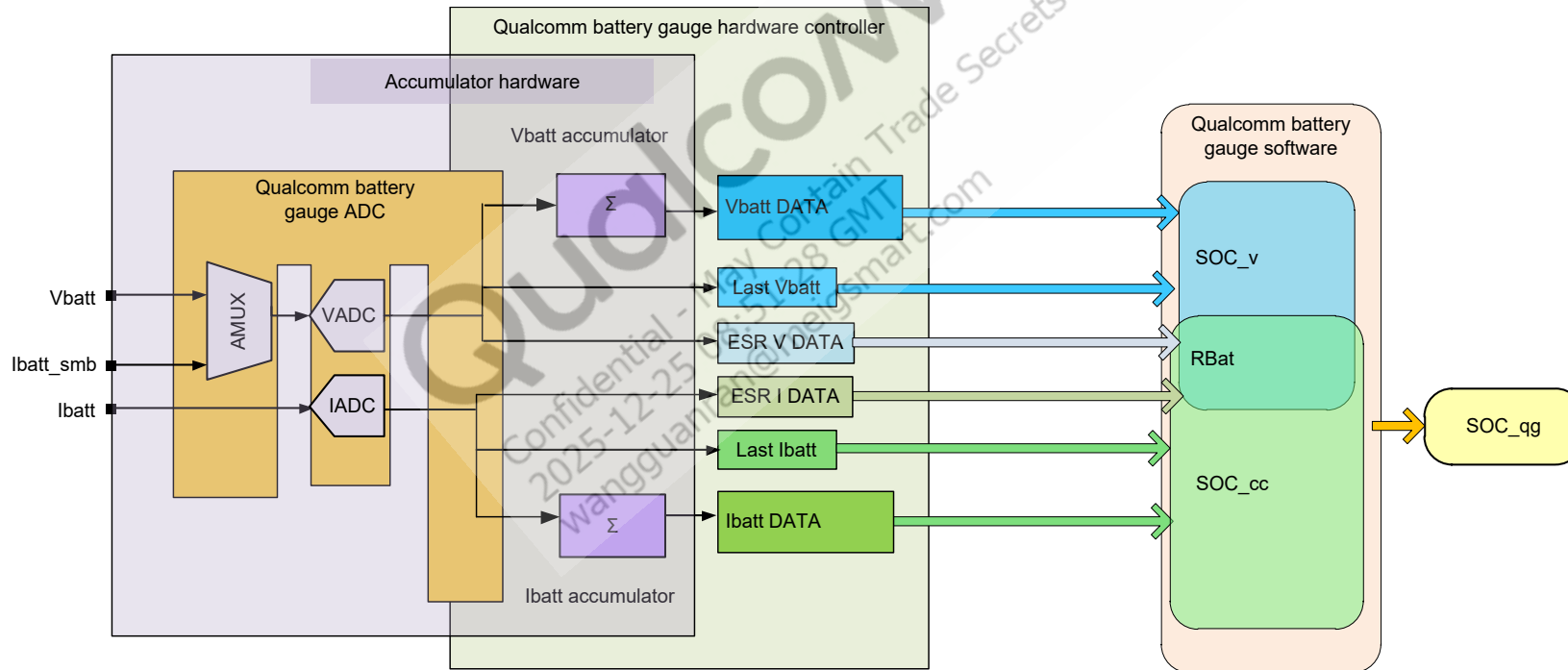
Qualcomm Battery Gauge Hardware Block Diagram

- Two ADCs; VADC for Vbatt and general purpose measurements and IADC for Ibatt value measurements.
- Current sensing using internal BATFET
- VADC and IADC voltage reference includes integrated temperature correction.
- Separate hardware module for battery interface management



Qualcomm Battery Gauge Accumulator

- The core of PM7250B Qualcomm battery gauge hardware consists of accumulator, which periodically samples, accumulates, and saves Vbatt and Ibatt values into PMIC register to read and process SoC.
- The same hardware is reused for an online battery ESR estimation, where V and I readings are run and saved to record pre-ESR and ESR pulse Vbatt and Ibatt values. The software reads the registers for ESR estimation.





Section 2.2.2

Qualcomm Battery Gauge State of Charge Estimation

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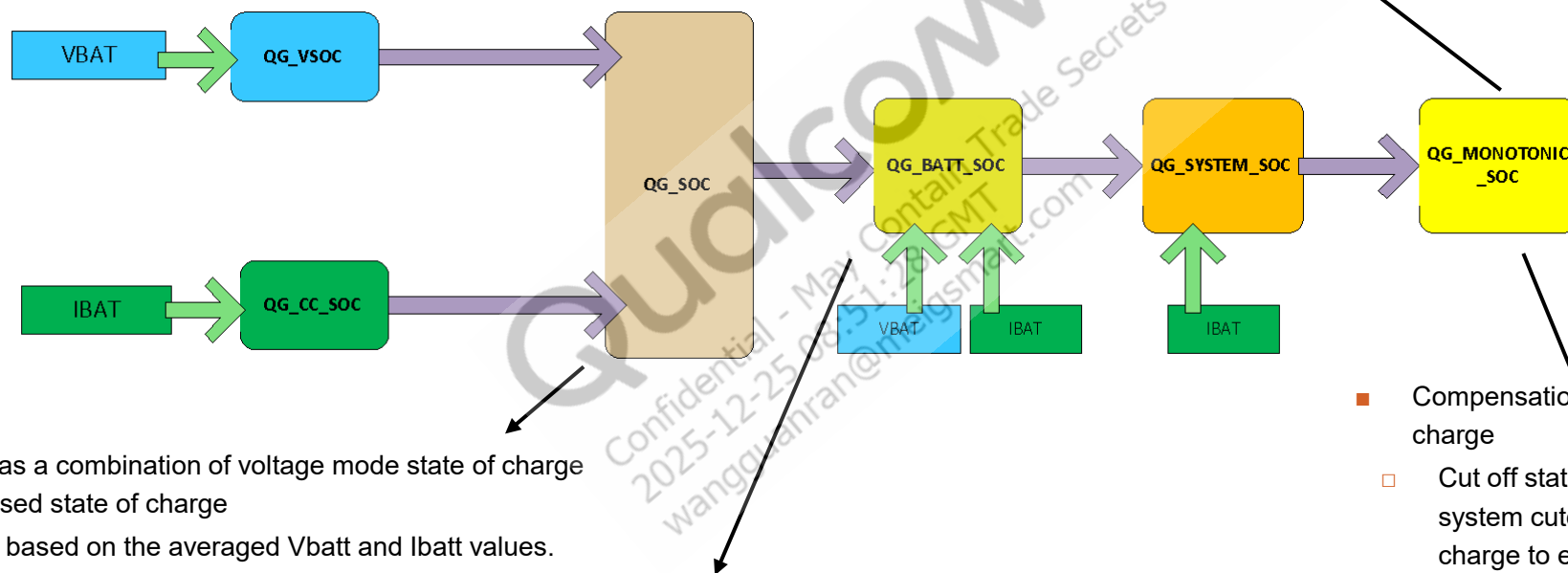
Qualcomm Battery Gauge State of Charge Estimation Flow

- Qualcomm battery gauge calculates the final reported state of charge, based on four stages of state of charge estimation.

- Qualcomm battery gauge state of charge
- Battery state of charge
- System state of charge
- Monotonic state of charge

- Guarantees that the system state of charge is monotonic

- Decreases only during discharging or supplemental mode
- Increases only during charging
- Implements the maximum allowed slope to guarantee a smooth user experience



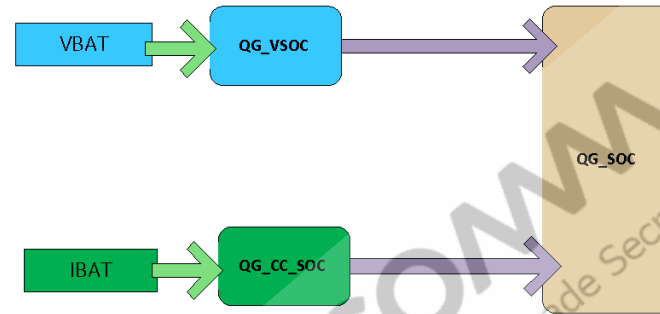
- Estimated as a combination of voltage mode state of charge and CC based state of charge
- Calculated based on the averaged Vbatt and Ibatt values.

- Estimated using actual ESR, voltage, current, temperature, and status values

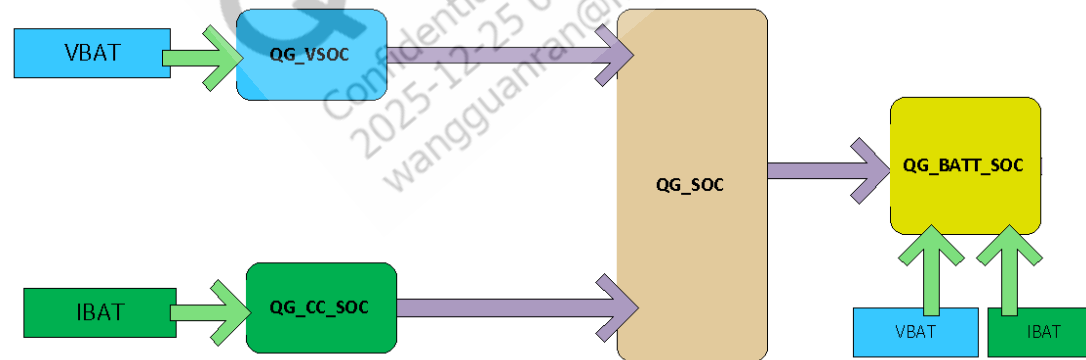
- Compensation for the cutoff state of charge and full state of charge
 - Cut off state of charge: a correction based upon the system cutoff voltage; it is applied to the battery state of charge to estimate the system state of charge.
 - Full state of charge: a correction based upon the charge float voltage and termination current; also applied to the battery state of charge to estimate the system state of charge.

Qualcomm Battery Gauge State of Charge and Battery State of Charge

- The Qualcomm battery gauge state of charge is a result of combination of coulomb counted state of charge and voltage mode state of charge.

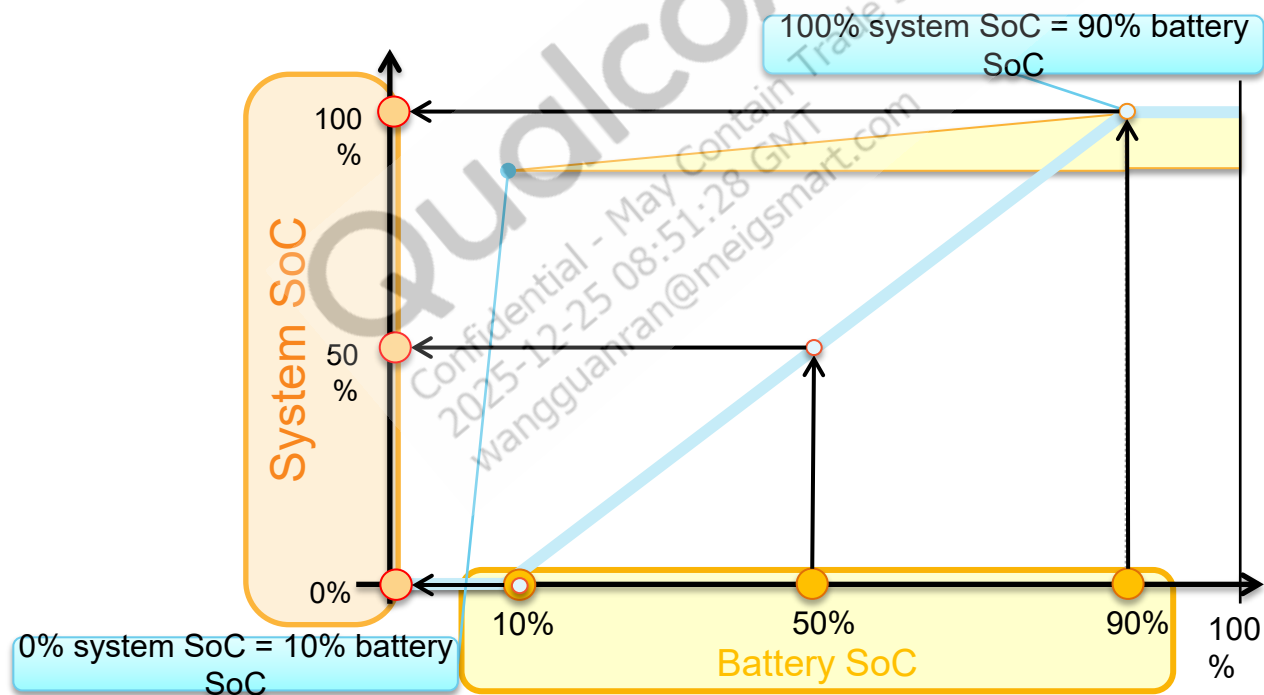


- The battery state of charge is estimated from Qualcomm battery gauge state of charge and by doing voltage loop feedback correction using ESR, temperature, voltage, current, and status values.



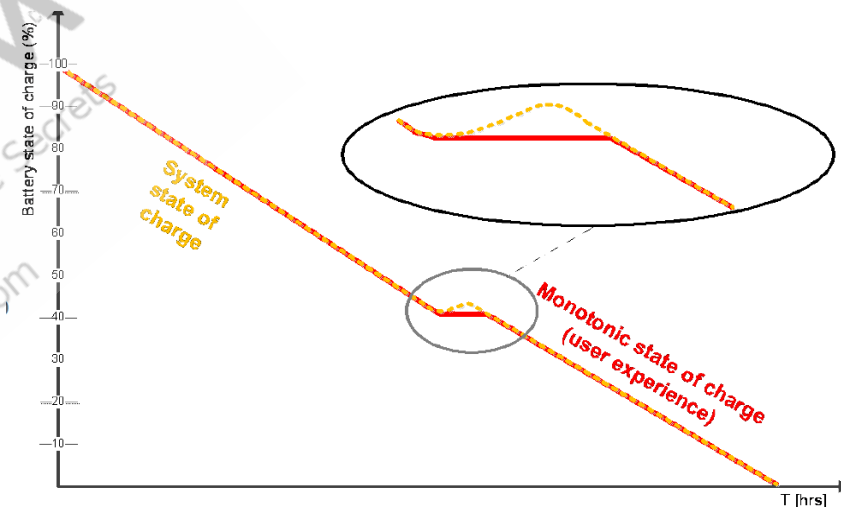
System Available State of Charge

- The reported 0% state of charge point is adaptively estimated based on:
 - Cutoff voltage: The cutoff voltage setting adjusts the 0% state of charge point to any battery voltage as appropriate. The Qualcomm battery gauge does this matching autonomously without software intervention.
- The 100% state of charge point is adaptively based on:
 - The programmable system termination current: This calculation predicts when 100% state of charge is displayed by using a programmable termination current threshold (system termination current). The system termination current must be set higher than the charger's end of charge termination current, or 100% state of charge will never be reached.



Monotonic State of Charge

- Because change in the operating conditions of the battery could result in an increase or decrease of the system available state of charge, value filtering is applied to obtain a desirable monotonic behavior.
- Monotonic filtering handles situations where the cutoff state of charge is increased and then decreased, which can happen due to:
 - Increases and decreases of the battery internal resistance value
 - Increases and decreases of the battery load current
- The default configuration also enforces a slope limiter on the direction allowed given the current sign.
- The slope limiter is needed when:
 - Certain event sequences where the monotonic filter is applied and then a change in the operating conditions occur that would otherwise cause the clamped system state of charge value to propagate directly to the monotonic state of charge read by the user thus creating an undesired step.
- The slope limiter prevents steps in the monotonic state of charge.
- Guarantees that the system state of charge is monotonic:
 - Decreases only during battery discharge or supplemental mode
 - Increases only during battery charge
- This implements the maximum-allowed slope to guarantee a smooth user experience.

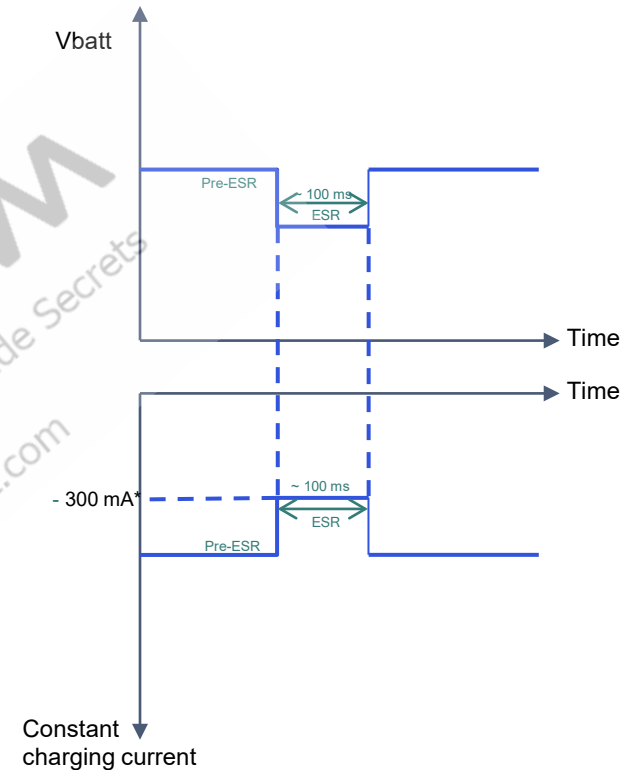


Online ESR Estimation

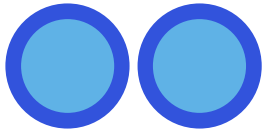
ESR estimation is enabled by the software when the device is in the constant current charging mode

- The charger reduces fast charging current (FCC) to 300 mA for ~100 ms to produce a current pulse.
- ADC takes V_{batt} and I_{batt} readings before and during the ESR pulse.

- Software reads the V_{batt} and I_{batt} registers for estimating ESR.
- This online ESR reading is used to track RBAT variation.



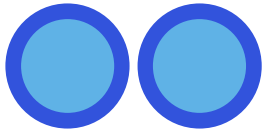
*Negative sign indicates that the battery is being charged during CC mode.



Section 2.3

Battery Interface Module (BIM)

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Section 2.4

Battery Current Limiting (BCL)

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BCL – Introduction (1 of 2)

- A typical Li-ion battery pack contains protection circuitry that has the ability to open-circuit the pack, due to overcurrent or undervoltage conditions.
- Under certain conditions, it is possible for a load transient to cause this protection circuit to activate, unexpectedly crashing the device.
- Potential causes of an unexpected system crash include:
 - Battery OCP
 - If OCP is triggered, the battery pack's protection hardware breaks the circuit.
 - This prevents overstress of the cell, but causes the system to crash due to undervoltage lockout (UVLO).
 - Battery UVP
 - The protection circuit of the Li-ion battery has the ability to break the connection when the cells' voltage drops too low.
 - This prevents overstress of the cell, but causes the system to crash due to UVLO.
 - PMIC UVLO
 - If the battery voltage drops below this chipset predefined threshold, the PMICs cause an immediate shutdown.

BCL – Introduction (2 of 2)

- BCL was designed to prevent an UVLO or unexpected system crash while permitting the maximum possible processor performance.
- Prevention of battery OCP events
 - Monitor battery current via IADC with submillisecond timing
 - Send a warning to SDM via LMh hardware when a battery overcurrent condition begins, to allow throttling of CPU speeds and hot plugging
 - If SDM is unable to mitigate an overcurrent event, open BATFET to protect the battery.
- Prevention of system UVLO/UVP
 - Monitor battery voltage via always-monitoring hardware comparators
 - Send a warning to SDM via limits management hardening (LMh) hardware when low battery voltage condition begins, to allow throttling of CPU speeds and hot plugging
 - If SDM is unable to mitigate the low battery voltage event, open BATFET to protect the battery.

NOTE: For more information about BCL, see the *Battery Current Limit (BCL) Overview and Tuning Presentation* (80-NM328-709).

BCL Hardware Block Diagram

- Measurements used for BCL:
 - Every $\sim 700\ \mu\text{s}$ in high-power mode (HPM) with the ADC
- Two VBATT comparators always measuring
 - Every $\sim 100\ \text{ms}$ in HPM with the ADC
- PMIC alerts:
 - SPMI interrupts for each detection
 - Hardware interrupts for direct mitigation
 - Normal interrupts for software action
 - BATTERY_ALARM_N (BAN) signal
 - Active low signal
 - Fault condition triggers transition
 - Cleared by the PMIC finite state machine (Qualcomm® FSM™)
 - BAN alarm causes mitigation in the following clients that are listening:
 - Charger
 - Flash
 - Backlighting



Section 3

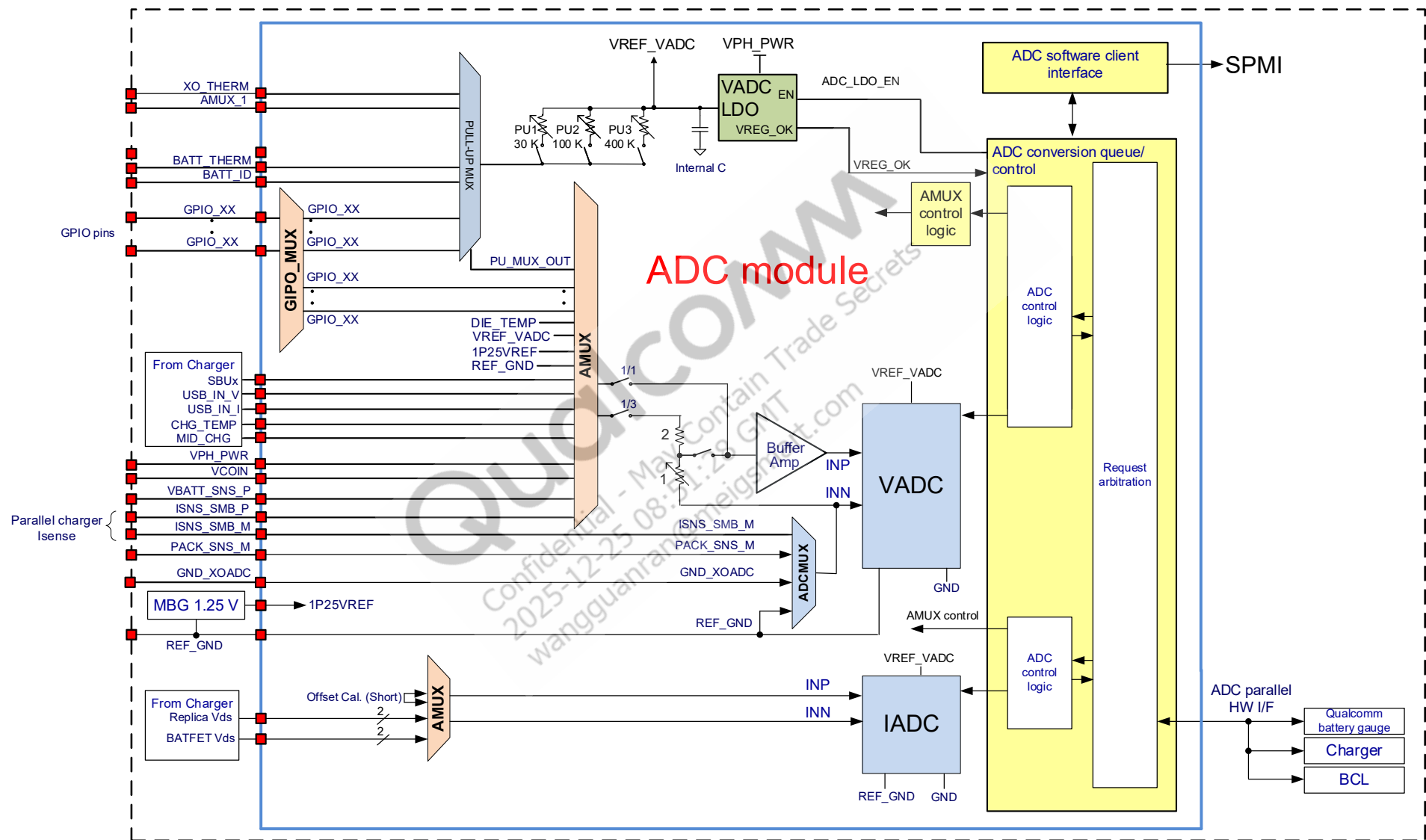
ADC Module

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ADC Module Features

- New stand-alone ADC module for all analog measurements
 - ADCs are no longer part of the Qualcomm battery gauge module.
 - One current ADC (IADC) and one voltage ADC (VADC)
 - Qualcomm battery gauge, BCL, and charger modules now have the capability to sniff ADC traffic.
 - This allows modules to acquire needed measurements without new requests.
 - This reduces redundancy and frees up ADC bandwidth.
- Multiplexer network connected to two 15-bit high-precision delta sigma ADCs
 - Multiplexer network provides 1/3 divider option for high voltage measurements
- Improved internal current measurement accuracy over previous fuel gauge modules
 - Improved current sensing circuit for BATFET current sensing
 - Hardware automatic ADC calibration every 125 ms in active mode
 - Current sensing via internal BATFET
 - Current sensing via ISNS_SMB_M, ISNS_SMB_P, and internal BATFET for parallel charging
 - Battery current sensing with the IADC: up to ± 10 A
- Gain and offset correction in hardware for all ADC channels
- BATT_ID no longer uses current sinks to take measurements, and instead cycles through three pull-up values: 30 k, 100 k, or 400 k
 - BATT_THERM no longer needs an external pull-up and uses same pull-up scheme
 - **Warning: DO NOT place any capacitors on the BATT_ID or BATT_THERM nets.**
 - Do not place them on the board or in the battery pack.
 - No capacitance is validated on these nets.
 - It is recommended that all OEMs ensure that their battery pack vendors do not include capacitors on either of these nets.

ADC Model Diagram





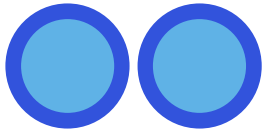
Section 4

GPIO Module

4.1 Configurable I/Os

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Section 4.1

Configurable I/Os

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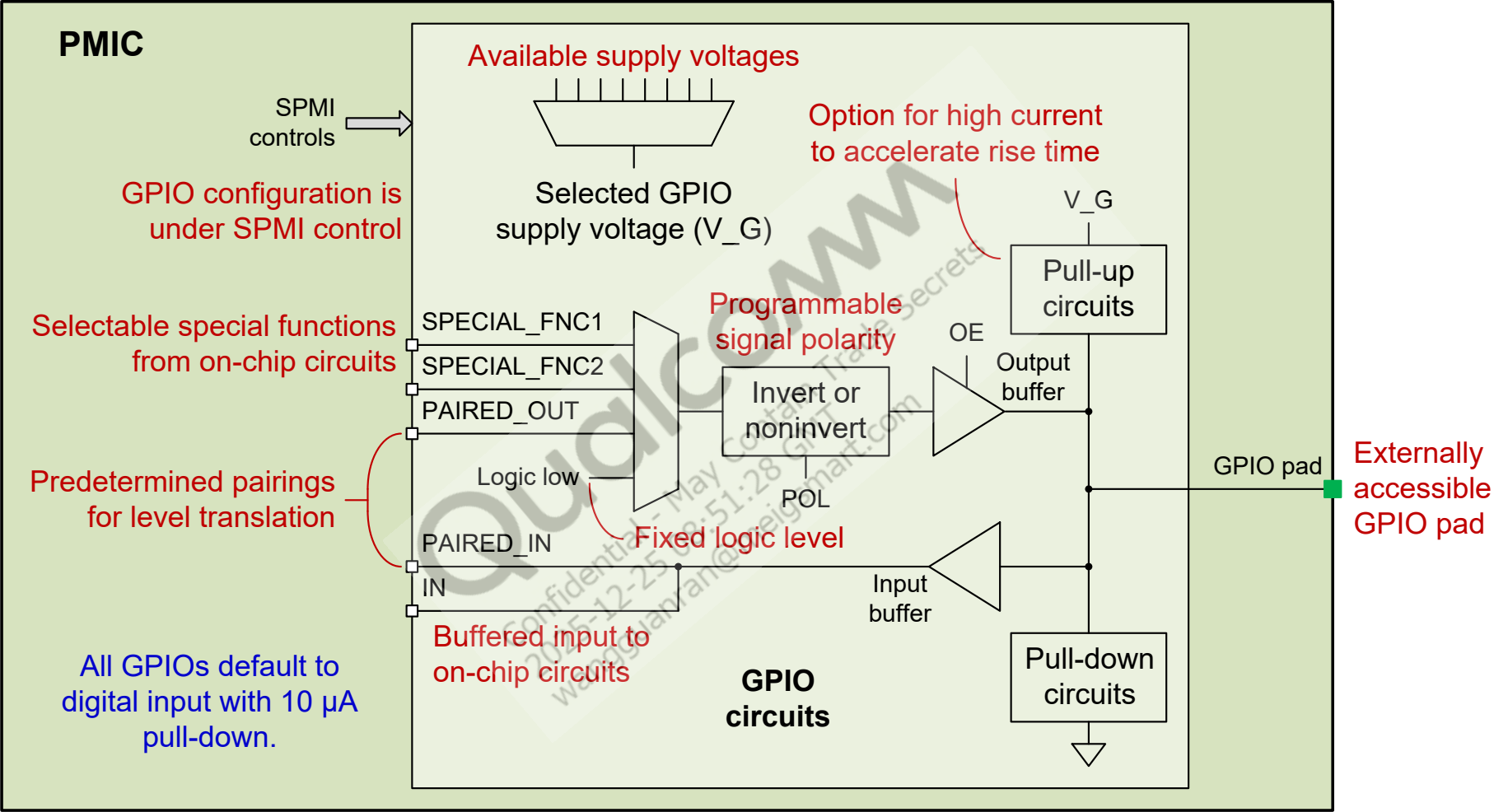
PMIC GPIO Pins

- The PM7250B device has 12 GPIOs.
 - GPIO_5 to GPIO_8 are MV GPIOs (VPH_PWR or 1.8 V).
 - GPIO_1 to GPIO_4 and GPIO_9 to GPIO_12 are LV GPIOs (1.8 V only).

Power source	Default	GPIO availability
VDD_MSM_IO	1.800 V	GPIO_1 to GPIO_4 and GPIO_9 to GPIO_12
VPH_PWR	3.600 V (typical)	GPIO_5 to GPIO_8

- PM7250B device custom GPIO features – special connections that can only exist in specific GPIO locations:
 - GPIO_01, GPIO_05, GPIO_08, GPIO_11, and GPIO_12: AMUX/ADC input
 - GPIO_09 and GPIO_10: factory mode boot signals

PMIC GPIO Pin Diagram



PM7250B GPIO Assignments

GPIO	Function	Hardware default configuration	SF1	SF2	SM7250 usage description	Type	Can be repurposed?
01	Spare	Digital input, 10 μ A PD				LV	Yes
02	Spare	Digital input, 10 μ A PD				LV	Yes
03	KYPD_HOME	Digital input, 10 μ A PD			Home key	LV	Yes
04	Spare	Digital input, 10 μ A PD				LV	Yes
05	BMR_S_THERM	Digital input, 10 μ A PD			Used for temperature monitoring of mmWave module	MV	Yes
06	SMB_STAT	Digital input, 10 μ A PD			Configurable; default digital input with 10uA pull-down SMB1355/SMB1393 status.	MV	Yes
07	Spare	Digital input, 10 μ A PD				MV	Yes
08	CAM_FLASH_THERM	Digital input, 10 μ A PD			Used for temperature monitoring of camera module	MV	Yes
09	FMB1	Digital input, 10 μ A PD	FMB1		Used in Type-C DAM mode to initiate EDL on MSM	LV	Yes
10	FMB2	Digital input, 10 μ A PD	FMB2		Used in Type-C DAM mode to control UART	LV	Yes
11	Spare	Digital input, 10 μ A PD				LV	Yes
12	BMR_W_THERM	Digital input, 10 μ A PD			Used for temperature of monitoring of mmWave module	LV	Yes



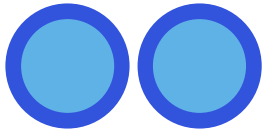
Section 5

User Interfaces

5.1 Haptics

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Section 5.1

Haptics

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Vibrator Driver

- The vibration is achieved by using an actuator that converts the electrical energy into mechanical force to be perceived as vibration
- Characteristics to consider for a vibrator
 - Response time – the start and stop time of the actuator
 - Vibration strength – the maximum acceleration achievable
 - Power consumption
- Common types of vibrators
 - The eccentric rotating mass (ERM) motor
 - The linear resonant actuator (LRA)
- PM7250B vibrator driver features:
 - Supports only ERM
 - SPMI/PBS control only
 - Input voltage range – VPH_PWR range (3.2 V to 4.75 V)
 - Programmable maximum output voltage between 1.504 V to 3.544 V
 - Overcurrent/short-circuit protection using the LDO OCP feature
 - Overdrive start-up feature
 - No external PWM or line input feature supported

Eccentric Rotating Mass Motor

- ERM is a simple permanent magnet DC brushed motor with an eccentric (off-centered) mass as a load on its shaft.
- This eccentric mass causes the vibration when this motor powered to rotate and is firmly housed on a phone
- The advantages of an ERM include:
 - Simple in construction makes it a cheap part
 - Requires only a DC input
 - Low cost

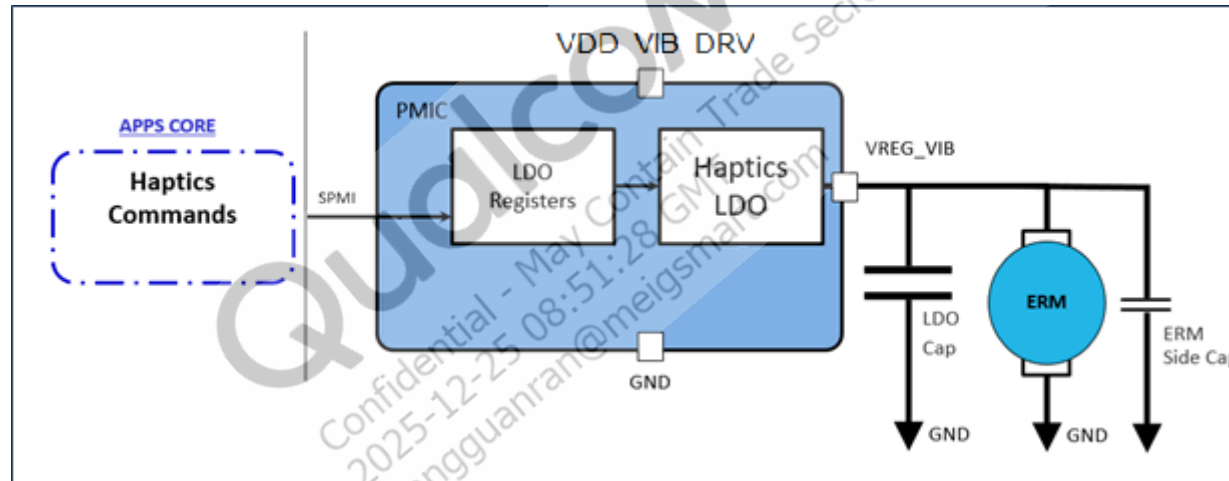


- The data shows that ERMs standard applicable voltage is ~3 V
- The peak current consumption is about 95 mA

Specification	Minimum	Typical	Maximum	Units
Input voltage	1.1	3	4.2	V
Actuator resistance at startup	10	30	100	Ω
Actuator resistance during running	12	45	175	Ω
Time to reach full G vibration			100	ms
Time to reach no vibration from full vibration			75	ms

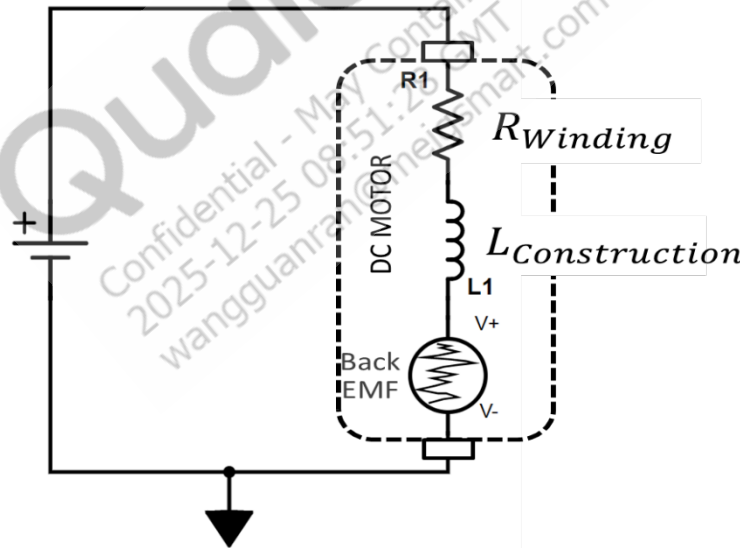
PM7250B Vibrator Driver

- An ERM is driven by a DC voltage applied differentially across the motor terminals.
- Each voltage level represents a combination of amplitude and frequency. The amplitude and frequency cannot be varied independently.
- Programmable through the SPMI.



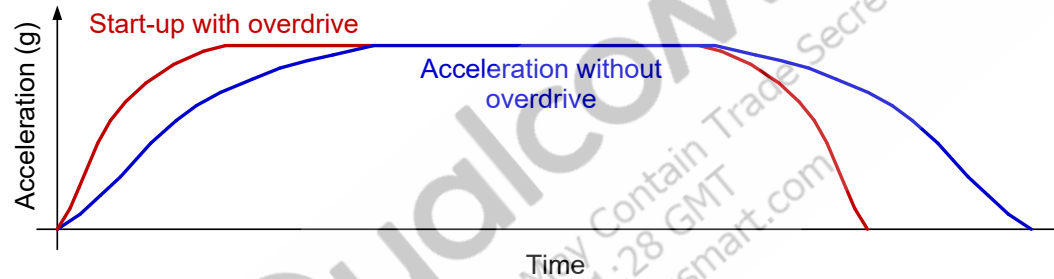
Modeling

- ERM is a simple permanent magnet DC brushed motor with an eccentric (off-centered) mass as a load on its shaft.
- This eccentric mass causes the vibration when this motor powered to rotate and is firmly housed on a phone.
- Two parasitic elements. First, the coil has a
 - Resistance ($R_{Winding}$)
 - Self-inductance ($L_{Construction}$)
- When the motor is rotating, it produces a voltage to oppose this rotation according to the Lenz's law. This voltage is called the Back-EMF.

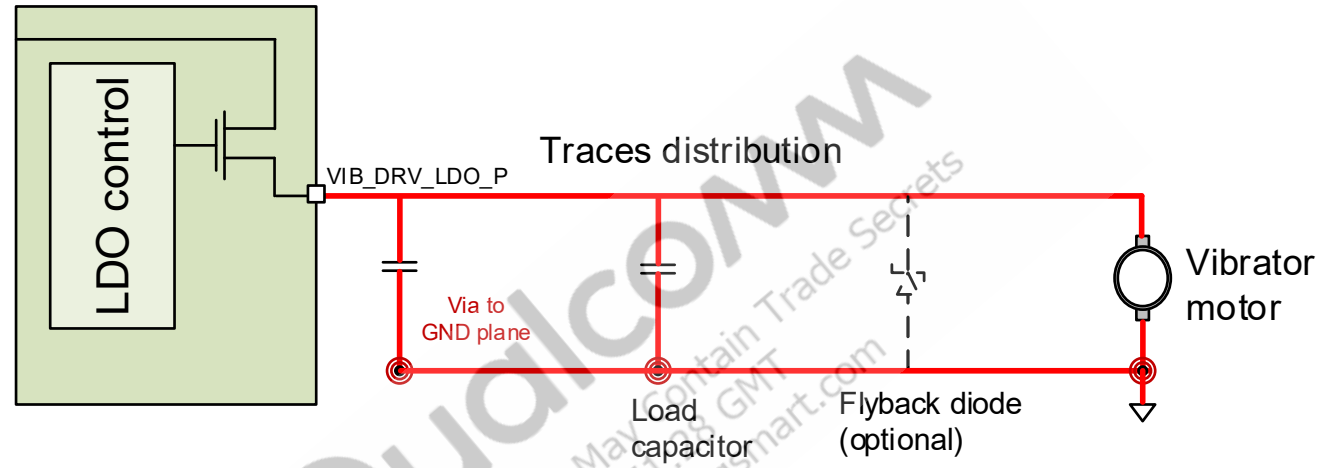


Overdrive

- ERM devices have sluggish start-up and stop characteristics.
- Achieved by applying more than the rated voltage across the VIB_DRV for a short duration.
- Shutdown overdrive is possible by enabling a strong pull down only (no reverse voltage is possible).



Haptics Schematic



Unused Pin Termination – PM7250B

Pin name	Unused pin termination (– indicates that the pin must be used per the reference schematic)
Charger and Battery gauge	
DC_IN_PON	NC
DC_IN_EN	NC
DC_IN_PSNS	NC
USB_IN	NC
MID_CHG	NC
VSW_CHG	NC
BOOT_PWR	NC
BOOT_CAP	NC
SYS_OK	–
PGND_CHG	GND
GND_CHG	GND
GND_PSUB_CHG	GND
REF_GND_CHG	GND
VPH_PWR	–
VBATT_PWR	VBATT
VBATT_SNS_P	VBATT
VBATT_SNS_M	GND
PACK_SNS_M	GND
USB_DP	NC
USB_DM	NC
CC1_ID	NC
CC2	NC
CC_OUT	NC
SBU1	NC
SBU2	NC

Pin name	Unused pin termination (– indicates that the pin must be used per the reference schematic)
Charger and Battery gauge (cont.)	
VDD_VCONN	–
SMB_EN	NC
IUSB_OUT	NC
ISNS_SMB_P/M	NC or GND
OPT_P/M	VBATT
VDD_PDPHY	–
BIM, ADC and BCL	
BATT_ID *	NC or 100 kΩ pull-down to GND
BATT_THERM *	NC or 100 kΩ pull-down to GND
CONN_THERM	NC
SMB_THERM	NC
AMUX_1	NC
BA_N	NC
GND_ADC	GND
Haptics	
VIB_DRV_P	NC
VDD_VIB_DRV	–
GPIOs - Any GPIO pin, if unused, can be left floating. See the PM7250B GPIO Assignments slide for additional details.	

Pin name	Unused pin termination (– indicates that the pin must be used per the reference schematic)
Infrastructure supplies	
VARB_CHG	–
AVDD_BYP	–
PVDD	–
VDD_IO	–
Other pins	
KPD_PWR_N	–
FAULT_N	–
TEST_EN_VPP	GND

* Unused pin termination for BATT_ID and BATT_THERM are given in context of battery not being used in the system such as in a battery less system.

To fake presence of battery and have charging enabled, use a 100 kΩ pull-down to GND on BATT_ID. To fake presence of battery and have charging disabled, use 2 kΩ to 14 kΩ pull-down to GND on BATT_ID.

To fake a good battery temperature, connect a 100 kΩ pull-down to GND on BATT_THERM.

Ship Mode

- PM7250B charger supports ship mode to completely isolate battery from VPH. This helps during phone shipping/storage to prevent battery discharge through VPH leakage.

Ship mode entry:

- If software writes '1' to SHIP_MODE_EN bit in BATIF_SHIP_MODE (0x1240) register, PM7250B Charger enters Ship Mode in ~2 seconds.
- After SHIP_MODE_EN = 1 command, PM7250B ignores the Ship Mode exit conditions for ~13 seconds, allowing the operator time to box the phone without accidentally trigger Ship Mode exit.

Ship mode exit:

PM7250B Charger exits Ship Mode and resumes normal operation upon any of the following events:

1. Power key pressed (KPD_PWR_N pin pulled low) for > ~2 sec;
2. USB_IN unplug (< VCOARSE_DET) and then plug-in (> VCOARSE_DET for 20ms);
3. DCIN_PON unplug (< VCOARSE_DET) and then plug-in (> VCOARSE_DET for 20ms);

NOTES:

- For ship mode exit cases 2 and 3, note that any existing input power must be unplugged (that is discharged) below the threshold and then a new plug-in event should happen.
- Avoid USB_IN or DC_IN to remain connected after 13 seconds.



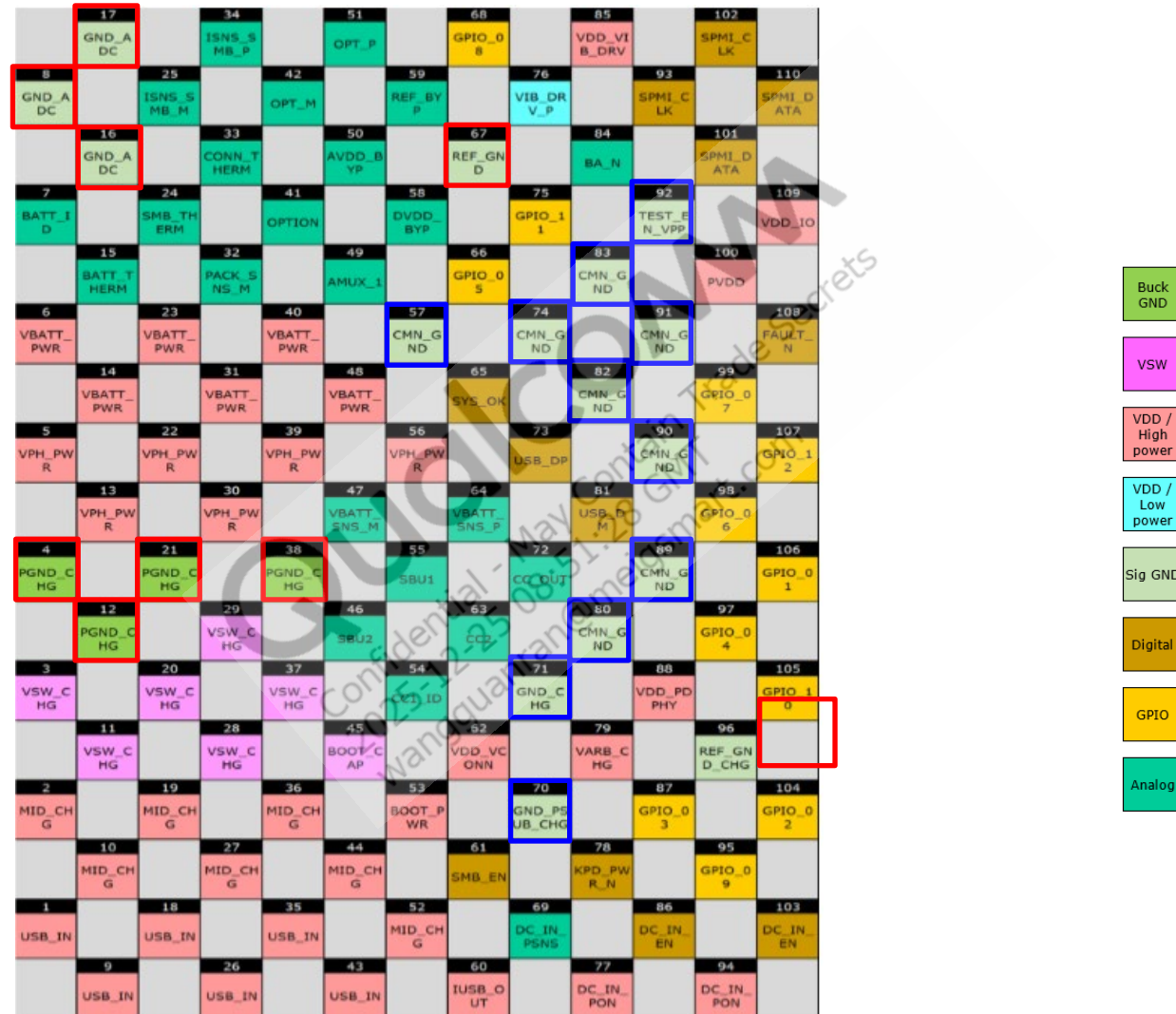
Section 6

PM7250B Layout Guidelines

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Special Grounding Rules

The **BLUE** outlined grounds are common and can be connected together. The **Red** outlined grounds have special requirements, see the checklist for details.

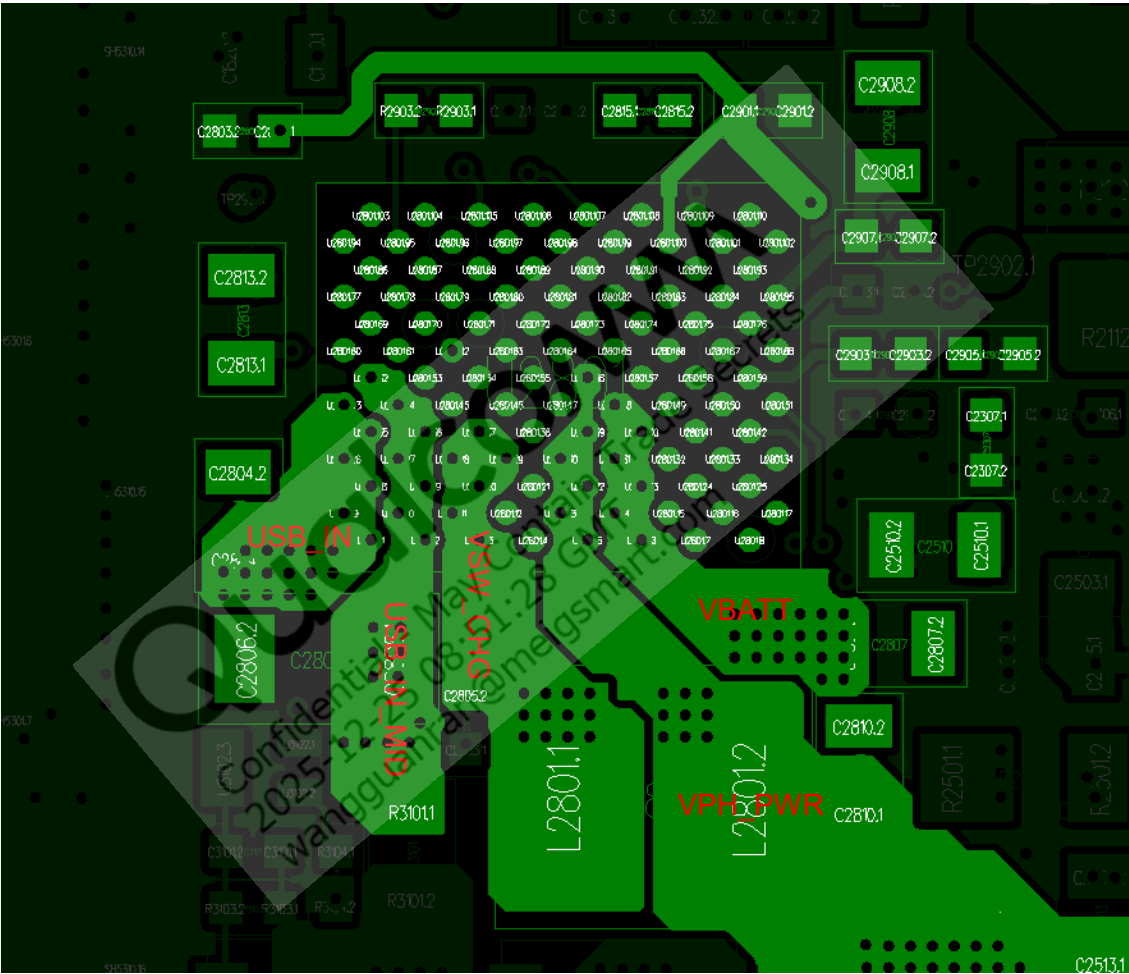


The component number indicators shown in the figure do not have any reference to any schematic and are only used to indicate the type of component (that is, L = inductor, and so on).



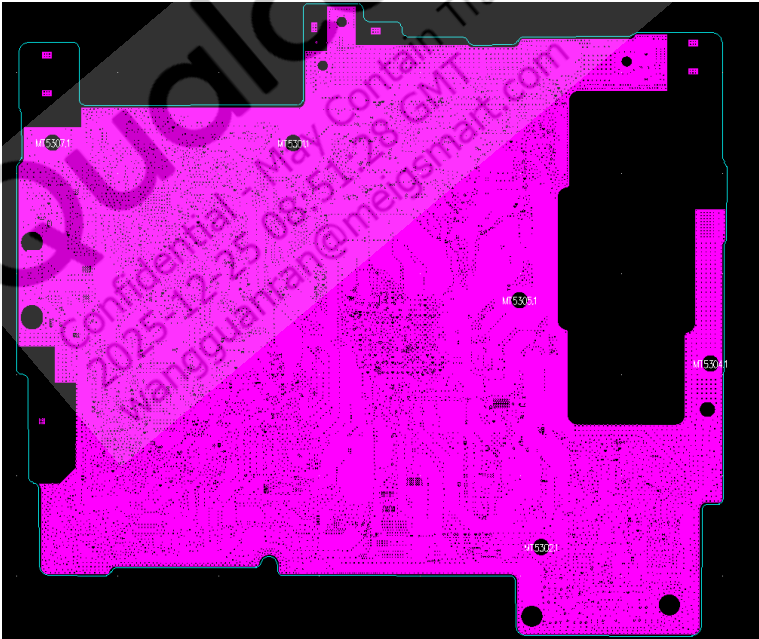
Routing

Outer Layer



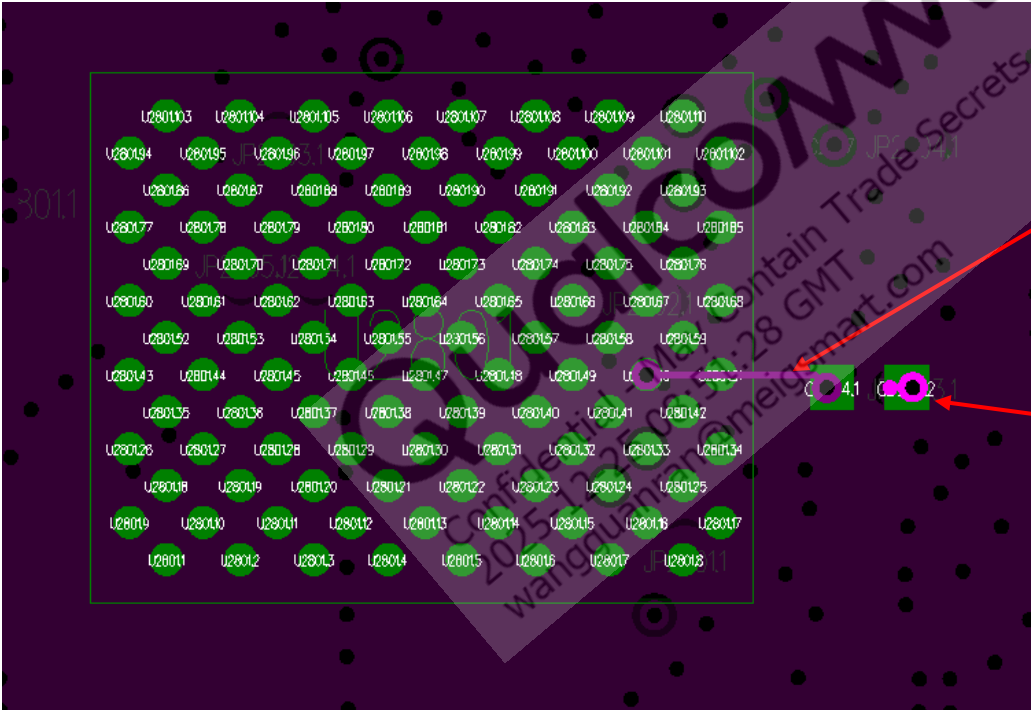
Main GND Plane

Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
2	High Risk	Main GND Plane	This is the most important of all requirements. "Main gnd plane" MUST be a VERY low impedance ground plane, from ANY one point to another. 1) It must NOT have too many via hole clearances (making it look like swiss cheese) and must NOT have ANY signals routed on it (which breaks up the ground plane). 2) 2) If multiple ground planes are used, they must be "stitched" together at frequent intervals so as to appear as a single solid plane. 3) 3) If a signal has a "dedicated via" requirement, it must be connected to the main GND plane(s) ONLY.	All the routing requirements that mention connection to main GND plane rely on the ground plane to have very low impedance from any one point to another. Violating this requirement can result in many problems that are extremely difficult to debug.



AVDD_BYP Routing

Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
3	High Risk	AVDD_BYP Pin: 50	Trace from pin to cap and a dedicated via from cap to main ground plane. DCR < 200 mΩ and ESL < 10 nH Do NOT connect to any other ground.	AVDD is the supply to various analog circuits in the PMIC and any noise picked up on this trace can cause a host of issues.

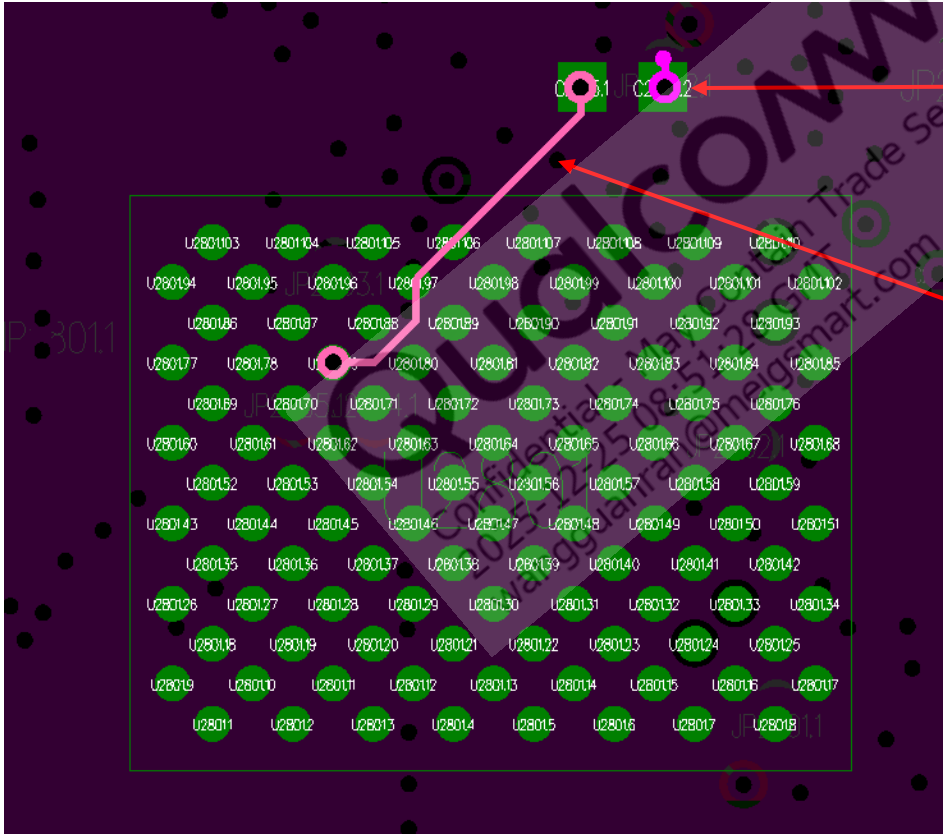


AVDD_BYP routed on layer 9, no overlap with any noisy trace on layer 10 and layer 8.

GND pad of cap is connected to main GND plane with a dedicated via without sharing to any GND patch or island on intermediate layer.

VARB_CHG Routing

Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
4	High Risk	VARB_CHG: Pin 79	Place the VARB_CHG cap as close as possible to the PMIC. Trace from the pin to the cap and a dedicated via from the cap ground to the main ground plane. Do NOT connect to any other ground.	Internal supply to sensitive circuits; noise on this supply can affect performance of the charger, FG, and other modules.

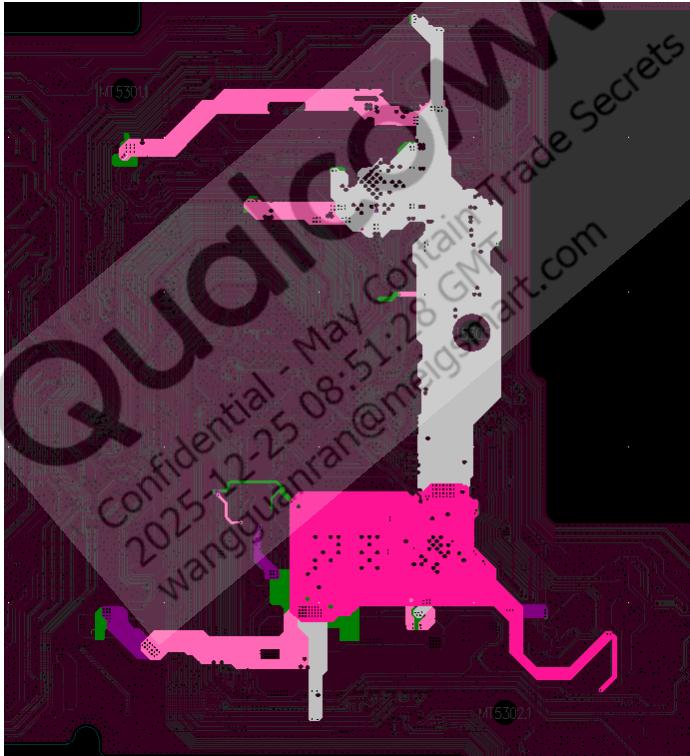


GND pad of cap is connected to main GND plane with a dedicated via without sharing to any GND patch or island on intermediate layer.

VARB_CHG routed on layer 8, with Layer 7 as GND shielding and no overlap with any noisy trace on layer 9.

VPH_PWR Routing

Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
6	High Risk	VPH_PWR for PMICs	VPH_PWR plane(s) must ensure power requirements of RF PA devices and PMICs don't adversely affect each other. A single good VPH plane is usually sufficient, but some cases may require separate plane shapes for PMIC and RF PA devices.	AC and DC voltage drop along PA power route can be large due to large PA load current. If the route is shared, the PMIC will experience large VPH droop when PA is transmitting. This has caused RF performance issues on past chipset.



USB_IN Routing

Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
7	High Risk	USB_IN Pin: 1, 9, 18, 26, 35, 43	Trace from USB connector to USBIN pin must be able to electrically and thermally handle max charger input current. Place the cap as close as possible to the USB_MID cap. Connect bottom of USBIN cap to bottom of USB_MID cap. Do NOT connect with any other ground.	Internal supply to sensitive circuits; noise on this supply can affect performance of the charger, FG, and other modules.



VBATT_PWR Routing

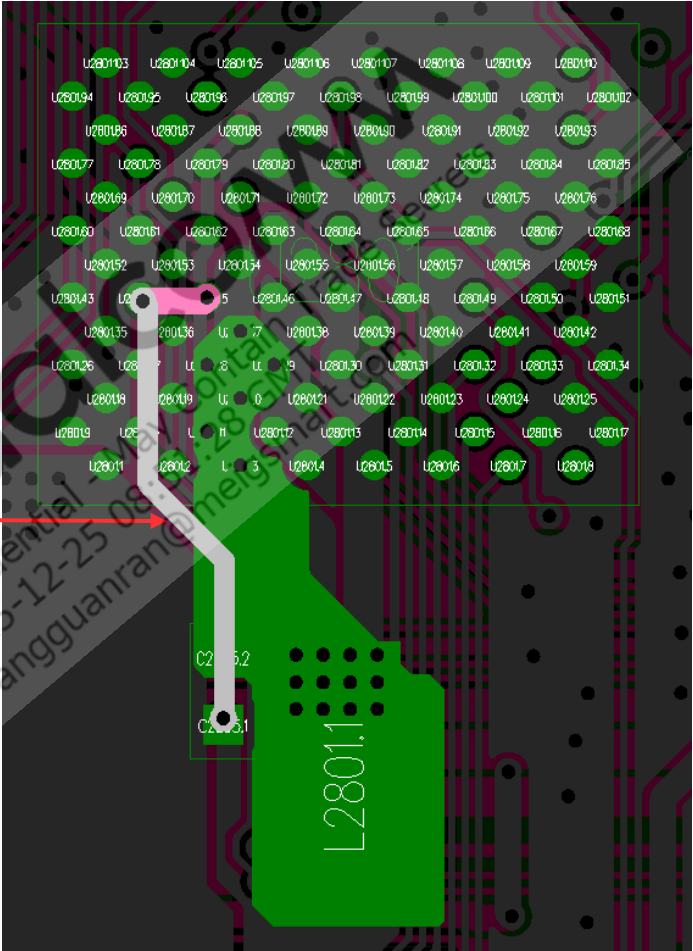
Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
12	High Risk	VBATT_PWR Pin: 6, 14, 23, 31, 40, 48	Trace from pin to the battery connector must be able to electrically and thermally handle max system load. Do NOT share this route with VBAT_SNS_P.	High current might damage the trace or create large voltage drop and increase charge time
	High risk	OPT_P pin: 51, OPT_M pin: 42	Isolated (star-route) from pin directly to the external sensing resistor (4Terminal recommended or follow reference schematics part). Route as a pair preferred. Do NOT share with other routes.	These pin as used for battery current measurement when using external sense resistor. Any IR drop due to shared routing will result in current sensing error. Rule of thumb, 1mV error in voltage means 100mA error in sensed current.



BOOT_CAP Routing

Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
13	Medium Risk	BOOT_CAP Pin: 45	Place the cap as close to the PMIC as possible. Trace from pin to cap. Short trace from cap to VSW_CHG. Trace inductance from BOOT_CAP to Cap must be < 5 nH, < 15 mΩ.	Results in improper buck functionality that can lead to part damage.

Short trace is routed from PMIC pin to the cap to meet the parasitic specs.

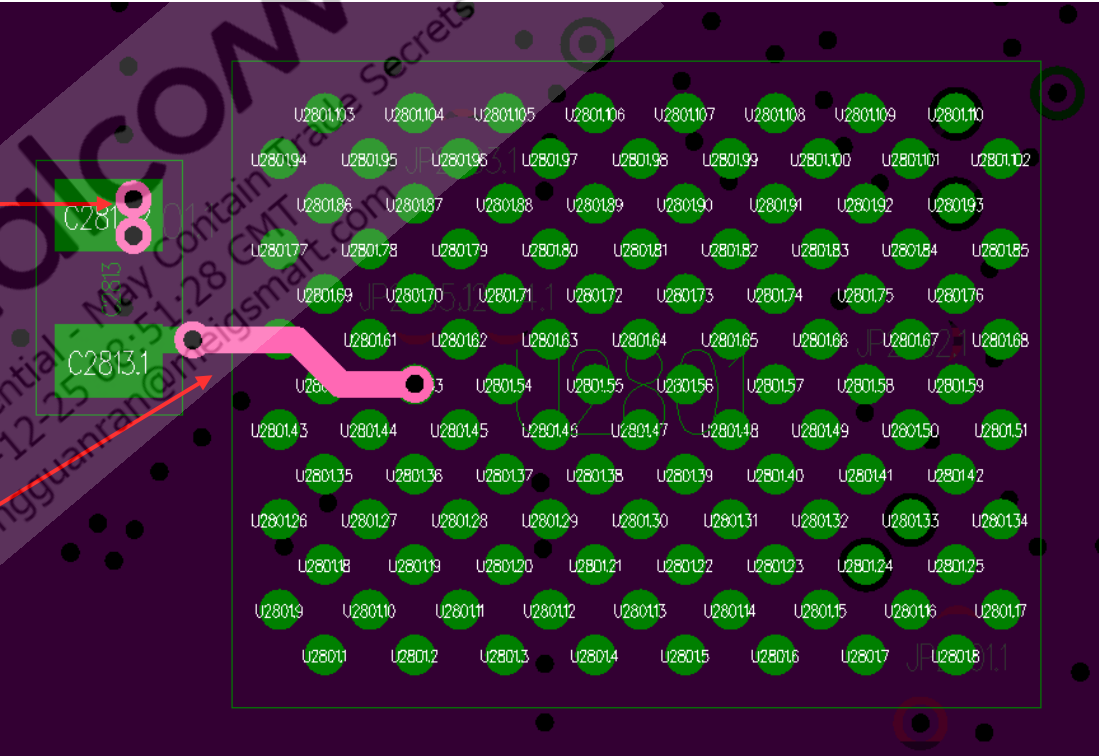


BOOT_PWR Routing

Item	Priority	Net	Routing requirement	Outcome of not meeting requirement
14	High Risk	BOOT_PWR Pin: 53	Place cap as close to the PMIC as possible. Trace from pin to cap. Dedicated via from cap to the main ground plane. Loop inductance from BOOT_PWR to Cap must be < 5 nH, <15 mΩ	Results in improper buck functionality that can lead to part damage

GND pad vias connected directly to main GND plane without sharing GND patched on intermediate layers

Cap placed close to the PMIC to meet parasitic specs





Section 7

Appendix 1 – Powering Schemes for Non-Mobile Phone Applications

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Overview

- Unlike mobile phones, few IoT applications do not require a battery and can be powered directly from DC source. This section explains various input powering schemes that can be implemented in IOT products.
- PM7250B supports three input power sources: MID_CHG, USB_IN, and VBATT. Based on the available input source, the use cases are listed below.

NOTE: The use cases mentioned in this section should be considered only as a guidance and **are not validated by QTI**. It is recommended that customers validate the schemes early in the product development.

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Use Cases

Use case	USB_IN	MID_CHG	VBATT	OTG reverse boost	Description	Example application	Simplified block diagram
1	Yes	No	Yes	PMIC internal reverse boost	No OTG support while charging the battery	Typical mobile phone	
2	No	No	Fake battery	PMIC internal reverse boost/external boost	Powered by third-party DC-DC switcher as fake battery	Always ON DC powered devices (PoE IP camera)	
3	Yes	No	No	No	Power the entire system using PMIC charger buck without battery	USB powered camera	
4	Yes	Yes	Yes	PMIC internal reverse boost	Power the entire system using DC source or USB_IN through PMIC charger	Device with USB and wireless charger	
5	Yes	Yes	Yes (hot swap featured)	PMIC internal reverse boost	No OTG support while battery charging or battery hot swap	Device which needs battery hot swap by keeping the system alive	
6	Yes	No	Yes	PMIC internal reverse boost	External boost OTG when charging with DC_IN input.	Device with USB + wireless charger.	

Pins Involved in Input Power Management

USB charger	DC charger	Battery I/F
USB_IN	MID_CHG	BATT_ID
USB_EN	DC_IN_EN	VPH_PWR
USB_SNS	DC_IN_PSNS	BATT_THERM
USB_DP	DC_IN_PON	VBATT_SNS_P
USB_DM		VBATT_SNS_M
CC1_ID		VBATT_PWR
CC2		
CC_OUT		

- See *PM7250B Power Management IC Device Specification* (80-PM008-1) for pin description.

Use Case 1: Typical Mobile Phone

- In mobile phone like applications, where the battery should charge from a USB port and the OTG reverse boost function is only required when the battery is not charging, customers can see the QTI reference design schematics; *SM7250 + PM7250 + PM7150A/PM7150L + PM7250B Reference Schematic (80-PL808-41)* for the implementation details.

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Use Case 2: Always ON DC Powered Devices

- In applications where the device is always powered by a DC connector (for example, a PoE powered device, DC wall adapter powered device, and so on) and a battery backup is not required, customers can follow the below two use cases:
 - A. Use case 2 A: OTG 5 V supply using the PMIC internal reverse boost
 - B. Use case 2 B: OTG 5 V supply using a third-party boost converter

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Use Case 2 A: OTG 5 V Supply using the PMIC Internal Reverse Boost

This use case utilizes the PMIC internal OTG 5 V reverse boost converter.

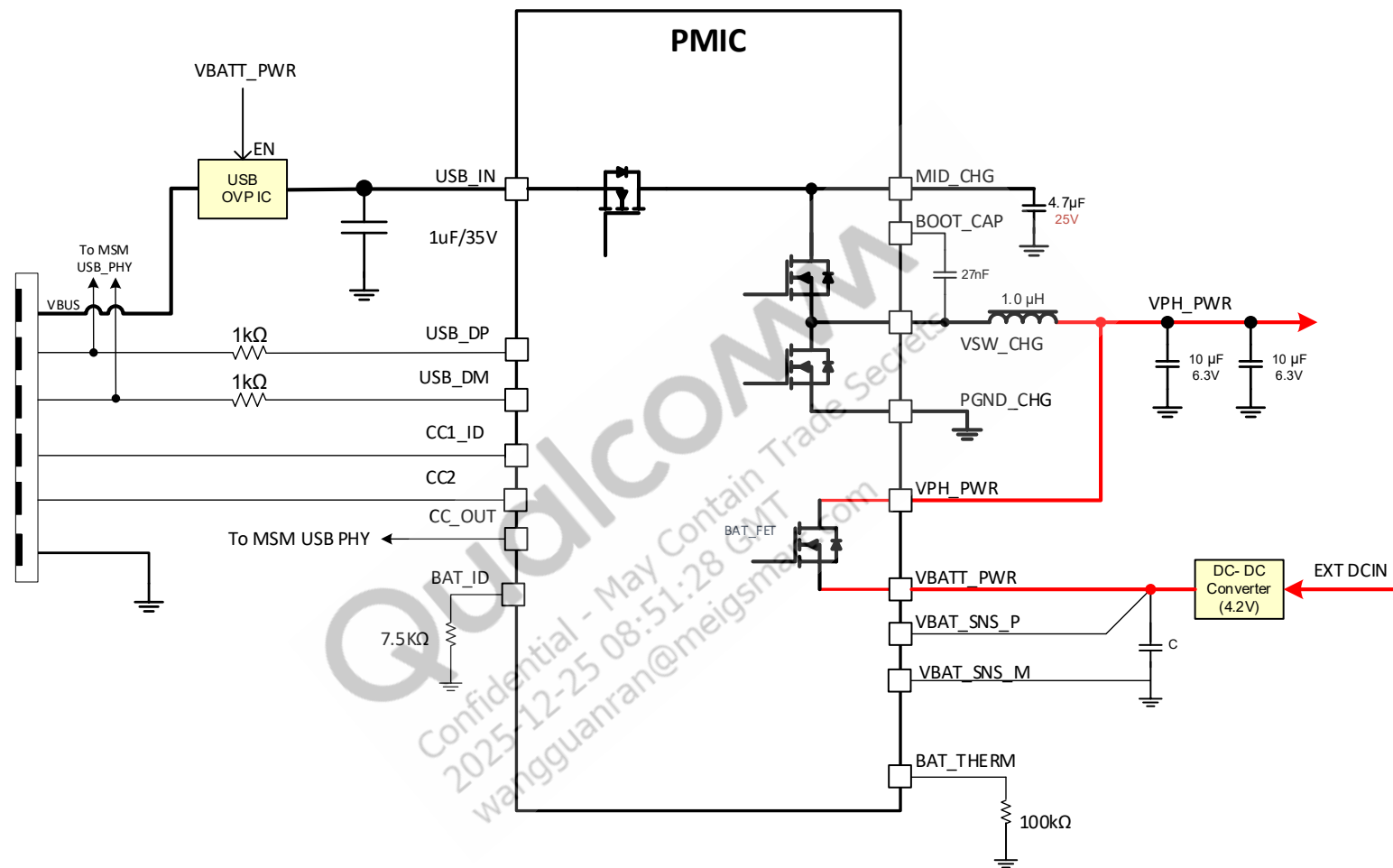
Hardware configuration

- Connect the DC source, EXT DCIN (2.7 V to 4.8 V) to VBATT_PWR. DCIN source and third-party DC-DC switcher must meet the transient system load demand on VPH_PWR. See [Guidance on Selecting a Third-party Buck](#) for transient current considerations on VPH node.
- Connect the USB_IN of PMIC to VBUS pin of the USB connector through an external OVP IC.
- Connect VBAT_SNS_P and VBAT_SNS_M to DC-DC converters output cap positive and negative terminals.
- Pull down BATT_ID with 7.5 K to disable the battery charging function in hardware.
- To avoid the PMIC PON trigger without EXT DCIN power supply (if USB cable is plugged in before EXT_DCIN supply) enable the external OVP/load switch through EXT DCIN or VBATT_PWR.
- When OTG device/UFP is connected on USB connector, PMIC will enable the reverse boost based on CC1_ID detection.
- For micro USB mode, connect CC1_ID pin to micro USB connector ID pin and float CC2. For Type-C mode, connect CC1_ID and CC2 pin to USB Type-C connector.
- Connect CC_OUT to SoC GPIO pin, similar to the QTI reference design.
- Pull down BAT_THERM with 100k to GND to fake a battery temperature.
- When OTG device/UFP is connected on USB connector, PMIC will enable the reverse boost based on CC1_ID.

Software configuration

- To disable the charger: SCHG_CHGR_CHARGING_ENABLE_CMD | 0x1042=0x00.

Use Case 2 A: OTG 5 V Supply using the PMIC Internal Reverse Boost Block Diagram



Use Case 2 B: OTG 5 V Supply using a Third-Party Boost Converter

- This use case utilizes the external third-party 5V output boost converter

Hardware configuration

- Connect the EXT DCIN (2.7 V to 4.8 V) to VBATT_PWR. DCIN source and third-party DC-DC switcher must meet the transient system load demand on VPH_PWR. See [Guidance on Selecting a Third-party Buck](#) for transient current considerations on VPH node.
- Use an external boost converter from VPH_PWR or EXT DCIN and enable it from the SoC GPIO pin.
- Connect the USB_IN of PMIC to VBUS pin of the USB connector through external OVP IC.
- Connect VBAT_SNS_P and VBAT_SNS_N to DC-DC converters output cap positive and negative terminals.
- Pull down BATT_ID with 7.5 K to disable the battery charging function in hardware.
- To avoid the PMIC PON trigger without EXT DCIN power supply (if USB cable is plugged in before EXT_DCIN supply) enable the external OVP/load switch through EXT DCIN or VBATT_PWR.
- When OTG device/UFP is connected on USB connector, PMIC will enable the reverse boost based on CC1_ID detection.
- For micro USB mode, connect CC1_ID pin to micro USB connector ID pin and float CC2. For Type-C mode, connect CC1_ID and CC2 pin to USB Type-C connector
- Pull down BAT_THERM with 100k to GND to fake a battery temperature.

Software configuration

- Disable the PMIC internal OTG reverse boost
 - SCHG_OTG_CMD_OTG | 0x1140 =0x00
 - SCHG_OTG_OTG_CFG | 0x1153 =0x20
- Implement the custom software logic to detect the connected OTG/UFP device through PMIC or SoC GPIO and enable the external OTG 5 V using SoC GPIO. Check with QTI USB software team for support.

Use Case 3: Power the Entire System using PMIC Charger Buck Without Battery

- The PMIC internal charger buck is designed only to charge the battery and it is not recommended to run the entire system powered only from charger buck converter.

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Use Case 4 A: Charging with a Generic DC Charger and USB_IN Input

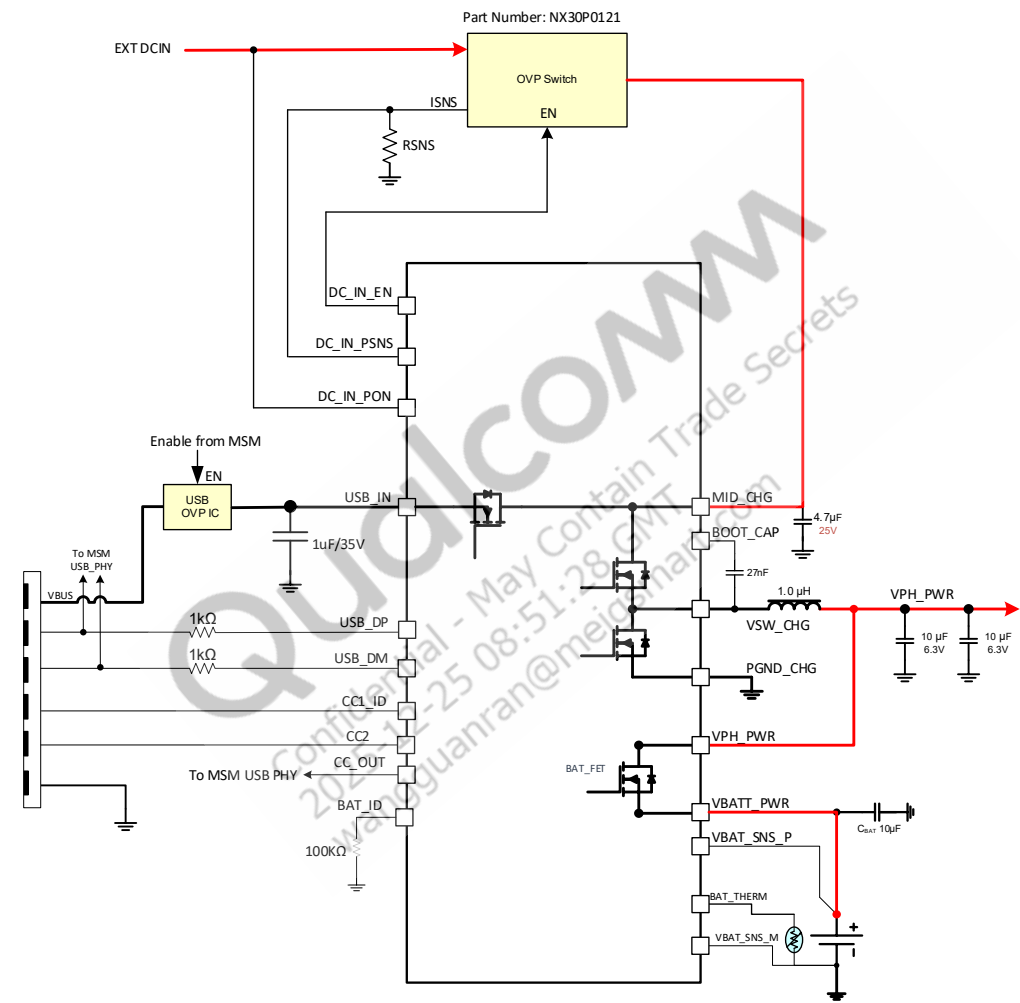
The PM7250B charger can also work with a generic DC charger input with an external B2B FET. DC input and USB charging paths are joined at the MID_CHG node, eliminating a DC FP FET or external PMUX.

Hardware configuration

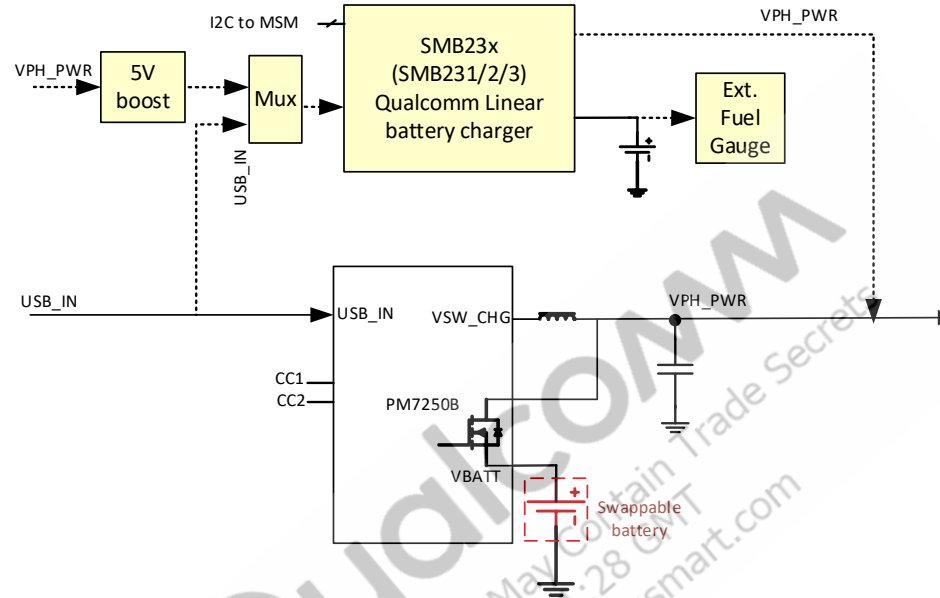
- Connect the DC source, EXT DCIN (3.7 V to 12.6 V) to MID_CHG through B2B FET. DC input and USB charging paths are joined at the MID_CHG node, eliminating a DC FP FET or external PMUX.
- Connect DC_IN_PSENS to ISNS of the NX30P0121. The B2B FET outputs a scaled version of the input current. The voltage across the RSNS resistor is fed to PM7250B DC_IN_PSENS regulation loop.
- For VBAT_SNS_P, VBAT_SNS_M, BATT_ID and BATT_THERM connection refers the QTI reference design.
- When OTG device/UFP is connected on USB connector, PMIC will enable the reverse boost based on CC1_ID detection.
- USB_IN takes priority unless software changes priority to DC_IN and DC_IN_EN remains low. When power is drawn from the USB_IN input, B2B FET (NX30P0121) is turned OFF by pulling DC_IN_EN to low.

Note: Implement external hardware logic to turn off the DC_IN_EN to low when the EXT DCIN falls below some threshold. Make sure that EXT DCIN falling threshold is greater than USB_IN max operating voltage.

Use Case 4 B: Charging with a Generic DC Charger Input and USB_IN Input Block Diagram



Use Case 4C: Battery Hot Swap Using Secondary Battery with SMB231



- Here battery hot swap implemented using QTI linear battery charger SMB231 with secondary backup battery.
- The secondary battery can be charged from SMB231 from either 5 V boost converter or USB_IN (~5 V only).
- During battery removal, the SMB231 will continue to power the system from the secondary battery.
- The 'make before break' rule must be followed while enabling/disabling the main battery and secondary battery.
- Ensure the device is in the low power mode during battery removal and the back up battery should be capable to keep the device alive.

Use Case 5A: Battery Hot Swap with USB Charger Connected

- This scheme can be used for applications where the battery needs to be replaced on the fly by keeping the device alive.
- In PM7250B, the battery hot swap feature allows detection of a plug-out and subsequent plug-in of the battery when the device is connected to USB charging port (DCP or CDP). SDP can be used, if it is sufficient to keep the system alive.

Here BAT_ID and BAT_THERM should be connected to the main battery connector

Mechanical consideration

- Ensure that the mechanical design of battery terminals and the battery compartment is such that the ground pin of battery always makes first contact before V+ or BAT_THERM do. This is to ensure that the battery interface pins do not experience any unwarranted potential difference.

Software configuration

During battery removal, if a backup supply is present then BAT_ID and BAT_THERM missing interrupt is handled by software. Once battery is plugged-in BAT_ID is re-detected and a new profile is loaded if required

Use Case 5 B: Battery Hot Swap Using Backup Power

- If the USB charger is not connected, then the battery hot swap can be implemented using the backup battery (example: super capacitor) as shown in the below image.
- Here BAT_ID and BAT_THERM should be connected to the main battery connector
- When the main battery is present, VPH_PWR will be stepped down using LDO to charge the backup battery
- During battery removal, the backup battery will be boosted up using boost converter IC to power VPH_PWR.
- The 'make before break' rule must be followed while enabling/disabling the LDO and boost converter IC.
- Ensure the device is in the lowest power mode during battery removal and the back up battery should be capable to keep the device alive.

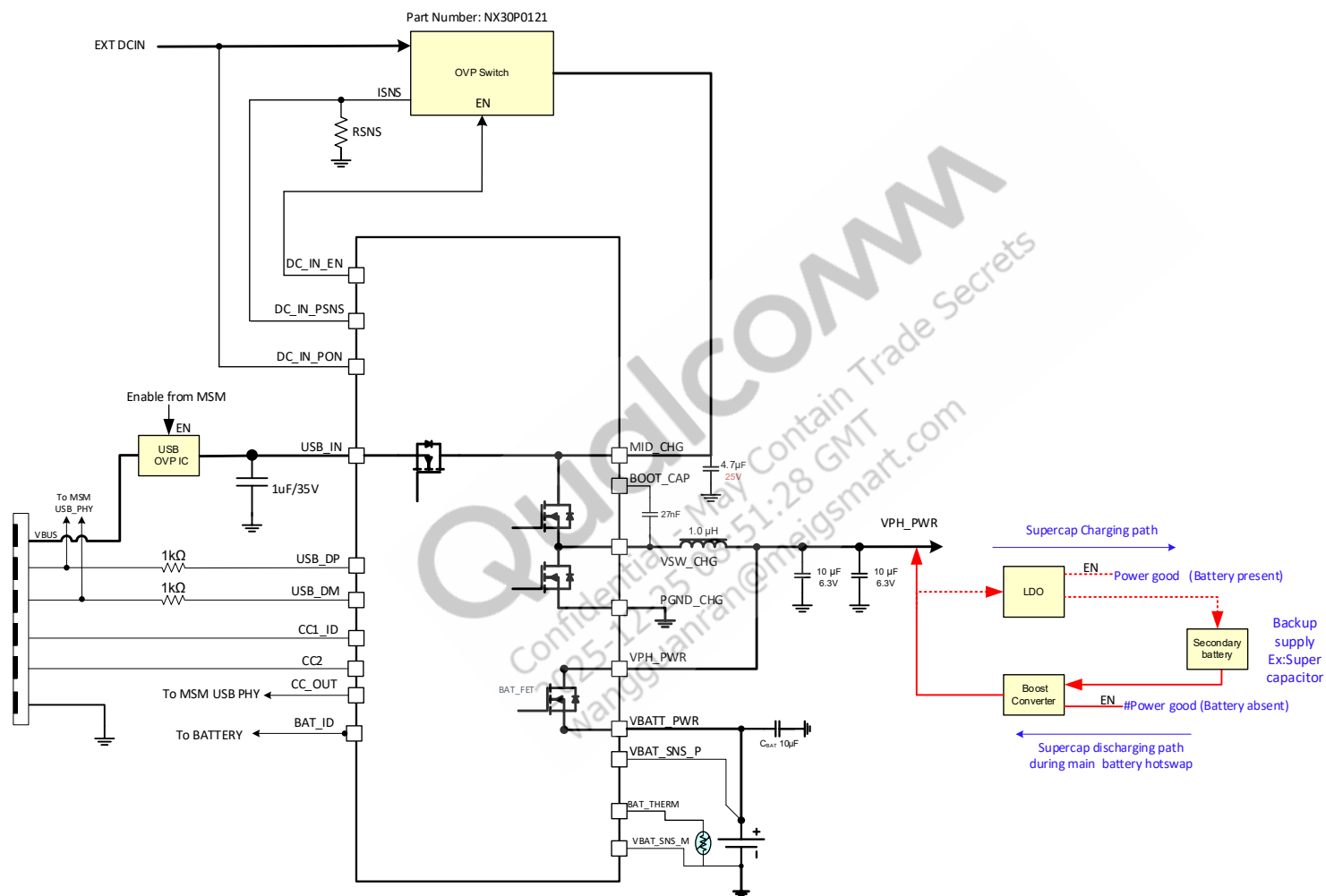
Mechanical consideration

- Ensure that the mechanical design of battery terminals and the battery compartment is such that the ground pin of battery always makes first contact before V+ or BAT_THERM do. This is to ensure that the battery interface pins do not experience any unwarranted potential difference.

Software configuration

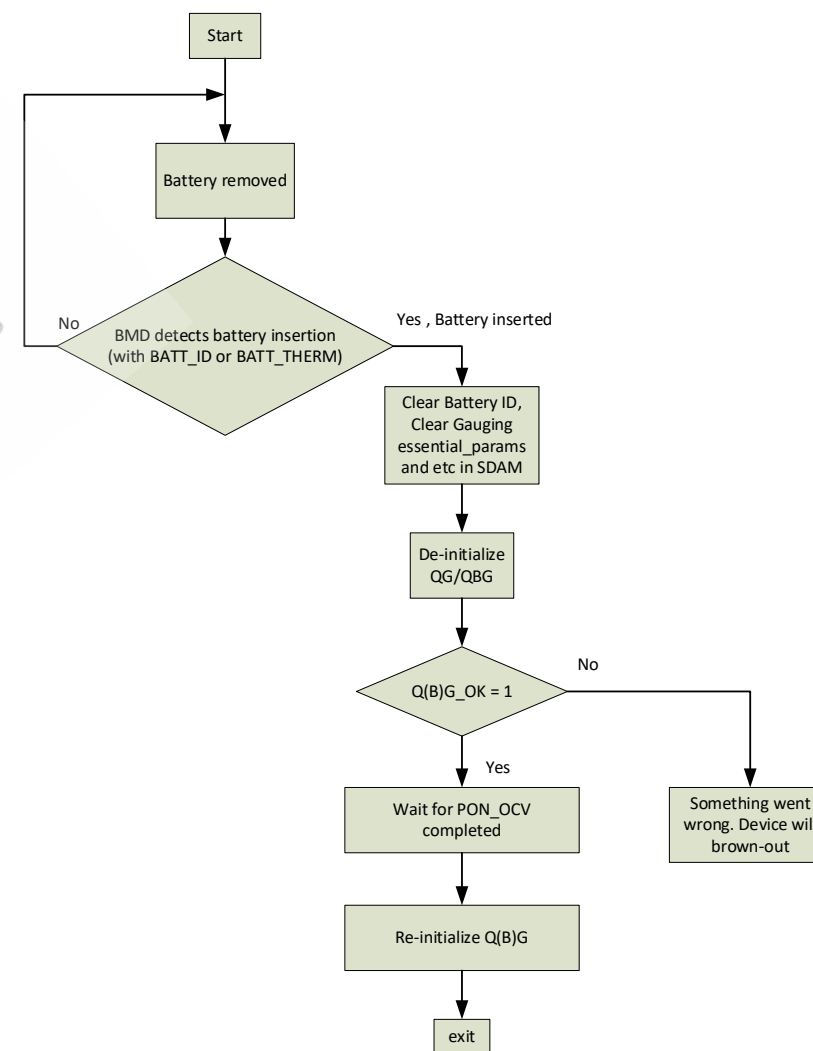
During battery removal, if a backup supply is present then BAT_ID and BAT_THERM missing interrupt is handled by software. Once battery is plugged-in BAT_ID is re-detected and a new profile is loaded if required

Use Case 5 B: Battery Hot Swap Using Backup Power Block Diagram



Battery Hot Swap and Reset Flow of Battery Gauging

- During the battery removal, the absence of BATT_ID or BATT_THERM will trigger the BMD (Battery missing detection) interrupt.
- When new battery is inserted, the presence of new BATT_ID or BATT_THERM will be detected, and the algorithm clears the FG registers.
- Based on the new BATT_ID, the battery profile will be loaded and waits for the PON_OCV (Power ON open circuit voltage) and then re-initializes the QBG.



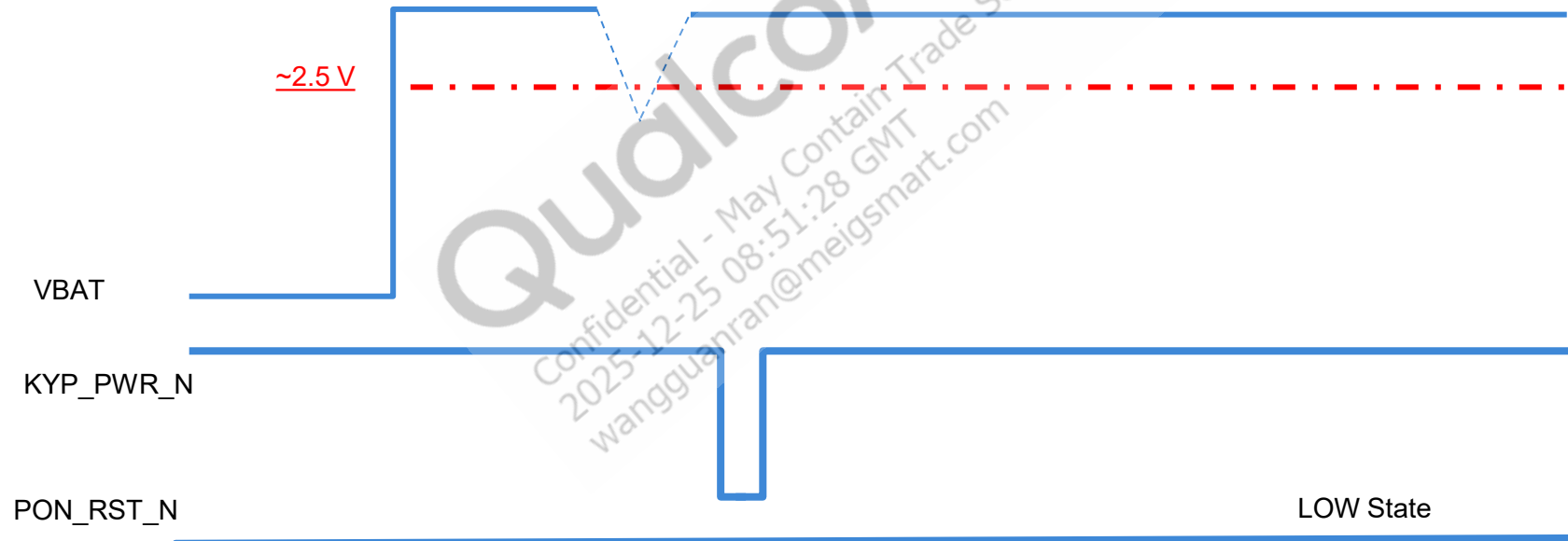
Considerations for Swappable Battery Application

Battery insertion/cold swap

- During first battery attach or battery cold swap press the KPD_PWR_N for ~2 s to boot the system properly (applicable for SM7325/QCM6490/QCM4490/QCS4490).

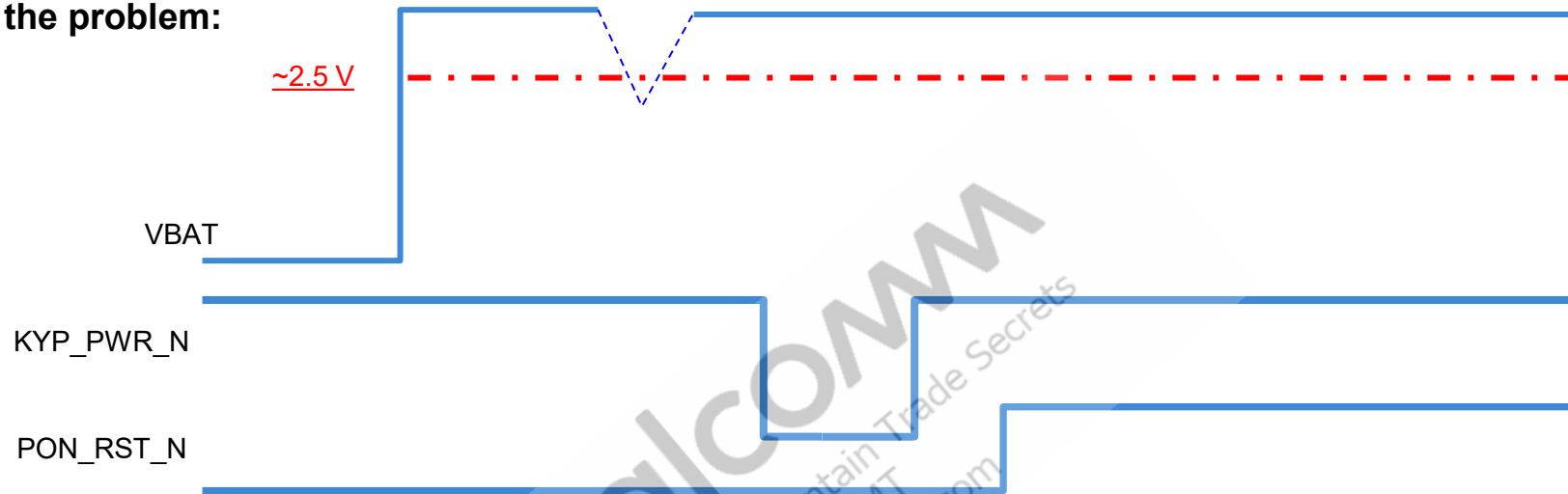
Reason:

- When the VBAT goes low momentarily to about 2.5 V or lower **before SBL is loaded**, PM7250B undergoes a silent reset without rest of the PMICs going through reset. This is because the dVDD reset threshold for PM7250B is 2.5 V.



Considerations for Swappable Battery Application (cont.)

Work around to fix the problem:



Why PON works with 2 sec KPD PWR N press?

- PM7250B has a default slave ID (SID) at reset. As part of DVDD config sequence (OTP), the SID for SM7325/QCM6490/QCM4490/QCS4490 platform is reconfigured by PMK7325/PMK4450.
- Once the silent reset happens on PM7250B, a PON attempt by KYPD_PWR_N will result in a fault condition. This will trigger a fault sequence in PMK7325/PMK4450 OTP, which will reconfigure PM7250B SID and the communication is restored. Another KYPD_PWR_N trigger or holding KYPD_PWR_N for more than 2 sec can successfully initiate the PON sequence.

Considerations for Swappable Battery Application (cont.)

Use case analysis:

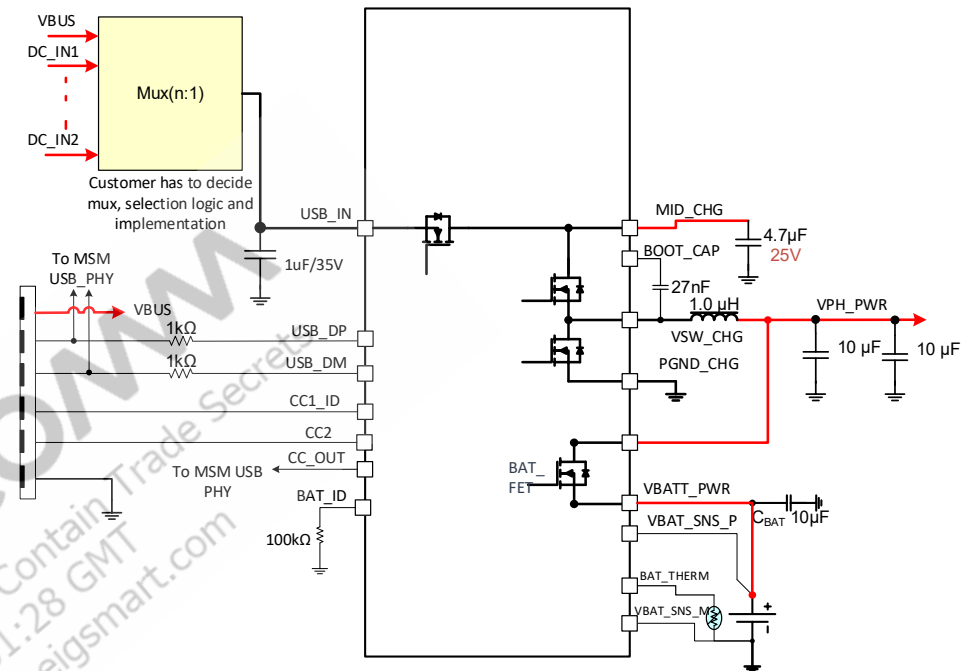
Use case	Details	Comments	Is device recoverable with KYPD_PWR_N?
Dead battery > charger insertion	Since charging is enabled by software, USB plugin would not initiate charging when battery voltage is low (< 2.5 V) and would require power key press (KPD_PWR_N)	This issue happens if the VBAT is within a range of ~2.5 V – 2.0 V. However, the understanding is that there is a deep discharge threshold for battery, the issue will show up only if battery is discharged below 2.5 V but above the battery discharge protection threshold, which is a corner case. If the battery protection gets activated, it will disconnect leading to VBAT falling below ~2.0 V. Once the user plugs a charger it will initiate a complete reset sequence for all PMIC and hence the issue will not be seen.	YES
Load attack at low VBAT	The voltage of VPH_PWR can be lower than ~2.5 V when heavy load happened suddenly, especially low battery.	This couldn't be a problem, because PM7250B dvdd supply is generated from VARB which is a maximum selector between VBUS, VPH and VBOB. So, while the device is ON, if VPH falls due to load attack at lower battery SoC, the dvdd supply is still maintained by VBOB.	N/A
Low temp battery ESR induced VBAT drop	Due to high ESR at extreme low temp (lower than -20°), there is a slim possibility that VBAT could drop below 2.5 V if the VBAT level was already low	Due to high ESR at extreme low temp (lower than -20°), there is possibility VBAT could drop below 2.5 V if the VBAT level was already low. Once the device is returned to room temperature, a PON attempt with charger insertion will fail. However, PON attempt with KPD_PWR_N will still work.	YES
SMPL event	Sudden momentary power loss due to battery mechanical contact	SMPL event during the device operation	YES

Use Case 6: Charging with Multiple Inputs (1 of 3)

The PM7250B charger supports USB_IN and MID_CHG only. If customer has more than two inputs, then customer should design external logic to support multiple inputs.

Hardware configuration

- DC Inputs without USB DP/DM considered as Float charger input.
- If float charger detected. Software set the ICL_MAX to 500 mA. Based on the input selection the ICL_MAX can be override by Software.
- If OTG required, the mux logic should be designed as bidirectional power path. It is better use external boost converter, in order to use OTG during charging via DC_IN.
- Pick the power mux input path with low impedance to avoid IR losses.
- **Note:** The mux should have ability to prioritize the inputs and have make-before-break feature to ensure smooth transition when more than one input is present. In addition, the mux output presents itself as USB-IN to the system so PMIC cannot comprehend the source capacity of various inputs unless ICL is forced to run. It is beyond the scope of this application note to discuss such use cases. Discuss with QTI hardware team with specific product level use cases.



Software configuration

- USB software changes are required to detect the connected UFP/OTG device through CC1_ID and CC2 status and enable the external OTG 5 V using SoC GPIO. Customer can modify this in USB driver and test it. Check with QTI USB software team for support.
- Charger input type can be found from the SCHG_P_USB_APSD_RESULT_STATUS register (0x1308).

Use Case 6: Charging with Multiple Inputs (2 of 3)

During dead battery or shutdown condition, the device will initiate the charging, only if PON is triggered.

Field name	Access	Description
SYSOK_OPTIONS (from Register SCHG_P_MISC_SYSOK_CFG 0x1694)	RW	Selects SYS_OK pin options; 00 = Option 1: SYS_OK high if (SHDN_CMD = 0 and Battery high) OR (DC_IN present) OR (USB_IN present and CC pin debounced) 01 = Option 2: SYS_OK high if (SHDN_CMD = 0 and Battery high) OR (DC_IN present) OR (USB_IN present) 10 = Option 3: SYS_OK high if (SHDN_CMD = 0 and Battery high) 11 = Option 4: Software Override

- Default SYSOK_OPTIONS = 0x00, hence it requires valid input either [USB_IN present and CC pin debounced] or [MID_CHG_IN]
- Float charger (VBUS only) insertion cannot trigger the PON since there is no CC lines.

For float charger PON trigger:

- Mux outputs can be connected to MID_CHG. It can trigger PON through DC_IN_PON.
- If Mux output is connected to VBUS, then it needs a trigger circuit as shown in figure 1.
It will assert the KPD_PWR_N during the float charger insertion.

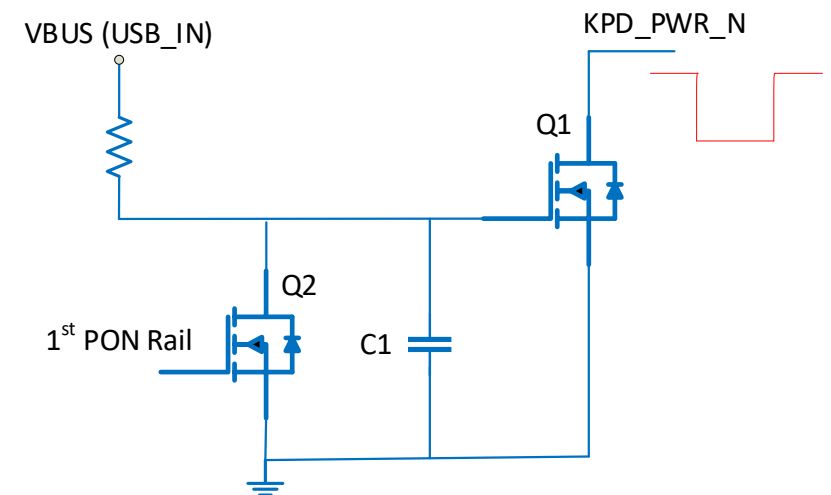


Figure 1: PON trigger circuit for float charger insertion.

Use Case 6: Charging with Multiple Inputs (3 of 3)

NOTES:

- Design a trigger circuit in such a way that it asserts the KPD_PWR_N by driving the Q1 FET.
- Disable the trigger circuit once the PON is initiated. It can be done by turning ON Q2 from the first PON rail, so that VBUS presence will not keep asserting KPD_PWR_N always.
- Ensure 1st PON rail is ON from PON triggered instance till device shutdown. So that it will not let Q1 to assert KPD.
- In external buck powered use cases, the PON trigger maybe required at buck power (that is VBATT power up, also called CBL_PWR_N trigger) in such cases the circuit can be used as shown in the figure 2.
- If a high pass filter logic trigger (figure 3) is used, ensure that the RC delay should assert the KPD_PWR_N for ~120 ms (typical).
- Ensure that VPH does not dip below 2.6 V (dvdd_reset) before/during PON due to inrush or any glitches.
- PON trigger circuits shown in figure 1 and figure 2 are closed loop logic. It is more effective to the open loop PON trigger logic (figure 3).

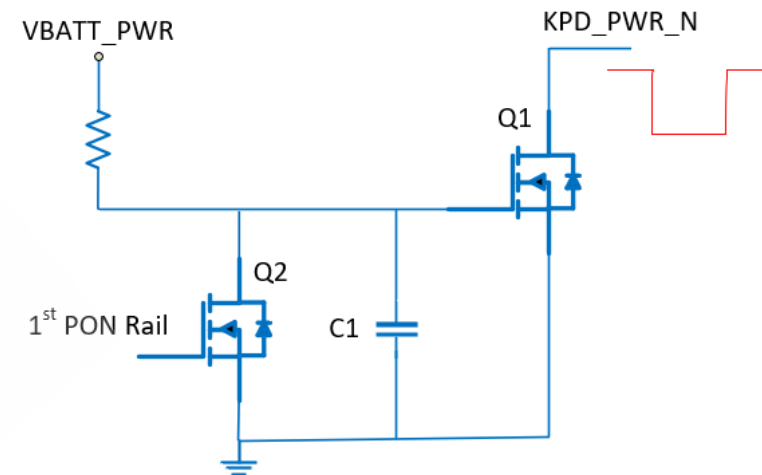


Figure 2. PON trigger circuit for DC Power insertion at VBATT

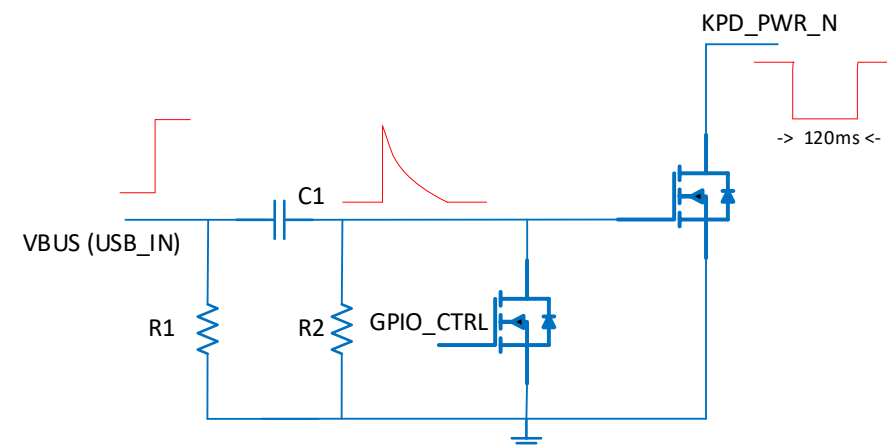


Figure 3. High pass filter logic trigger circuit

VPH_PWR behavior during battery insertion

- In PM7250B, during the first-time battery insertion, there is a voltage step expected in the VPH_PWR.
- VPH_PWR step level may vary depends on the battery voltage level.
- If any circuits which are powering from VPH_PWR should comply with this behavior.
- In QCM4490/QCS4490 chipset, the RGB driver IC with MPN# AW2016A requires a shorter power ON timing specification for proper operation. In that case, it is advisable to use an external load switch on the VPH_PWR.
- Connect VPH_PWR_RGB to VBAT pin of IC with MPN# AW2016A.
- Enable the load switch with one of PON sequence rail turn on early in the PON sequence. Ensure this rail always on even in RBSC state.
- Similarly, if any external circuits require such power on timing, it is advisable to use the load switch.

Note: This is for guidance and not validated by QTI. It is recommended that customers validate the schemes early in the product development.

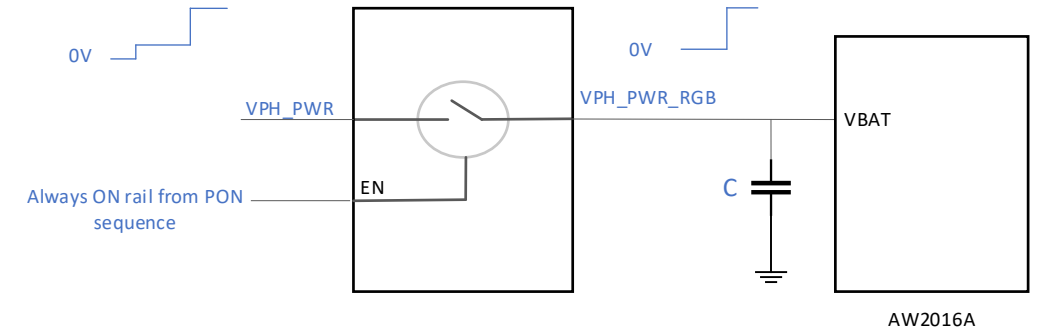
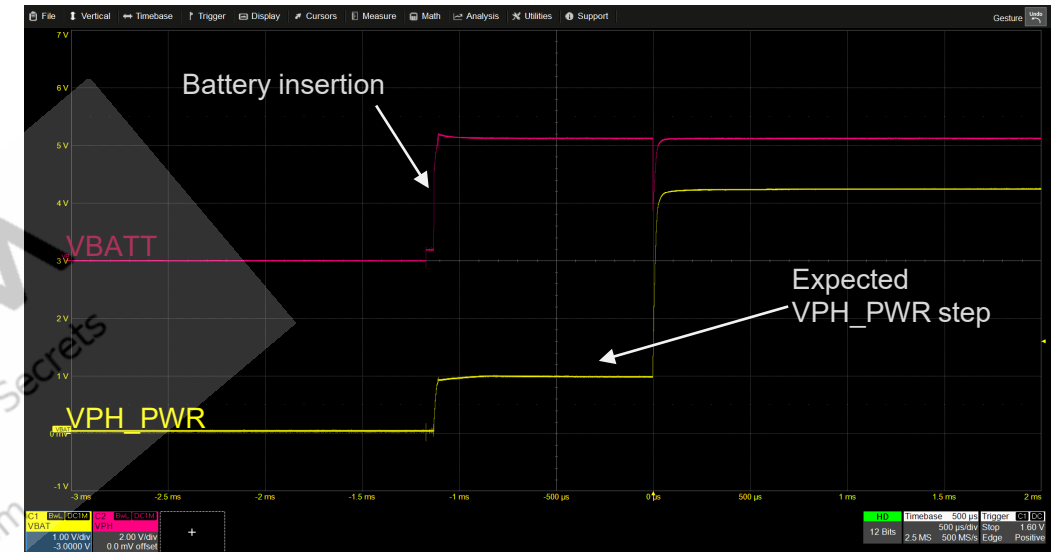


Fig. External load switch to avoid VPH_PWR step.

Guidance on Selecting a Third-party Buck

Transient current conditions on VPH_PWR node

- VPH_PWR (or simply, VPH) is the main power node to the entire chipset
- All PMIC SMPS's are powered from VPH
- VPH is fed either by charger buck (switching charger or SCHG) and/or by the battery
- The SCHG is primarily designed for battery charging. It is not meant to cater the transient current demands of VPH loads. Transient spec for SCHG is neither characterized nor validated by QTI.
- If there is a transient demand at VPH and if SCHG is unable to supply, then a charging battery is temporarily stopped and current is taken from the battery to prevent VPH dropping. This is called battery supplementary mode (BSM). Entry into and exit out of BSM is PMIC hardware autonomous.
- Transient current need at VPH depends upon user application and product use cases (for example, a demanding GPU task, use cases with lot of chipset level concurrencies).

Guidance on selecting a third-party buck

- The third-party buck that powers VPH mimics a battery
- Choose a buck whose transient current spec is at least that of a battery which would otherwise be used.
- OCP setting of such a battery's built-in PCM can be starting point
- Scale the third-party buck's transient capacity based on product's worst case operating conditions and use cases.
- It is also recommended that the extreme case tests be done early in the project cycle

Powering VPH directly vs. through VBAT

- Bypassing BATFET is possible, but fuel-gauging and protection features like Battery Current Limit (BCL) cannot be achieved.
- It is suggested that customer have a bypassing zero ohm (adequately sized for footprint) and decide it based on testing.

Power Grid Consideration

- Connect all the SMPS and LDO regulator output to the loads as per QTI reference schematics. It is not recommended to add any additional load or remove any load from the reference schematics power grid.
- If any changes are made on the power grid, explicitly mention the changes while submitting the schematics review case and get it approved by QCT.

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Questions?

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